National Fuel®
Supply Corporation

VOLUME 1 - PUBLIC INFORMATION

National Fuel Gas Supply Corporation
Prior Notice Application
Shippingport Lateral Project
Docket No. CP25-___-000

Filed at FERC on August 29, 2025

National Fuel®
Supply Corporation

Margaret D. Sroka
Senior Counsel
(716) 857-7066

August 29, 2025

Debbie-Anne A. Reese, Secretary
Federal Energy Regulatory Commission
888 First Street, N.E.
Washington, DC 20426

Re: National Fuel Gas Supply Corporation
Shippingport Lateral Project
Docket No. CP25-___-000

Dear Ms. Reese:

Pursuant to 18 CFR Sections 157.205, 157.208, 157.210 and 157.211 of the Federal Energy Regulatory Commission's ("Commission") regulations and its blanket certificate authority granted in Docket No. CP83-4-000, National Fuel Gas Supply Corporation ("National Fuel") hereby submits for filing this prior notice application requesting authorization to construct and operate approximately 7.26 miles of 24-inch pipeline, a delivery point, pipeline tie, and modifications to certain existing facilities. The facilities are located in Beaver and Mercer counties, Pennsylvania. The proposed project is referred to as the Shippingport Lateral Project.

This application contains Public, Privileged and Critical Energy Infrastructure Information ("CEII") and is organized as follows, consistent with the Commission's filing guidelines.

Volume 1 – Public
Transmittal Letter
Prior Notice Application
Form of Notice
Attachments 1 and 2
Attachment 4 – Concise Environmental Report

Volume 2 – Privileged (CUI//PRIV Contains – Do Not Release)
Resource Report 1 – Appendix B, Landowner List and Appendix K, Residential Site-Specific Construction Plans

Volume 3 – CEII (CUI//CEII – Do Not Release)
Attachment 3 – Flow diagram information (including computer models)

National Fuel requests that, pursuant to 18 CFR § 388.112, the information filed in Volume 2 be treated as privileged and confidential, and that it not be released to the public. National Fuel is requesting privileged treatment of the landowner list for the purpose of landowner privacy. In accordance with Commission policy, the Phase I Archaeological Survey Report is being filed as privileged. Volume 2 is labeled **"CUI//PRIV – Do Not Release"**. The undersigned should be contacted regarding the request for privileged and confidential treatment.

National Fuel further requests that the information in Volume 3 be treated as CEII as it contains information about proposed critical infrastructure that could be useful to any persons planning an attack on that infrastructure as outlined in 18 CFR § 388.113. Volume 3 is labeled **"CUI//CEII – Do Not Release"**.

Also enclosed is a verified statement by a senior officer of National Fuel verifying the information in the application.

In accordance with the Commission's "Filing Guide and Qualified Documents List" that was revised on December 17, 2022, National Fuel will not be providing staff paper copies. However, National Fuel will provide courtesy paper copies upon request.

Please direct any questions regarding this filing to the undersigned at (716) 857-7066 or by email at srokam@natfuel.com.

Respectfully,

/s/ Margaret D. Sroka

Margaret D. Sroka

Enclosure

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

National Fuel Gas Supply Corporation) Docket No. CP25-_____-000

REQUEST FOR AUTHORIZATION UNDER
THE NOTICE PROCEDURES OF §157.205

Shippingport Lateral Project

National Fuel Gas Supply Corporation ("National Fuel"), pursuant to Sections 157.205, 157.208, 157.210 and 157.211 of the regulations of the Federal Energy Regulatory Commission ("Commission"), hereby submits this request for authorization to construct, own, operate, and maintain: (1) approximately 7.26 miles of 24-inch pipeline and appurtenances; (2) a metering and regulating station/delivery point; (3) a pipeline tie-in; and (4) modifications to certain existing facilities and appurtenances, all in Beaver and Mercer counties, Pennsylvania (referred to as the "Shippingport Lateral Project" or the "Project"), all as more fully described below. In support of its request, National Fuel respectfully shows the following:

I.
APPLICANT

The exact legal name of National Fuel is National Fuel Gas Supply Corporation. National Fuel is a corporation organized and existing under the laws of the Commonwealth of Pennsylvania with its principal place of business located in Williamsville, New York. National Fuel is a natural gas company whose jurisdictional activities are subject to regulation by this Commission under the Natural Gas Act ("NGA"). National Fuel is

engaged in the business of transporting and storing natural gas by means of facilities located in the State of New York and the Commonwealth of Pennsylvania. National Fuel was granted a blanket certificate for routine transactions in Docket No. CP83-4-000.[1]

II.

COMMUNICATIONS

The name, title and mailing address of the persons who should be served with communications concerning this request are:

Margaret D. Sroka[2]
Senior Counsel
National Fuel Gas Supply Corporation
6363 Main Street
Williamsville, New York 14221
Telephone No. (716) 857-7066
Facsimile No. (716) 857-7206
srokam@natfuel.com

Janet R. Bayer[3]
Director, Certificates and Tariffs
National Fuel Gas Supply Corporation
6363 Main Street
Williamsville, New York 14221
Telephone No. (716) 857-7429
Facsimile No. (716) 857-7206
jrbferc@natfuel.com

Kenneth E. Webster[3]
General Counsel, Secretary
and Assistant Vice President
National Fuel Gas Supply Corporation
6363 Main Street
Williamsville, New York 14221
Telephone No. (716) 857-7067
Facsimile No. (716) 857-7206
websterk@natfuel.com

[1] *National Fuel Gas Supply Corporation*, 21 FERC ¶ 62,298 (1982).

[2] Designated as the responsible National Fuel officials under 157.205(b)(1) of the Commission's regulations who should be served with communications concerning this filing.

[3] Designated to receive service pursuant to Rule 2010 of the Commission's regulations. National Fuel respectfully requests that the Commission waive 18 CFR § 385.203(b)(3) in order to allow National Fuel to include three representatives on the official service list for this proceeding.

III.

BACKGROUND

In connection with the Project, National Fuel proposes to construct and operate 7.26 miles of new 24-inch diameter pipeline, a new delivery point interconnection facility, a pipeline tie and associated facilities, all as described more fully below and in Resource Report 1 of the Concise Environmental Report (see Attachment 4). Completion of these facilities will allow National Fuel to provide 205,000 dekatherms per day ("Dth/day") of incremental firm transportation service on its system, transporting these volumes from receipt points on National Fuel's Line N pipeline to a new interconnection with the Shippingport Power Station (the "SPS Facility").

The Project is needed to support the power generation needs at the SPS Facility, located in Shippingport Borough, Pennsylvania, at the former site of the Bruce Mansfield Power Plant. The redevelopment project is estimated to generate more than $6 billion of new economic activity, encompassing direct spending, indirect spending and induced economic output.[4]

In order to meet the in-service date requested by the Project Shipper, and to expeditiously advance this critical infrastructure project in the Commonwealth of Pennsylvania, National Fuel plans to begin clearing the proposed right-of-way during the winter season and anticipates completing all tree clearing activities prior to April 1, 2026.

[4] See July 15, 2025 press release, The Frontier Group of Companies to Transform Bruce Mansfield Power Plant into State-of-the-Art Natural Gas Power Plant, Supporting America's Energy Goals and Pennsylvania's Economic Growth, *available at* https://www.businesswire.com/news/home/20250715103342/en/The-Frontier-Group-of-Companies-to-Transform-Bruce-Mansfield-Power-Plant-into-State-of-the-Art-Natural-Gas-Power-Plant-Supporting-Americas-Energy-Goals-and-Pennsylvanias-Economic-Growth.

Construction of the Project is expected to commence in Spring 2026, with a target in service-date for the facilities of November 1, 2026.

IV.
DETAILED INFORMATION

A. Pursuant to Section 157.208(c), National Fuel submits the following information regarding the proposed pipeline:

(1) Purpose of the proposed facilities:

The purpose of the Project is to allow National Fuel to deliver 205,000 Dth/day of natural gas from its Line N20 to a new delivery point at the SPS Facility. The Project shipper, Shippingport Power Station, LLC ("Shipper"), has requested firm transportation service of 125,000 Dth/day of gas from National Fuel's existing Mercer Interconnection, and an additional 80,000 Dth/day from National Fuel's existing Market Area Pooling Point at Aliquippa, with all such volumes to be delivered to a new delivery point ("Shippingport M&R Station") at the SPS Facility.

(2) Description of the proposed facilities:

- Shippingport Lateral (Line N20B5) – Approximately 7.26 miles of 24-inch diameter, coated steel natural gas pipeline connecting National Fuel's Line N20 to the Shippingport Industrial Park. Line N20B5 will have a maximum allowable operating pressure of 1,440 psig and will have a pipe wall thickness ranging between 0.500 and 0.562 inch. Cathodic protection for Line N20B5 will be provided using magnesium anodes located entirely within the pipeline right-of-way.

- Service Creek Tie at Line N20 – Installation of a 24-inch bridle that will connect National Fuel's existing Line N20 to the proposed Shippingport Lateral (Line N20B5) and will include pig launchers, gas quality equipment, over pressure protection equipment, station fencing and security system, and ancillary equipment. The Service Creek Tie facility will be located adjacent to National Fuel's existing Service Creek pig launcher facility.

- Shippingport M&R Station (Delivery Point)[5] – Installation of a new measurement and regulation (M&R) station at the Shippingport Industrial Park to interconnect the Line N20B5 with the SPS Facility for delivery of natural gas.

- Minor Modifications at Mercer Compressor Station ("Mercer CS") – Installation of crossover valves and auxiliary facilities at National Fuel's existing Mercer CS.[6] No compression or horsepower upgrades are proposed as part of the Project, and no changes to the FERC certificated parameters for the Mercer CS are proposed.

(3) Location of the Proposed Facilities:

- Shippingport Lateral (Line N20B5) – the pipeline will traverse Independence Township, Raccoon Township, and Shippingport Borough, all in Beaver County, Pennsylvania.

- Service Creek Tie at Line N20 – at the east end of Line N20B5 in Independence Township, Beaver County, Pennsylvania.

- Shippingport M&R Station (Delivery Point) – at the west end of Line N20B5 in Shippingport Borough, Beaver County, Pennsylvania.

- Minor Modifications at Mercer CS – Jefferson Township, Mercer County, Pennsylvania.

(4) Maps showing the relationship of the proposed Project to National Fuel's existing facilities:

See Attachment 1 for a general project map and Attachment 2 for a topographic map of the Project locations.

(5) Flow Diagram:

See Attachment 3. National Fuel is filing the flow diagram information as Critical Energy Infrastructure Information and has labeled it "**CUI//CEII – Do Not Release**". This information is included in Volume 3.

[5] The Shippingport M&R Station is being installed pursuant to Section 157.211 and is discussed in more detail in Part B below.

[6] All work at the existing Mercer CS will be completed within the existing compressor station footprint. The work at the Mercer CS will be completed under 18 CFR § 2.55(a).

(6) The Estimated Cost and Financing:

The estimated capital cost of the Project facilities is $57.3 million[7]. The Project will be financed with internally generated funds and/or short-term bank loans.

(7) Public Convenience and Necessity:

National Fuel submits that this proposal is in the public convenience and necessity because it will provide capacity necessary to perform a requested firm transportation service. National Fuel states that it is willing and able to perform the activities proposed herein, and in so doing, conform to the applicable provisions of the Natural Gas Act and the Commission's regulations.

(8) Acquisitions of Facilities: Not applicable.

(9) Environmental Analysis:

A Concise Environmental Report is attached hereto as Attachment 4.

(10) Environmental Inspector's Report:

National Fuel will comply with this reporting requirement.

B. Pursuant to Section 157.211(a)(2), National Fuel submits the following information regarding the proposed new delivery point:

(1) The name of the end-user, the location of the delivery point, and the distribution company currently serving the end-user:

- End-user – Shippingport Power Station, LLC
- Location – Shippingport Borough, Beaver County, Pennsylvania
- Currently the end-user is not served by a local distribution company [8]

[7] The Project will be completed pursuant to the Commission's Order Granting Temporary Waiver of Regulations to Increase Blanket Cost Limits. See *Interstate Nat. Gas Ass'n of Am.*, 191 FERC ¶ 61,206 (2025).

[8] National Fuel will comply with any applicable notification requirements pursuant to 18 CFR § 284.13(e).

(2) A description of the facility and any appurtenant facilities:

The Shippingport M&R Station will be located at the western end of proposed Line N20B5, which will serve as a delivery point to the SPS Facility. The Shippingport M&R Station will include the installation of a 24-inch pig receiver, over pressure protection equipment, gas filtration equipment, control valves, measurement equipment, station fencing and a security system.

(3) A USGS topographic map showing the location of the proposed facilities:

See Attachment 2.

(4) The maximum quantity of gas to be delivered through the proposed facility:

205,000 Dth/d.

(5) A description, with supporting data, of the impact of the service rendered through the proposed delivery tap upon certificate holder's peak day and annual deliveries:

See Attachment 3. National Fuel is filing the flow diagram information as Critical Energy Infrastructure Information and has labeled it "**CUI//CEII – Do Not Release**". This information is included in Volume 3.

V.

AFFECTED LANDOWNERS

As set forth in the Concise Environmental Report (Attachment 4), National Fuel will comply with the Commission's landowner notification requirements set forth at Section 157.203(d)(2) of the Commission's regulations.

Pursuant to 18 C.F.R 388.112, National Fuel is filing the landowner list for the Project as confidential information as it contains landowner names and addresses which

should not be posted on the internet or made available in the Commission's Public Reference Room. The list is being furnished directly to Commission Staff and is labeled "**CUI//PRIV – DO NOT RELEASE"**. This information is included in Volume 2.

VI.

FORM OF NOTICE

A Form of Notice suitable for publication in the Federal Register is attached hereto.

VII.

CONCLUSION

Wherefore, National Fuel respectfully requests that the Commission grant it authority under its blanket certificate authorizing the activities described herein.

Respectfully submitted,

National Fuel Gas Supply Corporation

By: /s/ *Margaret D. Sroka*

Margaret D. Sroka
Senior Counsel
National Fuel Gas Supply Corporation
6363 Main Street
Williamsville, NY 14221
Telephone (716) 857-7066
Facsimile (716) 857-7206
srokam@natfuel.com

August 29, 2025

VERIFICATION

18 CFR § 157.205(b)(4)

Shippingport Lateral Project

STATE OF NEW YORK)
) ss:
COUNTY OF ERIE)

Matthew E. Frank, being duly sworn on his oath, deposes and says that he is Senior Vice President of National Fuel Gas Supply Corporation; that as such he is authorized to verify the foregoing filing; that he has read and is familiar with the contents thereof; and that the statements made therein are true and accurate to the best of his information, knowledge and belief; and that the activities proposed in the foregoing filing will comply with the requirements of Subpart F of Part 157 of the regulations of the Federal Energy Regulatory Commission.

Matthew E. Frank

SUBSCRIBED AND SWORN TO before me this 28th day of August, 2025.

COURTNEY AGIUS
Notary Public - State Of New York
No. 01AG6382086
Qualified in Erie County
My Commission Expires October 15, 2026

Notary

FORM OF NOTICE

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

National Fuel Gas Supply Corporation) Docket No. CP25-_____

NOTICE OF REQUEST UNDER BLANKET AUTHORIZATION AND ESTABLISHING INTERVENTION AND PROTEST DEADLINE

(September __, 2025)

Take notice that on August __, 2025, National Fuel Gas Supply Corporation (National Fuel) 6363 Main Street, Williamsville, New York 14221, filed in the above referenced docket, a prior notice request pursuant to sections 157.205, 157.208, 157.210 and 157.211 of the Federal Energy Regulatory Commission's regulations under the Natural Gas Act (NGA) and National Fuel's blanket certificate issued in Docket No. CP83-4-000, requesting authorization to construct, operate and maintain (1) approximately 7.26 miles of 24-inch pipeline and appurtenances; (2) a metering and regulating station/delivery point; (3) a pipeline tie; and (4) modifications to certain existing facilities and appurtenances, all in Beaver and Mercer counties, Pennsylvania. The Project will allow National Fuel to provide 205,000 dekatherms per day of firm transportation service on its system to the project shipper, Shippingport Power Station, LLC. The estimated capital cost for the project is $57.3 million, all as more fully set forth in the request which is on file with the Commission and open to public inspection. The proposed project is referred to as the Shippingport Lateral Project.

In addition to publishing the full text of this document in the Federal Register, the Commission provides all interested persons an opportunity to view and/or print the contents of this document via the Internet through the Commission's Home Page (http://www.ferc.gov). From the Commission's Home Page on the Internet, this information is available on eLibrary. The full text of this document is available on eLibrary in PDF and Microsoft Word format for viewing, printing, and/or downloading. To access this document in eLibrary, type the docket number excluding the last three digits of this document in the docket number field.

User assistance is available for eLibrary and the Commission's website during normal business hours from FERC Online Support at (202) 502-6652 (toll free at 1-866-208-3676) or email at ferconlinesupport@ferc.gov, or the Public Reference Room at (202) 502-8371, TTY (202) 502-8659. E-mail the Public Reference Room at public.referenceroom@ferc.gov.

Any questions concerning this application should be directed to Margaret Sroka, Senior Counsel, National Fuel Gas Supply Corporation, 6363 Main Street, Williamsville, New York 14221, by telephone at (716) 857-7066, or by email at srokam@natfuel.com.

PUBLIC PARTICIPATION

There are three ways to become involved in the Commission's review of this project: you can file a protest to the project, you can file a motion to intervene in the proceeding, and you can file comments on the project. There is no fee or cost for filing protests, motions to intervene, or comments. The deadline for filing protests, motions to intervene, and comments is 5:00 p.m. Eastern Time on **_____, 2025**. How to file protests, motions to intervene, and comments is explained below.

Protests

Pursuant to section 157.205 of the Commission's regulations under the NGA,[9] any person[10] or the Commission's staff may file a protest to the request. If no protest is filed within the time allowed or if a protest is filed and then withdrawn within 30 days after the allowed time for filing a protest, the proposed activity shall be deemed to be authorized effective the day after the time allowed for protest. If a protest is filed and not withdrawn within 30 days after the time allowed for filing a protest, the instant request for authorization will be considered by the Commission.

Protests must comply with the requirements specified in section 157.205(e) of the Commission's regulations,[11] and must be submitted by the protest deadline, which is **_____, 2025**. A protest may also serve as a motion to intervene so long as the protestor states it also seeks to be an intervenor.

Interventions

Any person has the option to file a motion to intervene in this proceeding. Only intervenors have the right to request rehearing of Commission orders issued in this proceeding and to subsequently challenge the Commission's orders in the U.S. Circuit Courts of Appeal.

To intervene, you must submit a motion to intervene to the Commission in accordance with Rule 214 of the Commission's Rules of Practice and Procedure[12] and the regulations under the NGA[13] by the intervention deadline for the project, which is **_____, 2025**. As described further in Rule 214, your motion to intervene must state, to the extent known, your position regarding the proceeding, as well as your interest in the proceeding. For an individual, this could include your status as a landowner, ratepayer, resident of an impacted community, or recreationist. You do not need to have property directly impacted by the project in order to intervene. For more information about motions to intervene, refer to the FERC website at https://www.ferc.gov/resources/guides/how-to/intervene.asp.

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[9] 18 C.F.R. 157.205.

[10] Persons include individuals, organizations, businesses, municipalities, and other entities. 18 C.F.R. 385.102(d).

[11] 18 C.F.R. 157.205(e).

[12] 18 C.F.R. 385.214.

[13] 18 C.F.R. 157.10.

All timely, unopposed motions to intervene are automatically granted by operation of Rule 214(c)(1). Motions to intervene that are filed after the intervention deadline are untimely and may be denied. Any late-filed motion to intervene must show good cause for being late and must explain why the time limitation should be waived and provide justification by reference to factors set forth in Rule 214(d) of the Commission's Rules and Regulations. A person obtaining party status will be placed on the service list maintained by the Secretary of the Commission and will receive copies (paper or electronic) of all documents filed by the applicant and by all other parties.

Comments

Any person wishing to comment on the project may do so. The Commission considers all comments received about the project in determining the appropriate action to be taken. To ensure that your comments are timely and properly recorded, please submit your comments on or before _____**, 2025**. **The filing of a comment alone will not serve to make the filer a party to the proceeding.** To become a party, you must intervene in the proceeding.

How to File Protests, Interventions, and Comments

There are two ways to submit protests, motions to intervene, and comments. In both instances, please reference the Project docket number **CP25-___-000** in your submission.

(1) You may file your protest, motion to intervene, and comments by using the Commission's eFiling feature, which is located on the Commission's website (www.ferc.gov) under the link to Documents and Filings. New eFiling users must first create an account by clicking on "eRegister." You will be asked to select the type of filing you are making; first select General" and then select "Protest", "Intervention", or "Comment on a Filing." The Commission's eFiling staff are available to assist you at (202) 502-8258 or FercOnlineSupport@ferc.gov.

(2) You can file a paper copy of your submission. Your submission must reference the Project docket number **CP25-___-000**.

To mail via USPS, use the following address:

Debbie-Anne A. Reese, Secretary
Federal Energy Regulatory Commission
888 First Street NE
Washington, DC 20426

To mail via any other courier, use the following address:

Debbie-Anne A. Reese, Secretary
Federal Energy Regulatory Commission
12225 Wilkins Avenue
Rockville, Maryland 20852

Protests and motions to intervene must be served on the applicant either by mail or email (with a link to the document) to Margaret Sroka, Senior Counsel, National Fuel Gas Supply Corporation, 6363 Main Street, Williamsville, New York 14221, or by email at srokam@natfuel.com. Any subsequent submissions by an intervenor must be served on the applicant and all other parties to the proceeding. Contact information for parties can be downloaded from the service list at the eService link on FERC Online.

TRACKING THE PROCEEDING

Throughout the proceeding, additional information about the project will be available from the Commission's Office of External Affairs, at **(866) 208-FERC**, or on the FERC website at www.ferc.gov using the "eLibrary" link as described above. The eLibrary link also provides access to the texts of all formal documents issued by the Commission, such as orders, notices, and rulemakings.

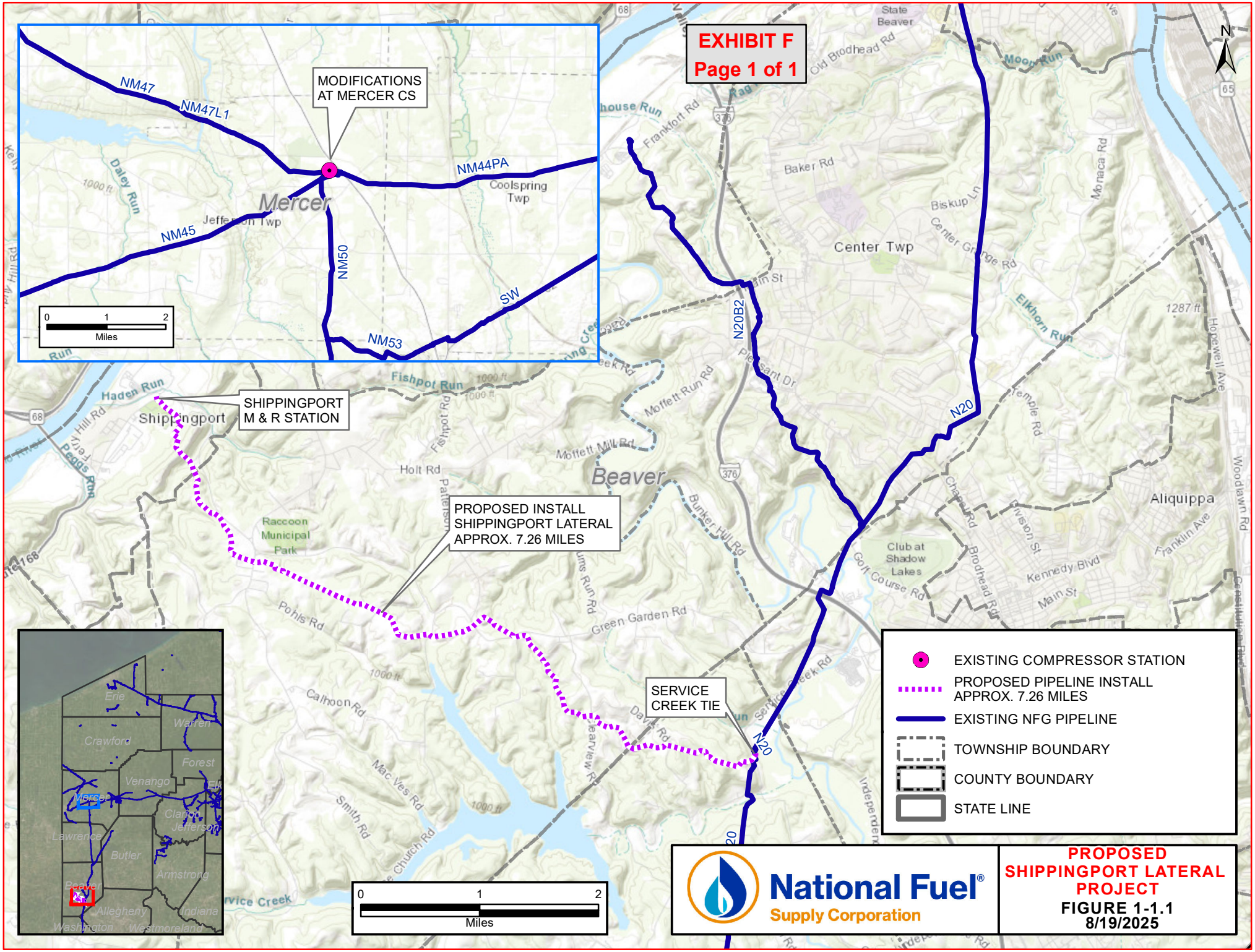
In addition, the Commission offers a free service called eSubscription which allows you to keep track of all formal issuances and submittals in specific dockets. This can reduce the amount of time you spend researching proceedings by automatically providing you with notification of these filings, document summaries, and direct links to the documents. For more information and to register, go to www.ferc.gov/docs-filing/esubscription.asp.

Debbie-Anne A. Reese,
Secretary.

List of Attachments

Attachment 1	System Map
Attachment 2	Topographic Map
Attachment 3	Flow Diagram (Filed as CEII)
Attachment 4	Concise Analysis of Environmental Resources (§ 380.12)

ATTACHMENT 1


EXHIBIT F
Page 1 of 1
MODIFICATIONS AT MERCER CS
NM47
NM47L1
NM44PA
NM45
NM50
NM53
SW
Mercer
Coolspring Twp
Jefferson Twp
Daley Run
0 1 2 Miles
SHIPPINGPORT M & R STATION
Shippingport
Haden Run
Fishpot Run
PROPOSED INSTALL SHIPPINGPORT LATERAL APPROX. 7.26 MILES
Raccoon Municipal Park
Pohls Rd
Holt Rd
Moffett Mill Rd
Moffett Run Rd
Beaver
Bunker Hill Rd
Green Garden Rd
Calhoun Rd
Mac Ves Rd
Smith Rd
Church Rd
SERVICE CREEK TIE
N20
N20B2
Center Twp
Baker Rd
Frankfort Rd
Old Brodhead Rd
Moon Run
Elkhorn Run
Aliquippa
Club at Shadow Lakes
Kennedy Blvd
Franklin Ave
Main St
Brodhead Rd
Golf Course Rd
Chapel Rd
Temple Rd
Monaca Rd
Hopewell Ave
Woodlawn Rd
Biskup Ln
Center Grange Rd
Pleasant Dr
Peggs Run Rd
Fishpot Rd
Patterson
Erie
Warren
Crawford
Forest
Venango
Clarion
Jefferson
Lawrence
Butler
Armstrong
Beaver
Allegheny
Indiana
Washington
Westmoreland
0 1 2 Miles
EXISTING COMPRESSOR STATION
PROPOSED PIPELINE INSTALL APPROX. 7.26 MILES
EXISTING NFG PIPELINE
TOWNSHIP BOUNDARY
COUNTY BOUNDARY
STATE LINE
National Fuel®
Supply Corporation
PROPOSED SHIPPINGPORT LATERAL PROJECT
FIGURE 1-1.1
8/19/2025
File Path: P:\Transmission\Projects\Expansion\2025\Supply\Line N - Shippingport Lateral\Map Requests\MXDs\ProposedNLateral_ExhibitF.mxd

ATTACHMENT 2

Shippingport M&R Station
Staging Area 2
Midland
Hookstown
Staging Area 3
Staging Area 5
Staging Area 4
Hookstown
Aliquippa
Midland
Beaver
Beaver
Aliquippa
Staging Area 6
Staging Area 7
Staging Area 8
Service
Creek Tie
RACCOON
CENTER
RACCOON



- Line N Lateral
- Permanent Access Road (PAR)
- Temporary Access Road (TAR)
- Staging Areas
- Service Creek Tie
- Shippingport M&R Station
- USGS Quadrangle Boundary

ATTACHMENT 2

0 670 1,340 Meters

0 2,000 4,000 Feet

Scale: 1:28,000

Source: ESRI US Topo 2013
Spatial Reference: NAD 1983 2011 StatePlane Pennsylvania South FIPS 3702 Ft US

N

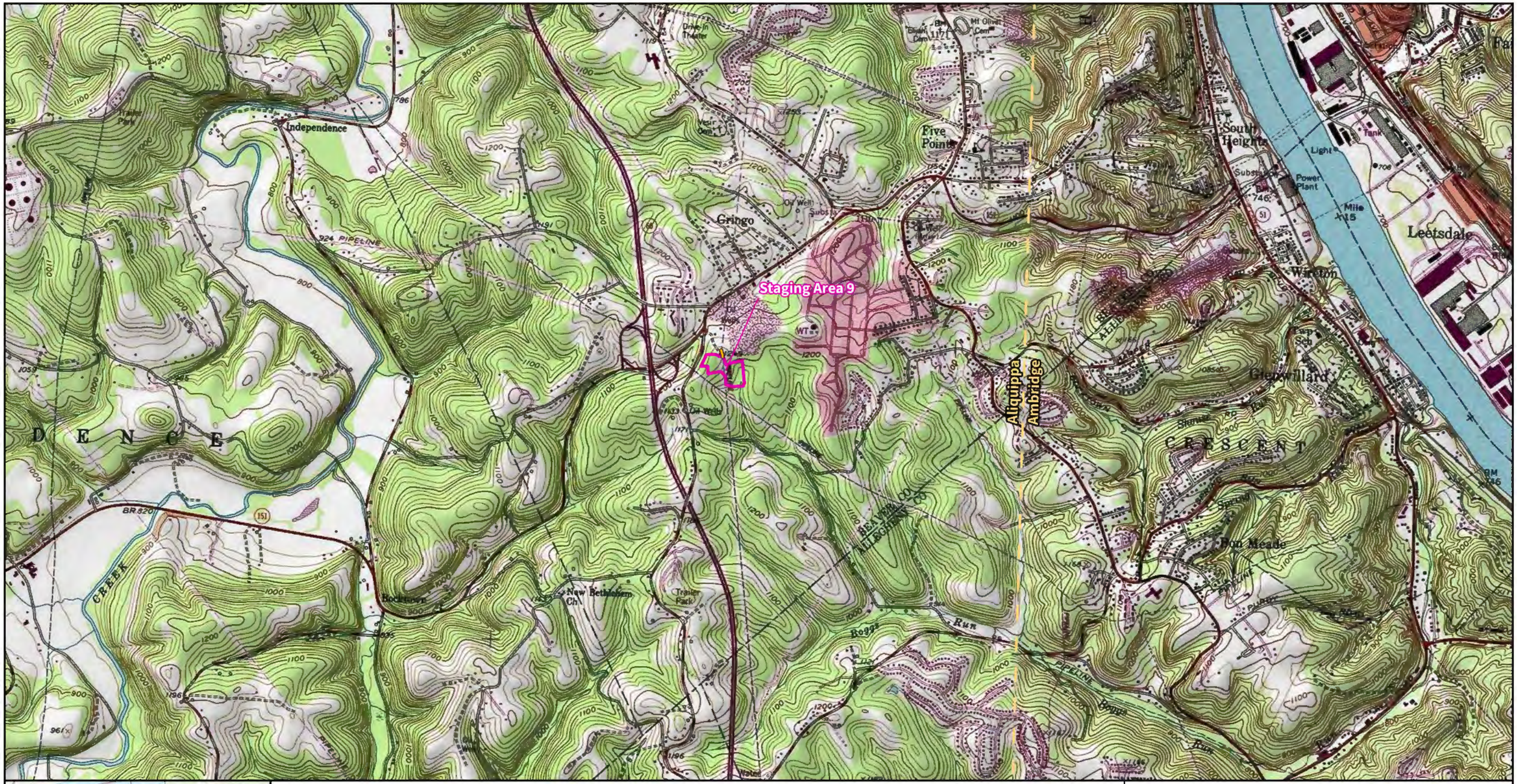
Shippingport Lateral Project

USGS Project Location
Beaver County, Pennsylvania

Prepared For: National Fuel Supply Corporation	Sheet 1 of 3
Prepared By: TETRA TECH	Date: 8/18/2025

Path: C:\Users\Julie.gibson\Tetra Tech, Inc\My Data Science - My PCP\212C-BF-00598 NFG Line N\3 Client Deliverables\P1B_Figures.aprx

Cartographer: JULIE.GIBSON





Temporary Access Road (TAR)
Staging Areas
USGS Quadrangle Boundary

ATTACHMENT 2

0 575 1,150 Meters

0 2,000 4,000 Feet

Scale: 1:24,000

Source: ESRI US Topo 2013
Spatial Reference: NAD 1983 2011 StatePlane Pennsylvania South FIPS 3702 Ft US

N

Shippingport Lateral Project

USGS Project Location
Beaver County, Pennsylvania

Prepared For: National Fuel Supply Corporation	Sheet 2 of 3
Prepared By: TETRA TECH	Date: 8/18/2025

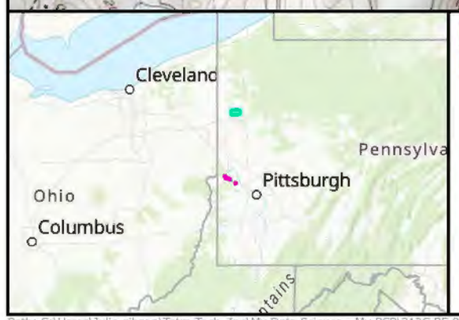
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Cartographer: JULIE.GIBSON

Mercer Compressor Station

Fredonia

Jackson Center



Mercer Compressor Station

USGS Quadrangle Boundary

ATTACHMENT 2

0 575 1,150 Meters

0 2,000 4,000 Feet

Scale: 1:24,000

Source: ESRI US Topo 2013
Spatial Reference: NAD 1983 2011 StatePlane Pennsylvania South FIPS 3702 Ft US

N

Shippingport Lateral Project

USGS Project Location
Mercer County, Pennsylvania

Prepared For: National Fuel Supply Corporation	Sheet 3 of 3
Prepared By: TETRA TECH	Date: 8/18/2025

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Cartographer: JULIE.GIBSON

National Fuel Gas Supply Corporation
Shippingport Lateral Project
Docket No. CP25-___-00

INFORMATION HAS BEN REMOVED FOR CEII TREATMENT
See Volume 3 marked "CUI//CEII – DO NOT RELEASE"

ATTACHMENT 3

Flow Diagram Data

Matters relating to CEII information should be directed to*

Margaret D. Sroka
Senior Counsel
National Fuel Gas Supply Corporation
6363 Main Street
Williamsville, New York 14221
Telephone No. (716) 857-7066
srokam@natfuel.com

ATTACHMENT 4

VOLUME 1 – PUBLIC

National Fuel®
Supply Corporation

CONCISE ENVIRONMENTAL REPORT

NATIONAL FUEL GAS SUPPLY CORPORATION
SHIPPINGPORT LATERAL PROJECT

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CONCISE ENVIRONMENTAL REPORT
FOR THE
NATIONAL FUEL GAS SUPPLY CORPORATION
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LIST OF ACRONYMS AND ABBREVIATIONS

ac	acre or acres
ACHP	Advisory Council on Historic Preservation
ACS	American Community Survey (U.S. Census)
amsl	above mean sea level
APE	Area of Potential Effect
AQCR	Air Quality Control Region
ATWS	additional temporary workspace
AWC	available water capacity
BMPs	best management practices
CAA	Clean Air Act
CH_4	methane
CO	carbon monoxide
CO_2e	carbon dioxide equivalent
CFR	Code of Federal Regulations
cm	centimeters
Commission	Federal Energy Regulatory Commission
CS	compressor station
CUI//PRIV	Controlled Unclassified Information/Privileged
CWA	Clean Water Act
CWF	Coldwater Fishes
D-Key	USFWS Determination Key (online system)
dB	decibel(s)
dBA	A-weighted decibels
E&SCP	Erosion and Sedimentation Control Plan
ESCAMP	National Fuel's Erosion and Sedimentation Control and Agricultural Mitigation Plan
°F	degrees Fahrenheit
FEMA	Federal Emergency Management Agency
FERC	Federal Energy Regulatory Commission
FERC Plan	Upland Erosion Control, Revegetation, and Maintenance Plan
FERC Procedures	Wetland and Waterbody Construction and Mitigation Procedures
Frontier	Frontier Group of Companies
FRS	Facility Registry Service
ft	foot or feet
GHG	greenhouse gases
gpm	gallons per minute
HQ	Pennsylvania-designated High Quality Waters
Hz	Hertz
GWP	global warming potential

HAP	Hazardous Air Pollutant
HUC	Hydrologic Unit Code
IBA	Important Bird Area
IPaC	Information for Planning and Consultation
ISO	Organization for International Standardization
L_{dn}	day-night sound level
L_{eq}	equivalent sound level
LOD	Project limits of disturbance
LNG	liquefied natural gas
L_{max}	maximum instantaneous sound level measured during a specific timepoint
M&R	measurement and regulation
MACT	Maximum Achievable Control Technology
MBTA	Migratory Bird Treaty Act of 1918
mi	mile or miles
MLRC	Multi-Resolution Land Characteristics Consortium
Monarch CCAA	Nationwide Monarch Butterfly Candidate Conservation Agreement for Energy and Transportation Lands
MP	milepost
N_2O	nitrous oxide
N/A	not applicable
NAAQS	National Ambient Air Quality Standards
National Fuel	National Fuel Gas Supply Corporation
NEPA	National Environmental Policy Act
NESHAPs	National Emission Standards for Hazardous Air Pollutants
NHA	Natural Heritage Area
NHPA	National Historic Preservation Act
NLCD	National Land Cover Database
NLEB	northern long-eared bat
NO_2	nitrogen dioxide
NO_x	oxides of nitrogen
NOAA	National Oceanic and Atmospheric Administration
NPS	National Park Service
NRCC	Northeastern Regional Climate Center
NRCS	Natural Resources Conservation Service
NRHP	National Register of Historic Places
NSA	noise sensitive area
NSPS	New Source Performance Standards
NSR	New Source Review
NWI	National Wetlands Inventory
OPP	over pressure protection

OTR	Ozone Transportation Region
PA	Pennsylvania
PA DCNR	Pennsylvania Department of Conservation and Natural Resources
PA DEP	Pennsylvania Department of Environmental Protection
PA DOT	Pennsylvania Department of Transportation
PA FBC	Pennsylvania Fish and Boat Commission
PA GC	Pennsylvania Game Commission
PA GD	Pennsylvania GeoData
PA NHP	Pennsylvania Natural Heritage Program
PA SDA	Pennsylvania Spatial Data Access
PA-SHARE	Pennsylvania Historic and Archaeological Resources Exchange
PA SHPO	Pennsylvania State Historic Preservation Office
PASS	Pennsylvania Archaeological Site Survey
PAR	permanent access road
PCBs	polychlorinated biphenyls
PCSM	post-construction stormwater management plan
PEM	palustrine emergent
PFO	palustrine forested
pga	peak ground acceleration
PHMC	Pennsylvania Historical and Museum Commission
PM	particulate matter
PM_{10}	particulate matter less than 10 microns in diameter
$PM_{2.5}$	particulate matter less than 2.5 microns in diameter
PNDI	Pennsylvania Natural Diversity Inventory
ppb	parts per billion by volume
PPC Plan	Preparedness, Prevention, and Contingency and Emergency Action Plan
ppm	parts per million by volume
Project	National Fuel Shippingport Lateral Project
PSD	Prevention of Significant Deterioration
PSS	palustrine scrub-shrub
PSIG	Pounds per square inch gauge
ROW	right-of-way
SIP	State Implementation Plan
SO_2	sulfur dioxide
SPS	Shippingport Power Station
SRBC	Susquehanna River Basin Commission
T&E	threatened and endangered
TAR	temporary access road
TDAT	Tribal Directory Assessment Tool
Tetra Tech	Tetra Tech, Inc.

THPO	Tribal Historic Preservation Office(er)
tpy	tons per year
TWS	temporary workspace
UNT	unnamed tributary
U.S.	United States
USACE	United States Army Corps of Engineers
USCB	United States Census Bureau
USDA	United States Department of Agriculture
USDOT	United States Department of Transportation
USEPA	United States Environmental Protection Agency
USFS	United States Forest Service
USFWS	United States Fish and Wildlife Service
USGS	United States Geological Survey
$\mu g/m^3$	micrograms per cubic meter
VOC	volatile organic compound
WWF	Warmwater Fishes

RESOURCE REPORT 1
GENERAL PROJECT DESCRIPTION

1.0 GENERAL PROJECT DESCRIPTION

This Concise Environmental Report addresses the information requirements pursuant to Code of Federal Regulations (CFR) sections 157.205, 157.208(b) and (c), 157.210, 157.211(b) and (c), and section 380.12 of the Federal Energy Regulatory Commission's (FERC or Commission) regulations for a Blanket Certificate Prior Notice Request of National Fuel Gas Supply Corporation (National Fuel) for authorization of the proposed Shippingport Lateral Project (Project) located in Beaver and Mercer counties, Pennsylvania (PA). A map depicting the Project locations is provided in Appendix A-1.

1.1 PROPOSED FACILITIES, DESCRIPTION, AND LOCATION

National Fuel proposes to construct and operate the following facilities:

- Shippingport Lateral (Line N20B5) - A new approximately 7.26-mile, 24-inch diameter, coated steel natural gas pipeline connecting National Fuel's Line N20 natural gas transmission pipeline to the Shippingport Industrial Park in Shippingport Borough, Beaver County, PA. The pipeline will traverse Independence Township, Raccoon Township, and Shippingport Borough in Beaver County, PA.
- Service Creek Tie on Line N20 - Installation of a 24-inch bridle that will connect National Fuel's existing Line N20 to the proposed Line N20B5, pig launcher, gas quality equipment, over pressure protection (OPP), station fencing and security systems, and ancillary equipment adjacent to National Fuel's existing Service Creek pigging facility at the east end of the proposed Line N20B5 in Independence Township, Beaver County, PA.
- Shippingport M&R Station (Delivery Point)- A new measurement and regulation (M&R) station (Shippingport M&R Station) at Shippingport Industrial Park that includes the installation of a 24-inch pig receiver, OPP, gas filtration equipment, control valves, measurement equipment, station fencing and security systems, and ancillary equipment at the west end of proposed Line N20B5in Shippingport Borough, Beaver County, PA.
- Modifications at Mercer Compressor Station (Mercer CS) - Installation of crossover valves and auxiliary facilities at National Fuel's existing Mercer CS in Jefferson Township, Mercer County, PA.
- Other:
    - Temporary use of nine (9) staging/storage areas totaling 14.21 acres (ac).
    - Temporary or permanent use of 21 access roads totaling 3.28 miles.

Line N20B5 will require a new 75-foot-wide permanent right-of-way (ROW). Line N20B5 will have a maximum allowable operating pressure of 1,440 pounds per square inch gauge (PSIG) and will have a pipe wall thickness ranging from 0.500 to 0.562 inch. Cathodic protection for the pipeline will be provided using magnesium anodes located

entirely within the pipeline ROW. The Service Creek Tie will be sited adjacent to National Fuel's existing Service Creek pigging facilities at the east end of Line N20B5 and will require expansion of the footprint of the existing facility. The Shippingport M&R Station will be sited on an existing pad at the Shippingport Industrial Park. The Service Creek Tie and the Shippingport M&R facilities' permanent footprint will be completely fenced for security, and the fencing will have an additional visual barrier such as that provided by vertical tube slats. All work at the existing Mercer CS will be completed within the existing compressor station footprint and will not require any new impervious area. No compression or horsepower upgrades are proposed as part of the Project, and no changes to the FERC certificated parameters are proposed. No valve sites are proposed along Line N20B5. As such, the work at the Mercer CS will be completed under 18 CFR § 2.55(a). Although not required, the modifications at the Mercer CS have been included throughout the Concise Environmental Report to assess the Project as a whole.

Nine (9) staging areas totaling 14.21 ac will be used temporarily for pipe, equipment, and materials storage, contractor yards, and parking areas during construction. These areas are generally open and unforested, or previously disturbed/graveled and/or have been used in the past/present for similar uses. National Fuel will restore the staging areas to their original condition after construction is complete.

Eleven (11) access roads will be used as permanent access roads (PARs) to access the pipeline as needed during construction and operations. These PARs are generally existing, well-developed access roads that will continue to remain in place after construction and will be used to access the ROW during operation of the new pipeline. No permanent modifications are proposed for the PARs. PARs will be returned to original conditions after construction is complete. Ten (10) temporary access roads (TARs) will also be used during construction but will not be needed for routine operation of the pipeline. National Fuel will restore the TARs to their original condition after construction is complete.

Tree clearing is proposed to occur during the winter season and be completed before March 31, 2026. Construction activities (including grading, trench excavation, pipeline removal and installation, and backfilling are proposed to start in Spring 2026.

1.1.1 PURPOSE AND NEED

The purpose of the Project is to provide a natural gas transportation service to support power generation needs at the Shippingport Industrial Park, the former site of the Bruce Mansfield Power Plant. The project shipper, Shippingport Power Station, LLC (SPS), has requested that National Fuel construct an extension of its pipeline and install other appurtenant facilities along its Line N20 pipeline system to enable the transportation and delivery of 205,000 dekatherms per day to the Shippingport Industrial Park. National Fuel and SPS have entered into a Precedent Agreement for the total project capacity.

1.1.2 LOCATION MAPS, DETAILED ROUTE MAPS, AND SITE-SPECIFIC CONSTRUCTION PLANS

Figures showing the site locations and the proposed facilities on location maps and construction plan drawings are provided in Appendix A. Appendix A-1 contains a Project overview map and Project location maps on United States (U.S.) Geological Survey (USGS) 7.5-Minute Series Topographic Quadrangle base maps. Appendix A-2 provides the Project map with National Wetlands Inventory (NWI) mapped wetlands and mapped soils. Appendix A-3 contains the Project construction alignment sheets, including workspaces required for construction of the Project. Appendix A-4 contains a plan drawing for Mercer Compressor Station. Appendix A-5 contains right-of-way (ROW) construction configuration drawings showing the cross-section of construction workspace and planned workspace management/activities. No major waterbodies greater than 100 feet wide will be crossed by the Project; therefore, no site-specific waterbody construction plans were necessary.

Appendix B, found in Volume 2, contains a list of affected landowners. Pursuant to 18 CFR §388.112, this information is included in Volume 2, labeled "CUI//PRIV – Do Not Release." This volume contains information that is customarily treated as privileged and confidential.

Appendix C contains copies of all relevant agency correspondence for the Project, specifically regarding threatened/endangered (T&E) species (Resource Report 3) and cultural resources (Resource Report 4). Appendix D contains the Aquatic Resources Survey Report. Appendix E contains National Fuel's Erosion and Sedimentation Control and Agricultural Mitigation Plan (ESCAMP), and Appendix F contains a Preparedness, Prevention, and Contingency and Emergency Action Plan (PPC Plan) for the Project. Appendix G holds space for a Phase I Archaeological Investigation Report for the Project. Appendix H contains an Unanticipated Discovery Plan for Cultural Resources and Human Remains, in the event previously unidentified cultural resources or human remains are discovered during construction. Appendix I contains National Fuel's Blasting Guidance Document. Appendix J contains a Geohazard Assessment that was completed for the Project. Appendix K, included in Volume 2 labeled "CUI//PRIV – Do Not Release, contains site-specific residential construction plans for use in areas where workspace is within 25 feet of residences. Appendix L contains air emissions calculation tables for the Project.

1.2 LAND REQUIREMENTS

The Project will require a total of approximately 131.01 ac to construct, including permanent easements and temporary workspace areas for the pipeline, aboveground facilities, staging areas, and access roads. A breakdown of land requirements by each facility and duration is provided in Table 1-1. Additional information concerning land use associated with the Project area is presented in Resource Report 8 – Land Use.

Table 1-1. Shippingport Lateral Project

Facility Type and Name	Length of Proposed Construction Workspace (Or length of Access Road) (mi)[a]	New Permanent ROW Width (ft)	Total Construction ROW Width (or maximum width of access road) (ft)[b]	Disturbed During Construction (ac)	New Permanent ROW/Facilities[c] (ac)	Description
Shippingport Lateral Pipeline (Line N20B5)						
Pipeline	7.26	75	Generally varies between 75, 100, and 125 feet; depending on need for and availability of space	80.84	64.95	A 75-foot-wide new permanent ROW will be established in the form of new land easements, to facilitate long-term pipeline monitoring and maintenance.
Subtotal Pipeline Facilities				**80.84**	**64.95**	
Aboveground Facilities						
Service Creek Tie						
Tie-in to existing Line N transmission pipeline (MP 7.26)	N/A	N/A	N/A	2.11	1.20	Installation of pig launcher, gas quality equipment, and ancillary equipment adjacent to National Fuel's existing Service Creek pigging facility, east end of Line N20B5 (Service Creek Road).
Shippingport M&R Station						
Interconnect M&R at gas delivery point (MP 0.00)	N/A	N/A	N/A	1.15	0.59	New M&R station, west end of Line N20B5.
Modifications at Mercer CS						
Existing CS (Approx. 45 mi north of pipeline)	N/A	N/A	N/A	13.38	0.00	Installation of crossover valves (risers/isolation valves and control equipment at existing Mercer CS. The 13.38 ac is the entire existing station property; actual land disturbance will be less than 1 ac.
Subtotal Aboveground Facilities				**16.64**	**1.79**	
Staging Areas						
Staging Area 1 (MP 0.00)	N/A	N/A	N/A	0.67	0	Staging Area 1, located at the northwest terminus of the pipeline at Shippingport Industrial Park, will be used for parking and storage of construction equipment and materials for Project construction.

Facility Type and Name	Length of Proposed Construction Workspace (Or length of Access Road) (mi)[a]	New Permanent ROW Width (ft)	Total Construction ROW Width (or maximum width of access road) (ft)[b]	Disturbed During Construction (ac)	New Permanent ROW/Facilities[c] (ac)	Description
Staging Area 2 (MP 0.03)	N/A	N/A	N/A	0.54	0	Staging Area 2, abutting MP 0.03 on the south side of Shippingport Road, will be used for parking and storage of construction equipment and materials for Project construction.
Staging Area 3 (MP 1.40)	N/A	N/A	N/A	0.83	0	Staging Area 3, located at MP 1.40 on the south side of Shippingport Road, will be used for parking and storage of construction equipment and materials for Project construction.
Staging Area 4 (Along TAR-3)	N/A	N/A	N/A	1.10	0	Staging Area 4, located approximately 0.10 mile north of MP 1.95, at the intersection of State Route 18 and Green Garden Road, will be used for parking and storage of construction equipment and materials for Project construction.
Staging Area 5 (MP 1.82)	N/A	N/A	N/A	0.94	0	Staging Area 5, abutting the north side of the ROW at MP 1.82 along Frankfort Road, will be used for parking and storage of construction equipment and materials for Project construction.
Staging Area 6 (MP 4.02)	N/A	N/A	N/A	1.05	0	Staging Area 6, abutting the north side of the ROW at MP 4.02 along Green Garden Road, will be used for parking and storage of construction equipment and materials for Project construction.
Staging Area 7 (MP 4.43)	N/A	N/A	N/A	0.33	0	Staging Area 7, abutting the north side of the ROW at MP 4.43 along Green Garden Road, will be used for parking and storage of construction equipment and materials for Project construction.
Staging Area 8 (MP 4.51)	N/A	N/A	N/A	0.24	0	Staging Area 8, abutting the south side of the ROW at MP 4.51, will be used for parking and storage of construction equipment and materials for Project construction.
Staging Area 9 (Along TAR-10)	N/A	N/A	N/A	8.51	0	Staging Area 9, located approximately 3.10 miles southeast of the eastern terminus of the pipeline, will be used for parking and storage of construction equipment, pipe, and materials for Project construction.
Subtotal Staging Areas				**14.21**	**0**	
Access Roads[d]						
PAR-1	0.03	Up to 50	Up to 50	0.20	0.20	Asphalt in an industrial area.
PAR-2	0.34	Up to 50	Up to 50	2.08	2.08	Asphalt in a residential area.
PAR-3	0.10	Up to 50	Up to 50	0.60	0.60	Gravel in residential area.
PAR-4	0.05	Up to 50	Up to 50	0.32	0.32	Gravel/grass in light industrial area (gravel storage

Facility Type and Name	Length of Proposed Construction Workspace (Or length of Access Road) (mi)[a]	New Permanent ROW Width (ft)	Total Construction ROW Width (or maximum width of access road) (ft)[b]	Disturbed During Construction (ac)	New Permanent ROW/Facilities[c] (ac)	Description
						lot)/ROW.
PAR-5	0.18	Up to 50	Up to 50	1.11	1.11	Gravel in residential area.
PAR-6	0.39	Up to 50	Up to 50	2.37	2.37	Dirt road in industrial area (junk car lot).
PAR-7	0.03	Up to 50	Up to 50	0.19	0.19	2-track gravel in residential area.
PAR-8	0.26	Up to 50	Up to 50	1.56	1.56	Gravel in residential area.
PAR-9	0.12	Up to 50	Up to 50	0.78	0.78	Gravel/grass in ROW.
PAR-10	0.43	Up to 50	Up to 50	1.97	1.97	2-track in a forested area.
PAR-11	0.02	Up to 50	Up to 50	0.12	0.00	Existing access road at National Fuel's existing pigging facility at Service Creek Tie location.
TAR-1	0.03	N/A	Up to 50	0.18	N/A	Gravel in forested residential area.
TAR-2	0.21	N/A	Up to 50	1.28	N/A	Gravel in forested residential area.
TAR-3	0.04	N/A	Up to 50	0.23	N/A	Gravel in commercial area (lot behind gas station).
TAR-4	0.07	N/A	Up to 50	0.41	N/A	Gravel in residential area.
TAR-5	0.03	N/A	Up to 50	0.19	N/A	Dirt road in forested area.
TAR-6	0.08	N/A	Up to 50	0.49	N/A	Dirt road in residential driveway/ROW.
TAR-7	0.16	N/A	Up to 50	1.01	N/A	Gravel/grass in residential area/open land.
TAR-8	0.24	N/A	Up to 50	1.39	N/A	Dirt road in forested area.
TAR-9	0.44	N/A	Up to 50	2.60	N/A	2-track in ROW/forested area.
TAR-10	0.03	N/A	Up to 50	0.15	N/A	Gravel in industrial area.
Subtotal Access Roads	**3.28**			**19.23**	**11.18**	
			TOTAL	**131.01**	**77.92**	

Notes:
ft = foot or feet, mi = mile or miles, MP = milepost, N/A = not applicable
Totals may differ slightly from sum of parts due to rounding.
[a] Length of the pipeline is provided here as 3-dimensional total length, which is accounts for topographic variation and civil surveys, resulting in actual pipe length.
[b] Total Construction ROW width equals Permanent ROW width plus Temporary Workspace (TWS)/Additional Temporary Workspace (ATWS) widths. Total width varies depending on site-specific need (such as for specialized construction techniques near stream, road, and utility line crossings, on steep slopes, and in agricultural areas (to allow for topsoil segregation).
[c] Acres of New Permanent ROW/Facilities reflect new permanent ROW for the pipeline and new land to be acquired for the Shippingport M&R Station and Service Creek Tie. All new permanent area will be acquired in the form of land easements.
[d] Access Roads: PAR = Permanent Access Road, TAR = Temporary Access Road. Roads may be widened as required to support construction needs, to a maximum of 50 feet wide. PARs are generally existing, well-developed access roads that will continue to remain in place after construction and will be used to access the ROW / aboveground facilities during operation of the new pipeline. TARs will be restored to preconstruction condition following construction.

1.3 FIELD SURVEYS AND AGENCY CLEARANCES

The area studied during environmental and cultural resources field surveys for the Project included (and exceeded) all proposed workspace areas, in order to identify any sensitive resources warranting protection within and adjacent to the proposed construction area. Surveys were completed on 100 percent of the Project area; there are no properties where surveys could not be conducted due to landowner permission or other issues.

Wetland/waterbody delineation and general habitat surveys were conducted between March and May 2025.

Consultation was initiated with environmental agencies pursuant to the Endangered Species Act, Section 106 of the National Historic Preservation Act, and similar Commonwealth of PA protection acts. On behalf of National Fuel, Tetra Tech, Inc. (Tetra Tech) submitted Project information to the PA Natural Diversity Inventory (PNDI) to identify potential species of concern within the Project area. To date, the PA Game Commission (PA GC), PA Fish and Boat Commission (PA FBC), PA Department of Conservation and Natural Resources (PA DCNR), and U.S. Fish and Wildlife Service (USFWS) have all provided clearances for the Project. T&E species are discussed in more detail in Resource Report 3 – Vegetation and Wildlife, and copies of all agency clearance letters and correspondence are provided in Appendix C.

The PA State Historic Preservation Office's (PA SHPO) PA Historic and Archaeological Resources Exchange (PA-SHARE) was accessed to search for archaeological sites and historic properties that may exist within the Project area. National Fuel submitted Project information, including the results of background review and site file search, to the PA SHPO on June 6, 2025, to initiate consultation. On June 27, 2025, PA SHPO responded with No Effect on aboveground historic properties, and requested a Phase I archaeological survey. The archaeological survey was completed in June 2025 and did not find any sites that would be considered potentially eligible for listing on the National Register of Historic Places. National Fuel anticipates a No Effect determination from PA SHPO and the concurrence letter will be provided to FERC when received.

1.4 CONSTRUCTION PROCEDURES AND SCHEDULE

1.4.1 GENERAL

The Project will be constructed in accordance with applicable government regulations, permits, and approvals. Construction methods will be those that are consistent with industry-recognized practices, National Fuel's company policies, and best management practices (BMPs). National Fuel plans to implement construction procedures consistent with guidelines and recommendations from the U.S. Army Corps of Engineers (USACE), the PA Department of Environmental Protection (PA DEP), and FERC, particularly the FERC's 2013 Upland Erosion Control, Revegetation, and

Maintenance Plan (FERC Plan) and Wetland and Waterbody Construction and Mitigation Procedures (FERC Procedures) (FERC 2013a, 2013b).

National Fuel has incorporated the FERC Plan and FERC Procedures, and BMPs from USACE, PA DEP, and County Soil and Water Conservation Districts within National Fuel's Erosion and Sedimentation Control and Agricultural Mitigation Plan (ESCAMP), which supports the Project's site-specific earth disturbance plan. In addition to the ESCAMP, National Fuel has developed a Project-specific Erosion and Sedimentation Control Plan (E&SCP) for the Project. These plans are designed to minimize erosion and sedimentation during soil disturbance activities, protect environmental resources during construction, and restore the Project area following construction. In accordance with PA DEP erosion and sedimentation control permitting, National Fuel has developed a Preparedness, Prevention, and Contingency and Emergency Action Plan (PPC Plan) that prescribes fueling and vehicle maintenance measures; use, handling, and storage procedures for specific materials; spill and leak prevention measures; emergency spill response; required spill kit contents; material and waste inventory and disposal procedures; approved disposal sites for specific materials; contractors available for assistance; and a list of emergency contacts. National Fuel and its contractors will follow the provisions of the ESCAMP, the Project-specific E&SCP, and the PPC Plan during soil disturbance and restoration activities associated with the Project. There are no proposed deviations from the FERC Plan and FERC Procedures in National Fuel's plans for the Project.

Land clearing is planned to occur between November 15, 2025 and March 31, 2026 to avoid potential impacts to T&E bat species protected under the Endangered Species Act of 1973 and migratory birds protected under the Migratory Bird Treaty Act of 1918 (MBTA). Construction, including installation of the new pipeline and work at new and existing aboveground facilities, is planned to begin in April 2026. The planned completion and in-service date is November 1, 2026, and any remaining final restoration activities are planned to be completed for the entire Project by the end of 2027. The crossing of environmentally sensitive resources (e.g., waterbodies with fisheries), if required, will be in accordance with PA DEP and PA FBC timing restrictions, unless a waiver is granted by the applicable agency.

National Fuel anticipates construction work occurring 6 days per week, Monday through Saturday, and 10 hours per day during daytime hours. Landowner requests, specialized construction activities, schedule, and unforeseen events may impact and influence the need to work additional hours and days such as Sundays. Activities, such as, but not limited to, hydrotesting, installation of resource/stream crossings, environmental control device maintenance/installations, and tie-in work may exceed typical daytime work hours (generally between 7:00 a.m. and 5:00 p.m.).

The following section provides information on National Fuel's proposed construction procedures.

General Construction Sequencing

The Project will be constructed using a single pipeline construction spread. A separate crew(s) will be dedicated to conduct stream and road crossings at discrete tie-ins, as needed to comply with specific stream and road crossing seasonal and durational timing restrictions. Following completion of stream and road crossings, this crew(s) will be absorbed back into the larger mainline construction spread until Project construction is complete. Separate construction crews will be utilized to fabricate and install the facilities at the Shippingport M&R Station and Service Creek Tie.

The following general construction sequence provides an overview of National Fuel's planned construction process. Wherever practicable, construction activities will occur simultaneously, and some steps may not occur in the exact order in which they are listed below.

1. Winter Clearing: Survey and stake/flag the limits of construction workspaces. Manually or machine-cut (provided all required permits acquired) and fell trees within the construction limits of disturbance (LOD).
2. Mobilize and set up staging areas, including installation of erosion/sedimentation controls, road entrance pads, and proper hazardous material storage.
3. Reconfirm and refresh previously surveyed and staked/flagged limits of construction workspaces. Locate other utilities to prevent accidental damage during construction.
4. Install entrance pads at all access points from paved roads and install appropriate erosion/sedimentation controls along edge of road where needed to protect adjacent wetlands or water resources.
5. Locate, sign, and improve, maintain, or construct temporary and permanent access roads where noted.
6. Flag/field mark wetland and waterbody areas.
7. Begin processing and clearing debris and brush along the ROW.
8. Install erosion and sedimentation control filters and barriers (filter fence, compost filter sock, and straw bales) in areas that are not anticipated to be disturbed by subsequent grading and installation of temporary equipment crossings.
9. Grade the workspace if necessary, including topsoil segregation in active agricultural (cultivated and pasture), residential, and wetland areas.
10. Install timber mats or prefabricated temporary bridges for access roads/equipment crossings at stream crossings.
11. Install timber mats for access roads/equipment stabilization at wetland crossings.
12. Install all remaining required erosion and sedimentation control devices.
13. Install temporary water bars/slope breakers where required.
14. String new pipe along ROW.
15. Weld pipe sections together.
16. Excavate pipeline trench in upland areas.

17. Implement BMPs for trench dewatering as necessary.
18. Lower the new pipeline into trench.
19. Install trench plugs.
20. Backfill trench.
21. Restore grade to preconstruction contours, stabilize slopes, and install permanent slope breakers and drains where warranted.
22. Implement stream crossings (set up temporary dams/pumps, excavate trench, install, and backfill new pipeline with trench plugs where required, install sedimentation controls as necessary for retaining temporary spoil piles), and restore/stabilize stream banks (including erosion control blanket where required).
23. Install wetland crossings with trench plugs where required and restore segregated topsoil on top.
24. Hydrostatically pressure test the pipeline segments and conduct a baseline internal line inspection.
25. Apply lime, fertilizer, seed, and mulch to all disturbed upland areas, and only the specified wetland seed mix in wetlands.
26. Install erosion control blankets or equivalent on steep slopes.
27. Monitor the adequacy of erosion control practices.
28. After permanent stabilization is achieved, remove temporary erosion and sediment controls.

Note: Steps 22 and 23 will be installed in the order that best suits site and scheduling conditions.

1.4.2 ABANDONMENT AND REMOVAL

No abandonment or removal of pipeline or facilities is proposed.

1.4.3 SOCIOECONOMIC CONSIDERATIONS

In accordance with FERC guidance provided in 18 CFR Part 380.12(c), Resource Report 5 is not required for this Project. In the absence of Resource Report 5 socioeconomic analysis, information regarding the average workforce requirements, estimated duration of construction activities, and number of personnel to be hired to construct the proposed Project is provided in this section.

Construction of the Project will require an estimated workforce of 200 personnel. No new permanent full-time staff positions will be generated as a result of this Project.

Overall, the Project will require approximately seven to eight months to complete. Tree clearing is proposed to occur between November 2025 and March 2026. Construction activities are proposed from April through October 2026.

1.4.4 ENVIRONMENTAL INSPECTOR TRAINING

National Fuel will hire at least one full-time qualified Environmental Inspector who will have responsibility for ensuring and documenting environmental compliance during construction and restoration of the Project.

The Environmental Inspector will monitor construction activities to ensure compliance with the FERC Plan and FERC Procedures (FERC 2013a, 2013b); specifications of the Project-specific E&SCP, ESCAMP, and PPC Plan; all applicable federal, state, and local environmental permits; any specifically required restoration or mitigation measures; and landowner agreements pertaining to environmental issues.

All Environmental Inspectors on site, starting in 2026, shall have completed the Qualified Visual Site Inspector Training Program administered by the Pennsylvania Department of Protection's Clean Water Academy.

1.5 OPERATION AND MAINTENANCE PROCEDURES

After completion of construction and restoration, National Fuel will maintain the pipeline in accordance with its standard system operating procedures, which are compliant with the FERC Plan and FERC Procedures.

1.6 FUTURE PLANS AND ABANDONMENT

National Fuel does not currently have any plans for future expansion of the Project ROW or Project facilities. National Fuel does not currently plan to abandon any Project facilities in the near future.

1.7 PERMITS AND APPROVALS

1.7.1 PERMITS, LICENSES, AND CERTIFICATIONS

National Fuel has already submitted applications for all applicable federal and state permits and approvals, as well as various other regional, and local permits for the Project, as provided in Table 1-2. National Fuel will obtain all required permits and approvals prior to commencing construction of the Project. Furthermore, National Fuel will require its contractor to become familiar with all required permits and approvals and comply with all applicable construction and restoration requirements and mitigation measures.

Table 1-2. Environmental Permits, Approvals, and Consultations for the Project

Permitting/Approval Agency	Permit, Approval, or Consultation	*Anticipated Filing Date* or Actual Filed Date	*Anticipated Receipt Date* or Actual Receipt Date
Federal			
Federal Energy Regulatory Commission	Natural Gas Act Authorization	August 29, 2025	*Anticipated 11/5/2025 (close of 60-day comment period)*
United States Army Corps of Engineers – Pittsburgh District	Clean Water Act (CWA) – Section 404 Permit – PA State Programmatic General Permit (PASPGP-6), Anticipate Qualifies for Category of Non-Reporting to USACE	8/19/2025 (submitted to PA DEP per agency protocol)	*Anticipated 1/28/2026*
United States Fish and Wildlife Service – PA Field Office	Section 7 of the Endangered Species Act Consultation and Migratory Bird Treaty Act	June – August 2025	May Affect – Not Likely to Adversely Affect received for two bat species 7/17/2025; No Effect for three mussel species received 8/5/2025; E-Mail of No adverse effects to Indiana bat received 8/13/2025. No further consultation required.
Tribal Consultation			
Tribal Consultation	Multiple Federally Recognized Native American Tribes (See Resource Report 4 – Cultural Resources for List of Tribes)	Consultation initiated 7/11/2025; follow-up correspondence sent 8/26/2025.	Response received from Shawnee Tribe (No concerns, Project may proceed, advise if any findings); No other responses received to date (8/26/2025)
State			
Pennsylvania Game Commission	State Threatened/Endangered Species Consultation (birds, mammals)	Consultation initiated 5/23/2025 (Mercer CS and Staging Area) and 6/2/2025 (Pipeline); Supplemental information submitted 6/2/2025; Project commitment letter submitted 8/6/2025.	Agency response/clearance/ recommendations on 7/22/2025 and 8/7/2025.
Pennsylvania Department of Conservation and Natural Resources	State Threatened/Endangered Species Consultation (plants)	Consultation initiated 5/23/2025 (Mercer CS and Staging Area) and 6/2/2025 (Pipeline).	Agency response “No Known Impact” on 5/23/2025 (Mercer CS and Staging Area) and 6/2/2025 (Pipeline).

Permitting/Approval Agency	Permit, Approval, or Consultation	*Anticipated Filing Date* or Actual Filed Date	*Anticipated Receipt Date* or Actual Receipt Date
Pennsylvania Fish and Boat Commission	State Threatened/Endangered Species Consultation (fish, mussels, amphibians, reptiles)	Consultation initiated 5/23/2025 (Mercer CS and Staging Area) and 6/2/2025 (Pipeline); Supplemental information submitted 6/2/2025.	Agency response "No Known Impact" for Mercer CS and Staging Area on 5/23/2025. Agency response "An element occurrence of a rare, candidate, threatened, or endangered species under our jurisdiction is known from the vicinity of the proposed project. However, given the nature of the proposed project, the immediate location, or the current status of the nearby element occurrence(s), no adverse impacts are expected to the species of special concern" on 6/2/2025. (No issues; no further action required)
Pennsylvania Department of Environmental Protection	CWA – Section 401 Water Quality Certification – will be included with approval of Chapter 105 General Permits 5 and 8 (below)	8/19/2025	*Anticipated 1/19/2026*
Pennsylvania Department of Environmental Protection	Chapter 105 Water Obstruction and Encroachment Permit (stream and wetland crossings): General Permit 5 – Utility Line Stream Crossings, and General Permit 8 – Temporary Road Crossings	8/19/2025	*Anticipated 1/19/2026*
Pennsylvania Department of Environmental Protection	PA Erosion and Sedimentation Control General Permit-4 (ESCGP-4)	8/19/2025	*Anticipated 1/19/2026*
Pennsylvania Historical and Museum Commission (PHMC) (PA State Historic Preservation Office)	Review of Historic and Archaeological Resources Under Section 106 of the National Historic Preservation Act (Cultural Resource Agency Consultation)	Consultation initiated 6/6/2025; Phase I Survey completed June 2025; *Phase I Archaeological Survey Report (no sites) anticipated to be submitted to PA SHPO 9/8/2025.*	Agency responses: 6/27/2025– "No Effect" on historic architectural properties; Phase I archaeological survey required. *Clearance anticipated: 10/8/2025*
Pennsylvania Department of Environmental Protection	Hydrostatic Test Water Discharge Permit (PAG-10)	National Fuel has existing Statewide (general) Permit that will be updated for this Project prior to discharge.	National Fuel has existing Statewide (general) Permit that will be updated for this Project prior to discharge.

1.8 REFERENCES

Federal Energy Regulatory Commission (FERC). 2013a. Upland Erosion Control, Revegetation, and Maintenance Plan. Washington, D.C. 20 pp.

FERC. 2013b. Wetland and Waterbody Construction and Mitigation Procedures. Washington, D.C. 22 pp.

RESOURCE REPORT 2
WATER USE AND QUALITY

2.0 WATER USE AND QUALITY

This Resource Report provides information on both surface water and groundwater resources in the vicinity of the Project and on construction-related water use. Section 2.1 describes groundwater availability, quality, and current uses. Section 2.2 describes surface water availability, quality, current uses, construction methods in the vicinity of surface waters, and possible uses during construction. Section 2.3 identifies wetland resources traversed by the Project. Section 2.4 lists references that provided information for this report.

National Fuel determined the nature and location of surface waters, springs, wells, groundwater hazards, and point and non-point pollution sources by using information obtained through consulting appropriate agencies, performing on-site field aquatic resource delineation surveys between March and May 2025, and data publicly available through internet resources.

2.1 GROUNDWATER RESOURCES

The Project is located within the Pittsburgh Low Plateau and Northwestern Glaciated Plateau sections of the Appalachian Plateaus Physiographic Province. The Project is within four (4) (three [3] contiguous and one [1] disparate) Hydrologic Unit Code (HUC) boundaries:

- Sixmile Run-Ohio River (HUC 050301010310),
- Ambridge Reservoir-Service Creek (HUC 050301010206),
- Raredon Run-Racoon Creek (HUC 050301010207), and
- Mercer CS is located within the Mathay Run-Shenango River (HUC 050301020403) (USGS 2025).

The availability and quality of groundwater resources within these regions is characterized below.

2.1.1 WATER AVAILABILITY

The geology of the Appalachian Plateaus Physiographic Province consists mostly of sandstone, siltstone, and shale. The pipeline portion of the Project area is situated within the Pittsburgh Low Plateau section and traverses the Casselman and Glenshaw Formations of the Conemaugh Geologic Group, with Casselman being the uppermost and Glenshaw the lowermost formations. The primary lithology of the Conemaugh Geologic Group aquifer is sandstone. The Mercer CS portion of the Project is located within the Northwestern Glaciated Plateau section. The geology of this section consists of sub-horizontal beds of sandstone, siltstone, and shale.

The USGS classifies the groundwater under the Project sites as "Other aquifers (N9999OTHER)", which is an area that is not a national aquifer. Sandstone aquifers, constituted of fractured sandstone and shale, commonly yield 5-60 gallons per minute (gpm), and sometimes exceed 600 gpm (Makuch and Ward 2022).

Two municipal authorities supply and service the public water systems that intersect the Project area: the Aliquippa Water Authority and the Shippingport Borough Water System. The former serves approximately 6,615 customers (Municipal Water Authority of Aliquippa 2025) whereas the latter has approximately 222 customers (Shippingport Borough Water System 2025). In Mercer County, no public water supply systems intersect the Project area. There are 7,794 recorded well locations in Beaver County, three (3) of which are within 150 feet of the Project area. There are 43,523 recorded well locations in Mercer County, four (4) of which are within 150 feet of the Project area. See section 2.1.6 for more detailed information about public and private water wells in the Project vicinity.

According to monitoring data USGS has collected from the BV 156 Beaver County Observation Well (Monitoring Location # 40300608025230, located approximately 7.5 miles south of the proposed pipeline, near Racoon Creek), statistics spanning 33 years show the depth to groundwater level fluctuating between a low of 9.45 feet (ft) and a high of 7.59 ft below land surface, with a mean of 8.71 ft (USGS 2024). This monitoring well is located approximately nine (9) miles from the Project area. See section 2.1.6 for more detailed information about public and private water wells in Project vicinity.

2.1.2 SOLE SOURCE AQUIFERS

There are no U.S. Environmental Protection Agency (USEPA)-designated Sole Source Aquifers underlying the Project area (USEPA 2019).

2.1.3 STATE-DESIGNATED AQUIFERS

Pennsylvania identifies no Primary Water Supply Aquifers in the vicinity of the Project (USEPA 2019).

2.1.4 GROUNDWATER HAZARDS

Based on review of geospatial and tabular data from the USEPA[1], five (5) facilities included in the Toxic Release Inventory (USEPA 2025a), two (2) facilities classified as Permitted Water Dischargers (USEPA 2025b), and one (1) facility classified as a Hazardous Waste Transporter (USEPA 2025c) are located within 1 mile of the Project area, as listed in Table 2-1.

[1] Categories reviewed: Toxic Release Inventory, Superfund Sites (National Priorities List and “SEMS”), Permitted Water Dischargers (Major and Minor National Pollutant Discharge Elimination System), Hazardous Waste Sites – Treaters, Storers, and Disposers, Hazardous Waste Sites – Transporters, Hazardous Waste Sites – Inactive, Hazardous Waste Sites – Large Quantity Generators, Hazardous Waste Sites – Active, Hazardous Waste Sites – (“RCRA”), Brownfield properties (“ACRES”).

Table 2-1. List of Potential Groundwater Hazards

Facility Name	Registry ID	PGM System ID	Facility Report URL
Toxic Release Inventory			
BRUCE MANSFIELD POWER PLANT	110056969721	15077FRSTNOFFRT	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110056969721
MODERNE GLASS CO INC	110012708043	15001MDRNG1000I	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110012708043
GREAT LAKES TERM & TRANS CORP OF PA	110000328976	15052GRTLK2601R	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110000328976
KINDER MORGAN ARROW TERMINAL	110031292535	15052RRWTRRD2XX	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110031292535
GOLD BOND - SHI PLANT	110017435925	15077NWNGC168SH	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110017435925
Permitted Water Dischargers			
BRUCE MANSFIELD POWER PLANT	110056969721	PA0027481	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110056969721
FIRSTENERGY NUCLEAR BEAVER VALLEY	110007801227	PA0025615	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110007801227
Resource Conservation and Recovery Act - Transporter			
CARDINAL WASTE OIL CO	110004827796	PAD000607929	https://ofmpub.epa.gov/frs_public2/fii_query_detail.disp_program_facility?p_registry_id=110004827796

In addition, the USEPA Facility Registry Service (FRS) Interests database was queried to within 0.25-mile of the Project LOD. Eighteen (18) facilities were listed, with 45 separate records (USEPA 2025d). Records with violations appeared to be due to compliance reporting, fees not being paid, and/or CWA "other violations;" however, there were no obvious soil, sediment, or groundwater contamination issues in the reviewed area.

2.1.5 PROJECT-RELATED GROUNDWATER USE

Groundwater is not proposed as a water source for either construction or operation of this Project.

2.1.6 PUBLIC AND PRIVATE WATER SUPPLY WELLS

Based on information from the Pennsylvania Geospatial Data Clearinghouse and the Pennsylvania Groundwater Information System, there are seven (7) records of public or private water wells within 150 feet of the Project area (PA DCNR 2022). Three (3) wells are located in Beaver County, in proximity to the Project pipeline workspace. Of those, two (2) are used for withdrawal for domestic use and the other lacks water and well use data but is associated with the Bruce Mansfield Power Plant. The remaining four (4) records are located in Mercer County. Of those, two (2) are used for withdrawal for domestic use and two (2) for mine monitoring (PA DCNR 2022). If any additional water supply wells or springs are identified within 150 feet of the

Project area during ongoing landowner communications, National Fuel will provide that information to FERC and offer to conduct preconstruction and post-construction well testing for each well, as per FERC guidance.

Table 2-2. Water Wells Within 150 Feet of Project Area

MP	Distance (ft)/ Direction	PA Well ID	Owner	Water Use	Well Use	Well Depth (ft)	Well Yield (gpm)
Shippingport Lateral Pipeline (Beaver County)							
0.00	54 ENE	419226	BRUCE MANSFIELD POWER PLANT	No Data	No Data	15	No Data
4.47	48 N	54773	CALERE	DOMESTIC	WITHDRAWAL	93	6
PAR-6 (where meets public road)	125 NE	54725	SARVEY	DOMESTIC	WITHDRAWAL	364	1
Modifications at Mercer CS (Mercer County)							
N/A	105S	131797	TENNESSEE GAS PIPELINE CO	OTHER	MONITORING, MINE, OBSERVATION	22	No Data
N/A	(within Mercer CS property)	131799	TENNESSEE GAS PIPELINE CO	OTHER	MONITORING, MINE	19	No Data
N/A	138 N	131804	TOWNLEY	DOMESTIC	WITHDRAWAL	90	14
N/A	(within Mercer CS property)	700727	NATIONAL FUEL GAS	DOMESTIC	WITHDRAWAL	100	30
Notes: N/A = not applicable							

2.1.7 GROUNDWATER IMPACT MITIGATION

Prior to construction, National Fuel will contact landowners to identify the locations of any private wells occurring within 150 feet of approved construction workspaces. Through National Fuel's ongoing coordination with landowners, additional wells may be identified before construction. National Fuel will exercise care during construction in order to protect the integrity of water wells, including those within the vicinity of the construction corridor. No blasting is anticipated for construction. In the event a well or spring is discovered within 150 feet of construction work areas during construction or landowner coordination, National Fuel will offer to evaluate the spring or well conditions prior to and after construction. In accordance with landowner permissions, National Fuel will record the flow capacity of the well or spring prior to and after construction. National Fuel will also offer to test the well water quality prior to and after construction to determine any potential impacts to the water. If it is determined that a well is impacted from the construction of the proposed facilities, National Fuel will be responsible for supplying the landowner with water, and for the appropriate level of mitigation and repair.

Although proposed pipeline construction activities could potentially affect groundwater resources, most potential impacts can be avoided or minimized using both standard and specialized construction techniques. Shallow aquifers could sustain minor impacts from changes in overland water flow and recharge caused by clearing and grading of the ROW. In forested areas that require clearing for the Project, water infiltration, which is normally enhanced by vegetation, will be reduced until vegetation has been re-established. In addition, near-surface soil compaction caused by heavy construction vehicles could reduce the soil's ability to absorb water. However, if it occurs at all, this effect will be temporary, minor, and localized to the Project LOD. The Project will be constructed in accordance with applicable environmental regulations, permits, and approvals. Additionally, construction methods will be consistent with industry-recognized practices.

Dewatering of the pipeline trench may be necessary in areas where there is a high-water table or during periods of excessive precipitation. However, any lowering of localized groundwater as a result of dewatering is expected to be temporary. In order to recharge the aquifer and prevent silt-laden waters from flowing into streams and wetlands, National Fuel proposes to discharge all water from dewatering activities into adjacent, stable, well-vegetated upland areas. If selected dewatering locations are not within or immediately adjacent to the construction ROW, they will be sited to minimize off-ROW impacts. If dewatering locations must occur within sensitive areas such as designated wetland areas, approved energy-dissipating and sediment filtering dewatering structures will be used. Under no circumstances shall heavily silt-laden water be directly discharged into wetlands or waterbodies.

Potential Project-related groundwater contamination sources could include accidental spills of heavy equipment fuel, lubrication oil, or hydraulic oil. During construction, National Fuel's PPC Plan will be implemented to avoid and minimize the effects of such spills. Accordingly, no major impacts to groundwater are anticipated as a result of the Project.

2.2 SURFACE WATER RESOURCES

As previously noted, the Project is within four (4) HUC boundaries:

- Sixmile Run-Ohio River (HUC 050301010310),
- Ambridge Reservoir-Service Creek (HUC 050301010206
- Raredon Run-Racoon Creek (HUC 050301010207), and
- Mercer CS is located within the Mathay Run-Shenango River (HUC 050301020403) (USGS 2025).

2.2.1 EXISTING RESOURCES

A field aquatic resources delineation survey was conducted within the Project area between March and May 2025 (see Appendix D for the Aquatic Resources Survey Report). The survey included a typical 300-foot-wide survey corridor, as well as some alternative routes/survey areas that have since been eliminated from the proposed route and workspace. Based on the survey, many resources have been avoided by route and workspace decisions, and several resources within the Project LOD are along the edges of the LOD and do not cross the proposed pipeline centerline.

Table 2-3 lists and details the waterbodies within the construction workspace/LOD of the Project, including proposed crossing methods.

Pipeline

A total of 20 waterbodies will be crossed by the pipeline ROW/construction workspaces, including 17 streams (seven [7] perennial streams, four [4] intermittent streams, and six [6] ephemeral streams), two (2) ditches, and one (1) swale. National Fuel anticipates that PA DEP will not require in-stream work timing restrictions because no streams have PA FBC-designated trout waters classifications, as listed in Table 2-3. Generally, crossings will be conducted under low flow conditions.

Access Roads

A total of 21 existing access roads have been proposed for use during construction. The majority of the lengths of these roads consist of existing roads. However, several roads will need minor extensions to reach the pipeline ROW or improvements to widen the roads (see Resource Report 8 – Land Use, Recreation, and Aesthetics for a list of access roads, existing conditions, and proposed modifications). A total of seven (7) waterbodies will be crossed by access roads, including one (1) perennial stream, two (2) intermittent streams, two (2) ephemeral streams, and two (2) ditches. The majority of these waterbodies are currently culverted or otherwise well conveyed under the existing road.

Staging Areas

One (1) roadside ditch is present in Staging Area 9.

Aboveground Facilities

There is a total of four (4) waterbodies within the Mercer CS site, including two (2) ditches and two (2) upland swales. No streams or waterbodies are present within the construction area for the Shippingport M&R Station or the Service Creek Tie.

Table 2-3. Waterbodies Crossed by the Project

MP	Feature ID	Waterbody Name[a]	Flow Regime	Crossing Width (ft)[b]	State Water Quality Classification[c]	Trout Waters Classification	Anticipated Construction Timing Restriction[d]	Crossing Method[e]
Pipeline								
0.33	S01	(Unnamed Tributary) UNT to Ohio River	Perennial	1	Drains to WWF	N/A	N/A	Dry Crossing
0.34	D03	N/A	N/A	1	N/A	N/A	N/A	Dry Crossing
0.42	S16	UNT to Ohio River	Intermittent	3	Drains to WWF	N/A	N/A	Dry Crossing
0.43	D07	N/A	N/A	1	N/A	N/A	N/A	Dry Crossing
0.43	WC-30	N/A	N/A	-	N/A	N/A	N/A	Matting/Water Management (Workspace only)
0.72	S02	UNT to Haden Run	Perennial	12	Drains to WWF	N/A	N/A	Dry Crossing
2.47	S07	UNT to Service Creek	Ephemeral	3	Drains to HQ-CWF	N/A	N/A	Matting/Water Management (Workspace only)
3.58	S12	UNT to Ambridge Reservoir	Perennial	7	HQ-CWF	N/A	N/A	Dry Crossing
4.18	S15	UNT to Ambridge Reservoir	Intermittent	6	WWF	N/A	N/A	Dry Crossing
4.36	S13	UNT to Ambridge Reservoir	Ephemeral	3	Drains to WWF	N/A	N/A	Dry Crossing
4.55	S47	UNT to Ambridge Reservoir	Ephemeral	2	Drains to HQ-CWF	N/A	N/A	Matting/Water Management (Workspace only)
4.70	S46	UNT to Ambridge Reservoir	Ephemeral	1	Drains to HQ-CWF	N/A	N/A	Matting/Water Management (Workspace only)
4.81	S45	UNT to Ambridge Reservoir	Ephemeral	1	Drains to HQ-CWF	N/A	N/A	Dry Crossing
4.82	S44	UNT to Ambridge Reservoir	Intermittent	4	Drains to HQ-CWF	N/A	N/A	Dry Crossing
4.97	S43	UNT to Ambridge Reservoir	Perennial	10	Drains to HQ-CWF	N/A	N/A	Dry Crossing
5.04	S41	UNT to Ambridge Reservoir	Ephemeral	2	Drains to HQ-CWF	N/A	N/A	Matting/Water Management (Workspace only)
5.04	S42	UNT to Ambridge Reservoir	Perennial	7	Drains to HQ-CWF	N/A	N/A	Dry Crossing

MP	Feature ID	Waterbody Name[a]	Flow Regime	Crossing Width (ft)[b]	State Water Quality Classification[c]	Trout Waters Classification	Anticipated Construction Timing Restriction[d]	Crossing Method[e]
5.43	S25	UNT to Ambridge Reservoir	Perennial	5	HQ-CWF	N/A	N/A	Dry Crossing
5.94	S36	UNT to Service Creek	Intermittent	4	Drains to WWF	N/A	N/A	Dry Crossing
6.10	S19	UNT to Service Creek	Perennial	6	WWF	N/A	N/A	Dry Crossing
Access Roads								
PAR-2	D03	N/A	N/A	1	N/A	N/A	N/A	Existing Culvert
PAR-3	S09	UNT to Haden Run	Intermittent	4	Drains to WWF	N/A	N/A	Existing Culvert
PAR-3	S10	UNT to Haden Run	Ephemeral	2	Drains to WWF	N/A	N/A	Existing Culvert
PAR-5	S38	UNT to Service Creek	Perennial	3	Drains to WWF	N/A	N/A	Existing Culvert
TAR-2	S39	UNT to Service Creek	Intermittent	5	Drains to WWF	N/A	N/A	Existing Culvert
TAR-2	S40	UNT to Service Creek	Ephemeral	1	Drains to WWF	N/A	N/A	Existing Culvert
TAR-10	D04	N/A	N/A	3	N/A	N/A	N/A	Existing Culvert
Staging Areas								
Staging Area 9	D05	N/A	N/A	1	N/A	N/A	N/A	(Avoided)
Aboveground Facilities								
Mercer CS	Swale01	N/A	N/A	-	N/A	N/A	N/A	Matted (or Avoided)
Mercer CS	Swale02	N/A	N/A	-	N/A	N/A	N/A	Matted (or Avoided)
Mercer CS	D01	N/A	N/A	2	N/A	N/A	N/A	Matted (or Avoided)
Mercer CS	D02	N/A	N/A	3	N/A	N/A	N/A	Matted (or Avoided)

[a] UNT = unnamed tributary

[b] Crossing width based on Ordinary High-Water Mark recorded during aquatic resources field surveys.

[c] CWF = Coldwater Fishes, WWF = Warmwater Fishes, HQ = High Quality Waters

[d] Waterbody crossing timing restrictions reflect periods when no in-stream work is permitted. Dates listed are *anticipated* and subject to change; National Fuel will comply with the final required timing restrictions as defined in the PA DEP Chapter 105 Water Obstruction and Encroachment Permit and any other applicable state agency approvals. "Waived" means the stream is eligible for a waiver from the need to obtain a Chapter 105 Water Obstruction and Encroachment Permit based on upstream drainage area less than 100 ac (25 PA Code Chapter 105.12). As there are no trout streams impacted by the Project, no crossing restrictions apply.

[e] Dry Crossing Method – either dam and flume or dam and pump method. If the stream is dry at the time of crossing, an open cut method may be used with materials and provisions on hand to quickly shift to a dry crossing method in the event stream begins to flow before completion of the crossing. All crossings will conducted under low flow conditions.

MP	Feature ID	Waterbody Name[a]	Flow Regime	Crossing Width (ft)[b]	State Water Quality Classification[c]	Trout Waters Classification	Anticipated Construction Timing Restriction[d]	Crossing Method[e]
Matting/Water Management (Workspace only) – means waterbody is in workspace but not crossed by the pipeline, so pipeline crossing method does not apply, but waterbody will be matted or avoided, any water flow will be managed if necessary to avoid adverse impacts to construction workspace and will be protected from sedimentation using erosion/sedimentation control measures. Existing Culvert – means waterbody along access roads is already channeled through an existing culvert under the existing access road. Source for state stream designations: PA DEP 2025. Source for trout stream information: PA FBC 2025.								

2.2.2 SENSITIVE SURFACE WATERS

In the vicinity of the Project area, one stream was identified as an Impaired Stream: Haden Run. The USEPA Waterbody Report (USEPA 2024) classifies this stream as "impaired" due to siltation caused by "Highway/road/bridge Runoff (Non-construction Related) and "Removal of Riparian Vegetation." However, this stream is adjacent to the northwestern end of the proposed pipeline, is not crossed by the proposed workspace, and will not be affected by construction. No other USEPA classified "impaired streams" intersect with or are adjacent to the Project construction workspace.

2.2.3 PUBLIC WATERSHED AREAS

The Project crosses four HUC boundaries: Sixmile Run-Ohio River (HUC 050301010310), Ambridge Reservoir-Service Creek (HUC 050301010206), Raredon Run-Racoon Creek (HUC 050301010207), and the Mercer CS is located within Mathay Run-Shenango River (HUC 050301020403) (USGS 2025).

The primary source of water in the region comes from two municipal authorities which supply and service the public water systems that intersect the Project area: the Aliquippa Water Authority and the Shippingport Borough Water System. Three wells were found to be presently recorded within 150 feet of the pipeline portion of the Project area in Beaver County. No other municipal water supplies in Beaver County were identified within 3 miles downstream of any waterbody crossed by the Project.

The Ambridge Reservoir is located approximately 0.40-mile south of the pipeline lateral. The Ambridge Reservoir serves approximately 30,000 individuals in eight boroughs and townships (Ambridge Water Authority 2025). The Ambridge Reservoir receives its water supply in part from the Service Creek Reservoir, located just east of the Ambridge Reservoir and approximately 0.75-mile south of the Project area. The Project will cross 11 tributaries to the Ambridge Reservoir (6 perennial and 5 intermittent/ephemeral, all less than 16 feet wide) and nine (9) tributaries to Service Creek. Based on the distance from the pipeline stream crossings to the Ambridge Reservoir (all are at least 0.40 miles from impounded water and 1.25 to 2.00 mi from the water intakes), the small size and flow of the streams to be crossed, the use of specialized dry stream crossing methods and restoration proposed, and the use of erosion/sedimentation controls and spill prevention measures, adverse impacts to water quality, water supply, and to the Ambridge Reservoir itself are not anticipated.

The Mercer CS is within 3 miles of a public water supply. Two man-made ephemeral ditches exist on site and neither waterbody conveys flow off of or away from the Project area.

2.2.4 FLOODPLAINS

The Project pipeline will be adjacent to floodplains mapped by the Federal Emergency Management Agency (FEMA) in four areas:

- A small portion (approximately 0.07 ac) of the Staging Area 1, located at Shippingport M&R Station (MP 0.00), is in a mapped 100-year floodplain of Haden Run. However, in this area the stream is enclosed/ contained in a large concrete box culvert where it flows adjacent to the proposed staging area and does not support a functioning FEMA (or PA DEP) floodway as the flow through the culvert is contained at all times. The Project use of this area is temporary for construction worker parking.
- At the Service Creek Tie site (end of pipeline, MP 7.26), the proposed construction workspace will be adjacent to the 100- and 500-year floodplains of Raccoon Creek. The Project will not cross Racoon Creek, and all construction activities and new developed area at Service Creek Tie will be outside of the floodplains.
- At MP 0.72, the pipeline will cross the 100-year floodplain of an Unnamed Tributary to Haden Run (S02) for approximately 207 feet. Disturbance of this floodplain area will be temporary and short term, and no permanent fill, aboveground structures, or new impervious areas will be constructed in the floodplain.
- At MP 3.58, the pipeline will cross the 100-year floodplain of another Unnamed Tributary to Ambridge Reservoir (S12) for approximately 165 feet. Disturbance of this floodplain area will be temporary and short term, and no permanent fill, aboveground structures, or new impervious areas will be constructed in the floodplain.

2.2.5 CONTAMINATED SEDIMENTS

No information was found related to the presence of contaminated sediments in any of the streams that are traversed by the Project. Contaminated sediments are not known to be present at the proposed stream crossing locations.

2.2.6 WATERBODY CONSTRUCTION PROCEDURES

The following stream crossing methods are options that National Fuel will consider for the crossing of waterbodies along this Project as appropriate based on field conditions. Generally, streams will be crossed during low flow conditions and in accordance with the FERC Procedures and applicable permit conditions.

Dry Stream Crossing Methods

The pipeline construction at all stream crossings for this Project will use dry trenched stream crossing methods for all streams that are flowing at the time of construction. Dry stream crossing methods consist of either the Dam & Flume or the Dam & Pump crossing methods, and the selection will be determined in the field at the time of crossing by the contractor and National Fuel's Environmental Inspector. Generally, crossings will be conducted under low flow conditions. The method selected will be that which provides the least disturbance and most expedient crossing to minimize overall impact. These methods are described in more detail below.

Dam & Flume Stream Crossing Method. A dam & flume crossing involves the installation of temporary dams (often constructed of sandbags) on the upstream and downstream sides of the work zone, surrounding culverts/flume pipes which convey streamflow cleanly through the trench line and construction workspace, keeping the work zone dry during pipe removal and installation. This allows for the trenching, pipe installation, and initial restoration to occur in dry conditions, under the flume set-up, while maintaining continuous downstream flow. Soil characteristics must be very stable and stream flow should be low to moderate for this method to be used successfully and safely. The flume pipe must be long enough to account for the possibility of the trench widening unexpectedly during the excavation (due to sloughing). An effective seal (using materials such as sandbags) must be created around the flume(s) so that water will not penetrate the work zone or possibly wash out the in-stream dam on both the inlet and outlet end. Once in place, the flumes should not be removed until the pipeline has been installed, and the streambed and banks have been restored.

Dam & Pump Crossing Method. This method may be used in situations where pumps are able to adequately capture and transfer water around the work area, and there are no concerns about sensitive species passage. The dam and pump crossing method ensures that downstream flow is maintained while keeping the work zone dry. Dams (often constructed of sandbags) on the upstream and downstream sides of the work zone function to keep the work zone dry during pipe installation. Potential impacts to aquatic species and water quality are minimized or avoided by screening the pump intake and using an energy dissipater to prevent streambed scour at the pump discharge location.

Wet/Open-Cut Crossing Method

If any stream is dry or has no perceptible flow at the time of construction, an open-cut crossing method may be used. For open-cut crossings, a backhoe, clam dredge, dragline, or similar equipment will be used for trench excavation. As a rule, the completion of all construction activities should not exceed 24 hours at minor stream crossings and 48 hours at intermediate stream crossings unless Project- or location-specific circumstances prevent completion within that timeframe. The pipe will be welded together and lowered into the trench, previously excavated material will be returned to the trench line for backfill, streambeds will be restored to their former elevations and grades, and all stream banks will be restored and stabilized with erosion controls.

2.2.7 CONSTRUCTION AND OPERATION IMPACTS

The construction workspace in waterbodies will be limited to 75 feet wide (permanent ROW). The construction of the proposed Project across waterbodies will result in minor, short-term impacts. These impacts will occur as a result of in-stream construction activities or construction on slopes adjacent to stream channels. These activities will result in a temporary localized increase in turbidity levels and downstream sediment

deposition. Sedimentation and turbidity may occur as a result of in-stream construction, trench dewatering, and soil erosion along the construction ROW. In slack or slowly moving waters, increases in suspended sediment may increase the biochemical oxygen demand and reduce levels of dissolved oxygen in localized areas during construction. Motile organisms may avoid these areas, but sessile and planktonic organisms may be adversely affected. Suspended sediments will also alter the chemical and physical characteristics of the water column (e.g., color and clarity) on a short-term basis. However, no foreign sediments will be introduced as all dredged or fill material will consist of on-site sediments.

To limit the time required for construction of a stream crossing, the ROW will be prepared on either side of the stream prior to the construction of the actual crossing. Where construction will be through wooded stream banks, care will be taken to preserve as many existing trees in the riparian area as possible. Stream crossings will be perpendicular to the flow, to an extent practical. If necessary, the pipe used for stream crossings and in floodplains will be weighted to prevent flotation. The pipe will be welded together and then carried or floated along the ROW into place. If the streambed is composed of unconsolidated material, the pipe will be pulled into place. In rock-bottomed streams, the pipe will be floated or lifted across and then lowered into place. After the pipe is lowered into the trench, previously excavated material will be returned to the trench line for backfill. Stream flow will be maintained at all waterbody crossings, and no alteration of the stream's capacity is anticipated as a result of pipeline construction.

At small streams encountered along the ROW, a trackhoe or similar equipment will be used for trench excavation. As a rule, the completion of all construction activities should not exceed 24 hours at minor stream crossings (<10 feet wide) and 48 hours at intermediate stream crossings (between 10 and 30 feet wide) unless Project- or location-specific circumstances prevent completion within that timeframe. Minimizing the duration of in-stream work will minimize the impacts to the waterbody.

After the completion of construction, streambeds will be restored to their former elevations and grades. Spoil, debris, pilings, cofferdams, construction materials, and any other obstructions resulting from or used during construction of the pipeline will be removed to prevent interference with normal water flow and use. Any excavated material not used as backfill will be disposed of in a manner and at locations satisfactory to the agencies having jurisdiction. Following grading, all stream banks will be restored to prevent subsequent erosion, in accordance with permit requirements.

During post-construction vegetation/ROW maintenance activities, National Fuel will comply with the FERC Procedures, including Section V.D.1 of the FERC Procedures (FERC 2013b), which states (for waterbodies):

> Limit routine vegetation mowing or clearing adjacent to waterbodies to allow a riparian strip at least 25 feet wide, as measured from the waterbody's mean high-water mark, to permanently revegetate with

native plant species across the entire construction right-of-way. However, to facilitate periodic corrosion/leak surveys, a corridor centered on the pipeline and up to 10 feet wide may be cleared at a frequency necessary to maintain the 10-foot corridor in an herbaceous state. In addition, trees that are located within 15 feet of the pipeline that have roots that could compromise the integrity of the pipeline coating may be cut and removed from the permanent right-of-way. Do not conduct any routine vegetation mowing or clearing in riparian areas that are between HDD entry and exit points.

The Ambridge Reservoir is located approximately 0.4-mile south of the proposed pipeline. The Project will cross 11 tributaries to the Ambridge Reservoir (6 perennial and 5 intermittent/ephemeral, and all are less than 16 feet wide) and nine (9) tributaries to Service Creek. The reservoir is managed by the Ambridge Water Authority (AWA) and, in accordance with the permit conditions, National Fuel will contact the local municipal water authority at least ten (10) days prior to construction. Based on the distance the pipeline stream crossings are from the Ambridge Reservoir (0.4-mile and greater from impounded water and 1.25 to 2.00 mi from the water intakes), the small size and flow of the streams to be crossed, National Fuel's implementation of the proposed specialized dry stream crossing methods, restoration measures, erosion and sedimentation controls, and spill prevention measures, adverse impacts to water quality, water supply, and the to the Ambridge Reservoir itself are not anticipated.

2.2.8 HYDROSTATIC TEST WATER

The new pipeline is planned to be hydrostatically tested in two separate sections before placing the Project into service. Water will be obtained from a municipal water source and transported by truck to the Project area. The water used in the first pipe section tested will then be transferred and re-used for testing the second section. The first section to be tested in the 24-inch-diameter pipeline (from MP 0.0 to approximate MP 4.0) will require approximately 457,531 gallons of water for the test; of (and after) that, the second section will reuse approximately 368,787 gallons for the test.

Following completion of hydrostatic testing, the water will be discharged at the Service Creek Tie at the eastern end of the pipeline in accordance with FERC Procedures and National Fuel's statewide hydrostatic test water discharge permit (PA DEP PAG-10). Water sampling, management, and reporting will be completed in accordance with permit requirements. The water will be discharged into an energy dissipation device and directed into a temporary filter basin constructed of silt fence and/or straw bales, located in a well-vegetated area. The energy dissipation device will reduce the velocity of the discharged water, reducing the potential for erosion and sedimentation. The discharged water will be detained for a short period in the filter basin, and then will filter through the straw bales, and flow overland through a well-vegetated upland area. As a result of National Fuel's adherence to these procedures, no adverse impacts to waterbodies are anticipated from hydrostatic test water discharge activities.

2.2.9 CONSTRUCTION PERMITS

National Fuel has submitted a registration application for a PA DEP Chapter 105 Water Obstruction and Encroachment General Permit 5 (Utility Stream Crossings) and General Permit 8 (Temporary Road Crossings). Approval of General Permits 5 and 8 conveys with it a CWA Section 401 Water Quality Certification.

CWA Section 404 authorization from the USACE will be required for dredge and fill activities associated with construction in wetlands and waterbodies that are federal-jurisdictional waters of the United States. National Fuel anticipates that the Project will receive USACE approval under the PA State Programmatic General Permit program.

National Fuel has also applied for a PA DEP Erosion and Sedimentation Control General Permit-4 and has prepared Project-specific erosion and sedimentation control plan drawings to implement during construction of the Project.

Finally, National Fuel will comply with its PA DEP PAG-10 permit for the discharge of hydrostatic test water, including the required agency notifications and compliance with management and performance conditions of the permit.

2.3 WETLAND RESOURCES

Information in this section is based on the results of a wetland delineation survey in the Project area conducted by environmental scientists from Tetra Tech between March and May 2025. Wetland delineation methodology was based on the USACE 1987 Wetland Delineation Manual (Environmental Laboratory 1987).

2.3.1 EXISTING RESOURCES

A total of eight (8) wetlands were identified within the proposed pipeline construction workspaces and above-ground facilities during the field survey of the Project area. All of these wetlands are classified wholly as palustrine emergent (PEM), and the two (2) wetlands at Mercer CS resulted from man-made berms created as part of a post-construction stormwater management plan (PCSM).

Table 2-4. Wetlands Crossed by the Project

Approximate MP	Wetland ID	Cowardin Classification	Approximate Crossing Length and Width (ft)	Area Affected During Construction (ac)	Area Affected During Operation[a,b] (ac)
Pipeline					
0.80	W05_PEM	PEM	7 x 2.5	<0.01	0
1.91	W02_PEM	PEM	32 x 14	0.01	0
2.24	W03_PEM	PEM	(Avoided)	<0.01 (will be avoided/located on edge of LOD)	0
2.40	W06_PEM	PEM	25.5 x 13	0.01	0

Approximate MP	Wetland ID	Cowardin Classification	Approximate Crossing Length and Width (ft)	Area Affected During Construction (ac)	Area Affected During Operation[a,b] (ac)
4.36	W08_PEM	PEM	(Avoided)	<0.01 (will be avoided/edge of LOD)	0
6.55	W12_PEM	PEM	53 x 28	0.03	0
Pipeline Subtotal				**0.052**	
Aboveground Facilities					
Mercer CS	W14_PEM/PCSM	PEM	(Avoided)	0.11 (will be avoided)	0
Mercer CS	W15_PEM/PCSM	PEM	(Avoided)	<0.01 (will be avoided)	0
TOTAL				**0.00**	**0**

Totals may differ slightly from sum of parts due to rounding.

[a] Area Affected During Operation intends to represent acreage of vegetation cover type that will be permanently converted from palustrine forested (PFO) or palustrine scrub-shrub (PSS) to a lower successional cover type (e.g., PSS or PEM) because of vegetation maintenance procedures within the new permanent ROW.

[b] Although wetlands will be present within the new permanent ROW, the acreage affected during operation is 0 because all wetlands will be restored and allowed to return to their preconstruction PEM wetland condition, will function as wetlands, and there will be no loss or filling of wetlands for the Project.

2.3.2 CONSTRUCTION AND OPERATION IMPACTS AND MITIGATION

A total of six (6) wetlands will be affected during construction of the pipeline. Two (2) man-made PCSM features that have developed wetland characteristics are present at the Mercer CS and will be avoided. The construction workspace in the wetlands will be limited to 75 feet wide, resulting in a total acreage affected during construction of 0.05 ac. During construction, all disturbed wetland areas will be restored as wetlands, resulting in no permanent loss of wetlands as a result of the Project.

Wetlands in the permanent ROW will continue to be maintained as PEM wetlands as part of National Fuel's annual vegetation maintenance. No PSS or PFO wetlands will be affected by construction or permanent vegetation maintenance along the Project ROW. Therefore, compensatory mitigation for permanent impacts to wetlands due to either the permanent fill (loss) or conversion of woody wetland types to lower vegetative succession types will not be required.

During post-construction vegetation/ROW maintenance activities, National Fuel will comply with the FERC Procedures, including Section VI.D.1 of the FERC Procedures (FERC 2013b), which states (for wetlands):

> Do not conduct routine vegetation mowing or clearing over the full width of the permanent right-of-way in wetlands. However, to facilitate periodic corrosion/leak surveys, a corridor centered on the pipeline and up to 10 feet wide may be cleared at a frequency necessary to maintain the 10-foot corridor in an herbaceous state. In addition, trees within 15 feet of the pipeline with roots that could compromise the integrity of

pipeline coating may be selectively cut and removed from the permanent right-of-way.

The following subsections describe National Fuel's wetland construction and restoration measures, which are consistent with the FERC Procedures.

Construction in Wetlands with Unsaturated Soils

The construction technique used to cross wetlands with stable, unsaturated soils at the time of construction will be like those used in dry upland areas. Soils may be dry and stable enough to support equipment without additional timber mat equipment support, and pipe may be strung along the ROW on skids through the wetland. Vegetation will be cut just above ground level, leaving root systems in place. Grading (where required) will be minimized to ensure safe operation of construction equipment to facilitate regrowth of existing root systems after construction. Wetland topsoil will be segregated from subsoil in the trench line area and stored in separate piles while the trench is open. The segregated soils will be backfilled in the proper order, with topsoil on top, and the preconstruction surface contours will be restored. Trench breakers will be placed in the trench at the base of slopes near the wetland boundaries prior to backfilling to prevent draining of the wetland along the trench line. No upland soil or fill material will be backfilled or imported into the wetland. The wetland will be seeded with annual ryegrass and/or a wetland seed mix to quickly establish a vegetative cover and allow the wetland's native seed and rhizomes (contained in the topsoil) to reestablish dominance over time. No lime, fertilizer, or mulch will be applied in wetlands.

Construction in Wetlands with Saturated Soils

In wetlands with wet, saturated soils at time of construction, topsoil will be segregated over the trench line if possible. Equipment will work on timber mats to distribute weight evenly, and to reduce the potential for rutting, soil compaction, and mixing of wetland topsoil and subsoils. Construction in saturated wetland areas may involve conventional pipeline construction procedures or a "drag section" or "push/pull" technique.

The drag section technique involves equipment carrying a prefabricated section of pipe into the wetland for placement into the excavated trench if soil conditions permit. This technique requires the installation of equipment support along the working side of the trench to provide a stable work surface and minimize soil disturbance and rutting. Clean-up and restoration procedures will be similar to those previously described for wetlands with unsaturated soils with the additional step of removing the equipment support from the wetland.

The push/pull technique is generally used only in wetlands with standing water or soils that are saturated to the surface. The trench may be excavated using either a backhoe (working on equipment support in the wetland) or a dragline or clamshell dredge (working either in the wetland or from the edge of the wetland, depending on wetland

size and extent of soil saturation). A prefabricated pipe is pushed from the edge of the wetland and/or pulled (e.g., with a winch) from the opposite bank of the wetland into the excavated trench. Floats may be attached to the pipe to give it positive buoyancy, allowing it to be "floated" into place over the excavated trench. Once the pipe is positioned, these floats will be removed, and the pipe will settle to the bottom of the trench and the trench will then be backfilled. The push/pull technique enables the pipeline to be installed with minimal equipment operating in the wetland.

Impacts and Mitigation Procedures

The proposed construction procedures are designed to ensure that potential impacts to wetlands are minimized to the fullest extent practicable. National Fuel will implement wetland crossing procedures and wetland protection measures outlined in the FERC Procedures, its ESCAMP, and PA DEP's Chapter 105 Water Obstruction and Encroachment General Permits 5 and 8, to further minimize impacts to wetlands crossed by the proposed Project.

Temporary construction impacts in wetlands may include loss of herbaceous, shrub, or forested vegetation; wildlife habitat disruption; soil disturbance associated with grading, trenching, and stump removal; sedimentation and turbidity increases; and hydrological profile changes. To minimize vegetation disturbance, National Fuel has reduced the width of temporary workspace where possible in wetlands. Disturbance will be further minimized by restricting equipment access in sensitive wetlands to machinery needed for actual pipeline installation, and by limiting the number of crossings. To minimize impacts on wetlands, National Fuel will implement erosion and sediment control measures to prevent soils disturbed by construction activities from leaving the construction area and entering wetlands. This will include implementing spill prevention and response procedures to avoid impacts from refueling of equipment and fuel storage within the vicinity of wetlands.

Confining stump removal in wetlands to the trench line (unless safety or access considerations require stump removal) will minimize soil disturbance and retain sources for re-sprouting and re-growth of wetland vegetation. Erosion control techniques, including deployment of silt fences, slope breakers, trench plugs, riprapping, terracing, netting, restoration, and revegetation will be used in upland areas to restrict sediment runoff into adjacent wetlands.

Preconstruction wetland conditions in the temporary ROW will be restored to the extent possible to promote revegetation by natural succession. Topsoil segregation in unsaturated wetlands will segregate and restore topsoil, preserving the native seed source, which will facilitate regrowth of herbaceous vegetation once pipeline installation is complete. In addition, wetlands will be restored and allowed to revert to naturally indigenous vegetation. The revegetation process will be monitored annually in accordance with the FERC Procedures and federal and state wetland permits. If excessive erosion occurs, these areas will be stabilized and revegetated.

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RESOURCE REPORT 3
VEGETATION AND WILDLIFE

3.0 VEGETATION AND WILDLIFE

This Resource Report describes the existing fish, wildlife, and vegetation resources in the vicinity of the Project and potential impacts of construction and operation on these resources. Section 3.1 describes fishery resources in the Project area. Section 3.2 describes the vegetation types currently found in the vicinity of the Project. Section 3.3 describes wildlife resources, including habitat use and unique ecosystems. Section 3.4 describes the occurrence of T&E species that have been reported in the vicinity of the Project. Finally, Section 3.5 provides a list of references.

3.1 FISHERIES

This section identifies and discusses the fisheries potentially affected by the Project. Fisheries habitats were determined based on PA Code stream water quality designations, PA FBC stream classifications, field environmental surveys conducted between March and May 2025, and supplemented by aerial photographs and site plan interpretations.

3.1.1 FISHERIES CLASSIFICATIONS

The Project area is not located within any National Oceanic and Atmospheric Administration (NOAA) Fisheries Essential Fish Habitat (NOAA Fisheries 2025). The streams in the Project LOD are all designated under PA Code Chapter 93 as High-Quality Coldwater Fishes (HQ CWF), Warmwater Fishes (WWF), drains to HQ CWF, or drains to WWF. Streams designated as HQ-CWF or CWF maintain and/or propagate fish species (including those of family Salmonidae) and additional flora and fauna which are indigenous to a cold-water habitat (25 PA Code § 93.3). Streams designated as WWF maintain and/or propagate fish species and additional flora and fauna which are indigenous to a warm water habitat (25 PA Code § 93.3).

The PA FBC provides trout water classifications for streams in PA. The Project does not cross or indirectly affect any trout streams, including Class A Trout Streams, Stocked Trout Streams, Special Regulation Trout Streams, Natural Reproduction Trout Streams, Keystone Select Stocked Trout Streams, or Wilderness Trout Streams (PA FBC 2025).

Please refer to Table 2-3 for PA Code Chapter 93 Designations and PA FBC classifications for all Project waterbodies.

3.1.2 EXISTING RESOURCES

Based on the results of the field environmental surveys, 31 waterbodies will be crossed by this Project (see Table 2-3 in Resource Report 2).

Pipeline

The pipeline construction workspace will cross 20 waterbodies, including 17 streams, two (2) roadside ditches, and one (1) upland swale (see Table 2-3 in Resource Report 2). The waterbodies include six (6) ephemeral streams, four (4) intermittent streams, and five (5) perennial streams (all UNTs to the Ohio River, Service Creek, Haden Run, and Ambridge Reservoir).

Of the streams within the pipeline construction workspace, seven (7) are designated WWF or drains to WWF, and 10 are designated HQ-CWF or drain to HQ-CWF.

National Fuel does not anticipate that PA DEP will require seasonal in-stream construction timing restrictions as no streams with trout water designations are crossed by the Project (see Table 2-3 in Resource Report 2).

Mercer CS

There are four (4) waterbodies present within the construction area for the Mercer CS, including two (2) ditches and two (2) upland swales.

Service Creek Tie

There are no waterbodies present within the construction area at the Service Creek Tie.

Shippingport M&R Station

There are no waterbodies present within the construction area for the Shippingport M&R Station.

Access Roads

A total of 21 permanent and temporary access roads have been proposed for use during construction. The majority of the lengths of these roads consist of existing roads. However, several roads will need minor extensions to reach the pipeline ROW or improvements to widen the roads (see Resource Report 8 – Land Use, Recreation, and Aesthetics for a list of access roads, existing conditions, and proposed modifications). These access roads will cross seven (7) waterbodies (see Table 2-3 in Resource Report 2), including two (2) ephemeral streams, two (2) intermittent streams, one (1) perennial stream (all UNTs to Haden Run or Service Creek), and two (2) ditches. The majority of these waterbodies are already culverted or otherwise well conveyed under existing roads and will not be directly impacted.

All of the streams and ditches within the access road construction workspace are designated drains to WWF.

Staging Areas

One (1) roadside ditch is present in Staging Area 9. It will be avoided, and no impacts are anticipated.

3.1.3 CONSTRUCTION AND OPERATION IMPACTS

National Fuel will observe timing restrictions as required in its PA DEP 105 Water Obstruction and Encroachment Permit for streams, if any require seasonal timing restrictions. Observance of timing restrictions will ensure sensitive fisheries resources located within the Project streams are not impacted during critical times of the year. However, in-stream work timing restrictions are not anticipated to be required, based on the absence of trout fisheries and aquatic T&E species issues.

National Fuel will employ a dry crossing method to cross the waterbodies crossed by the Project pipeline unless there is no perceptible flow at the time of construction, as described in Resource Report 2. After the pipe is lowered into the trench, previously excavated material will be returned to the trench line for backfill, streambeds will be restored to their former elevations and grades, and all stream banks will be restored and stabilized with erosion controls.

Additionally, National Fuel will implement their ESCAMP and PPC Plan during construction and restoration activities. As a result, no adverse effects to fisheries are anticipated from construction of the Project.

The pipeline will be hydrostatically tested following construction. Details describing the hydrostatic test water withdrawal and discharge methods are included in Resource Report 2. As described, hydrostatic testing activities (including water withdrawal and discharge) will not negatively impact fishery resources.

3.2 VEGETATION

This section identifies and discusses the terrestrial habitats impacted by the Project. Terrestrial habitats were determined based on field environmental surveys conducted between March and May 2025, as supplemented by aerial photographs and site plan interpretations.

3.2.1 EXISTING RESOURCES

The major vegetation types that occur in the Project area are forest/woodland, open land – herbaceous, open land – scrub-shrub, agricultural land, and PEM wetlands. Additionally, the following land use types are found in the Project area: industrial/commercial land, paved roads, and residential land.

The following section describes the vegetation resources in the Project area.

3.2.1.1 Forest/Woodland

Hardwood species such as red maple (*Acer rubrum*), white oak (*Quercus alba*), black cherry (*Prunus serotina)*, American beech (*Fagus grandifolia*), white ash (*Fraxinus americana*), and hickory (*Carya* spp.) species are common in the Project area. Softwood species are less prevalent and include Norway spruce (*Picea abies*). Common shrubs include multiflora rose (*Rosa multiflora*), honeysuckles (*Lonicera* spp.), and Japanese barberry (*Berberis thunbergii)*. Species dominance varies depending on location.

3.2.1.2 Open Land – Herbaceous

Herbaceous open land, which includes fallow fields and utility ROWs, is typically vegetated with grasses, forbs, and wildflower species. Perennial ryegrass (*Lolium perenne*), cat grass (*Dactylis glomerata*), smooth brome (*Bromus inermis*), reed canary grass (*Phalaris arundinacea),* white clover (*Trifolium repens*), plantain (*Plantago major, P. lanceolata*), dandelion (*Taraxacum officinale*), asters (*Aster* spp.), and goldenrod (*Solidago* spp.) dominate these areas. Most of the herbaceous open land in the Project area is maintained and regularly mowed.

3.2.1.3 Open Land – Scrub-Shrub

Scrub-shrub open land is similar to herbaceous open land but also includes a shrub stratum with scattered woody vegetation, including honeysuckles (*Lonicera* spp.) and multiflora rose (*Rosa multiflora)*.

3.2.1.4 Agricultural Land

Agricultural areas consist of a pasture. This field is bordered by open land and forest/woodland.

3.2.1.5 Palustrine Emergent Wetlands

PEM wetlands are freshwater wetlands primarily dominated by herbaceous vegetation. Representative species found in the Project PEM wetlands include sedges (*Carex* spp.), soft rush (*Juncus effusus*), cattail (*Typha* spp.), sensitive fern (*Onoclea sensibilis*), spotted jewelweed (*Impatiens capensis*), reed canary grass, creeping-jenny (*Lysimachia nummularia*), and bulbous bittercress (*Cardamine bulbosa*).

3.2.1.6 Industrial/Commercial Land and Residential Land

Industrial/commercial land and residential land consist of existing access roads, parking areas, other graveled ground surfaces, and areas of commercial and residential development. Vegetation community types within developed land vary from residential lawn and ornamental plantings to grass/wildflower and brush cover along roadsides,

and parking surfaces. These land uses typically do not provide high quality habitats for native plants or animal species.

3.2.1.7 Vegetation Communities of Special Concern

Communities of special concern include sensitive or protected vegetation types, natural areas, and plant communities. The PA Natural Heritage Program (PA NHP) identifies those plants, natural communities, and habitats most at risk of extinction at the global or local level. The Statewide Natural Heritage Conservation Explorer Map includes Core Areas. A review of PA NHP Natural Heritage Conservation Explorer Map (PA NHP 2021) showed that the Project does not intersect any Core Areas.

3.2.2 CONSTRUCTION AND OPERATION IMPACTS

Table 3-1 summarizes and quantifies vegetation types and land uses in the Project area. A total of 131.01 ac will be disturbed by Project activities. As presented in Table 3-1, this includes disturbance of a total of 101.7 ac of vegetated land, including forest/woodland (50.04 ac), open land – herbaceous (46.89 ac), open land – scrub shrub (4.36 ac), agricultural land (0.24 ac), and PEM wetlands (0.05 ac). The remaining construction acres (29.31 ac) consist of developed land (i.e., industrial/commercial, paved roads, and residential land). A significant portion of impacts (39.62 ac) is within temporary workspace areas and vegetation and land use that are temporarily impacted will be restored and return to preconstruction conditions. The remaining impacts (91.38 ac) are located within the new permanent ROW, within which most vegetation types will be restored and allowed to revert to preconstruction conditions, with the exception of forest/woodland areas (42.88 ac within the 75-foot-wide permanent ROW), which will generally not be allowed to revert within the permanently maintained portions of the ROW.

Eight (8) wetlands occur within the LOD of this Project: six (6) wetlands are located in the pipeline construction LOD, and two (2) wetlands (formed as a result of stormwater management measures) are located in the Mercer CS and will be avoided. No habitats of significant vegetation species or communities were identified.

Following completion of the Project, National Fuel will restore all temporarily disturbed areas in accordance with their ESCAMP, which is consistent with the FERC Plan and FERC Procedures (FERC 2013a, 2013b), and the Project-specific E&SCP. The vegetation types affected by construction will be restored and allowed to revert to their preconstruction cover type after completion of the Project, with the exception of forested and scrub shrub areas in the proposed permanent ROW, which will be converted to herbaceous open land.

No vegetative communities of special concern or habitat for rare, threatened, or endangered plant species will be affected by the Project (see section 3.4.1.2).

Table 3-1. Vegetation Types and Land Uses Affected by the Project

Project Component	Agricultural Land		Forest/ Woodland		Open Land – Herbaceous[a]		Open Land – Scrub-Shrub[a]		Palustrine Emergent Wetlands[a]		Paved Road		Industrial/ Commercial Land		Residential Land		Total		
	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Temp (ac)	Perm (ac)	Grand (ac)
Shippingport Lateral	0.11	0.13	3.86	37.18	9.33	22.32	0.04	0.09	0.00	0.05	0.10	0.30	0.57	0.73	1.88	4.15	**15.89**	**64.95**	**80.84**
Service Creek Tie	-	-	0.68	0.66	0.23	0.54	-	-	-	-	-	-	-	-	-	-	**0.91**	**1.20**	**2.11**
Shippingport M&R Station	-	-	-	-	-	-	0.28	0.06	-	-	-	-	0.28	0.53	-	-	**0.56**	**0.59**	**1.15**
Modifications at Mercer CS	-	-	0.00	0.94	0.00	7.65	-	-	0.00	0.12[b]	-	-	0.00	4.67	-	-	**0.00**	**13.38**	**13.38**
Staging Areas	-	-	0.02	0.00	0.86	0.00	3.88	0.00	-	-	-	-	8.16	0.00	1.29	0.00	**14.21**	**0.00**	**14.21**
Access Roads	-	-	2.60	4.10	3.50	2.46	0.00	0.01	-	-	-	-	0.39	2.04	1.57	2.65	**8.06**	**11.26**	**19.32**
Totals	**0.11**	**0.13**	**7.16**	**42.88**	**13.92**	**32.97**	**4.20**	**0.16**	**0.00**	**0.17**[2]	**0.10**	**0.30**	**9.40**	**7.97**	**4.74**	**6.80**	**39.63**	**91.38**	**131.01**
Grand Total	**0.24**		**50.04**		**46.89**		**4.36**		**0.17**[2]		**0.40**		**17.37**		**11.54**		**131.01**		

Notes:
ac = acre or acres, Temp = temporary, Perm = permanent
Totals may differ slightly from sum of parts due to rounding.

[a] This vegetation type is not discussed in Resource Report 8 – Land Use, Recreation, and Aesthetics. Any discrepancy in these acreages is due to the vegetation types being included in broader land use categories.

[b] 0.12 acre of manmade PCSM has resulted in the development of wetland characteristics at two locations within the Project's footprint at Mercer CS. However, impacts to these wetlands will be avoided, so the actual total impact to PEM wetlands will be 0.05 acre in the permanent ROW of the Shippingport Lateral pipeline.

3.3 WILDLIFE

This section identifies and discusses the wildlife species associated with the vegetation types identified in section 3.2.1, including forest/woodland, open land – herbaceous, open land – scrub-shrub, agricultural land, and PEM wetlands. Because industrial/commercial land and residential lands do not typically provide high quality habitat for animal species, these types are not discussed in this section.

3.3.1 EXISTING RESOURCES

3.3.1.1 Forest/Woodland

The forested portions of the Project area provide habitat for a number of wildlife species; in particular, for those species adapted to live in fragmented woodlands. The different vegetation layers present from the canopy, understory, herbaceous, and the leaf litter layers provide a variety of habitats. Forest/woodland habitat can support black bear (*Ursus americanus*), bobcat (*Lynx rufus*), coyote (*Canis latrans*), eastern chipmunk (*Tamias striatus*), fox squirrel (*Sciurus niger*), gray squirrel (*Sciurus carolinensis*), white-tailed deer (*Odocoileus virginianus*), great horned owl (*Bubo virginianus*), and turkey (*Meleagris gallopavo*) (PA GC 2011a, b).

3.3.1.2 Open Land – Herbaceous

Open land may provide habitat for mammals such as the bobcat, white-tailed deer, red-tailed hawk (*Buteo jamaicensis*), and turkey (PA GC 2011a, b). Maintained areas (i.e., routinely mowed) are regularly disturbed habitats and the high frequency of disturbance tends to limit resources for wildlife.

3.3.1.3 Open Land – Scrub-Shrub

Scrub-shrub may provide habitat for mammals such as the bobcat, coyote, white-tailed deer, and common raven (*Corvus corax*) (PA GC 2011a, b).

3.3.1.4 Agricultural Land

Agricultural crops in the Project area consist largely of hay, corn, and soybeans. Representative wildlife species include bobcat, coyote, eastern cottontail (*Sylvilagus floridanus*), white-tailed deer, red-tailed hawk, and turkey. (PA GC 2011a, b). Cultivated farm fields are considered of minor to moderate importance as wildlife habitat (relative to the other habitats identified) due to the lack of food source diversity that results from the monocultures of crops found in agricultural fields.

3.3.1.5 Palustrine Emergent Wetlands

PEM wetlands provide a source of water, food, and nesting, and cover for a variety of wildlife. These habitats can support bobcat, muskrat (*Ondatra zibethicus*), white-tailed deer, great blue heron (*Ardea herodias*), northern harrier (*Circus cyaneus*), red-winged blackbird (*Agelaius phoeniceus*), and various waterfowl (PA GC 2011a, b).

3.3.1.6 Wildlife Areas of Special Concern

Sensitive habitats are areas that provide breeding, rearing, nesting, or calving areas; migration routes; or high-quality cover or forage areas. Sensitive wildlife habitat typically includes existing or proposed National Wildlife Refuges, state wildlife management areas, or privately owned management areas or preserves. The Project does not cross any protected areas from the following agencies/entities: Bureau of Land Management, U.S. Census Bureau, U.S. Department of Defense, USFWS, National Park Service (NPS), Natural Resources Conservation Service (NRCS), U.S. Forest Service, or any state, local, or nonprofit fee and easement protected areas (USGS 2025).

A review of Pennsylvania Natural Heritage Inventory (PA NHP 2021) showed that the approximately 0.32 mile of southeasternmost portion of the pipeline, including the Service Creek Tie, intersects the Lower Raccoon Creek Natural Heritage Area (NHA). The Lower Raccoon Creek NHA encompasses a large section of steep slopes and tributary stream valleys, many of which are recovering from past disturbance. The most ecologically significant portions of the site are generally steep slopes and tributary valleys, which are generally inaccessible and will not be impacted by the Project. This NHA is designated as a site of global significance due to the presence of five species of concern, none of which are likely to be impacted by the Project. Raccoon Creek provides habitat for a rare fish species, bluebreast darter (*Etheostoma marurm*) and an unidentified species of sensitive concern; in addition, a population of white trout-lily (*Erythronium albidumn*) grows within the floodplain of Racoon Creek. Neither Raccoon Creek nor its floodplain will be crossed by the Project and these species were not identified as T&E species concerns in the PNDI results for the Project. Two additional species of concern rely on remnant patches of mature forest; however, the forest affected by the pipeline and Service Creek Tie is mid-successional, rather than mature, and the Project does not traverse any mature forest areas.

The majority of the Project is located with the Raccoon Creek Valley & State Park Important Bird Area (IBA). Racoon Creek Valley spans approximately 100 square miles of land between the Ohio River, Route 22, the Pennsylvania/Ohio border, and Route 79, and covers parts of southern Beaver, western Allegheny, and northern Washington counties. Habitat in the area consists predominantly of second and third-growth deciduous woodland, fields in advanced stages of succession, lakes, stream valleys and pine plantations. Raccoon Creek Valley represents one of the last relatively undeveloped areas in the Pittsburgh region and acts to funnel migratory birds north and south. Conservation concerns for this IBA include development (National Audubon Society 2025).

An Information for Planning and Consultation (IPaC) official species list was generated on June 2, 2025, as part of the USFWS consultation. Specific migratory bird species information was not available at that time. The IPaC species list was re-run on August 19, 2025, and specific migratory bird information continued to be absent (USFWS 2025; see Appendix C).

3.3.2 CONSTRUCTION AND OPERATION IMPACTS

The total acreage of impacts to wildlife habitats (vegetation cover types) is presented in Table 3-1. In general, construction of the Project will have minor, short-term, long-term, and permanent impacts on habitat, resulting in localized impacts to wildlife populations. During construction, the clearing and grading of the pipeline ROW and aboveground facility sites will result in a loss of vegetative cover and could result in some mortality to less mobile wildlife that are unable to avoid construction equipment/machinery (e.g., small rodents, reptiles, and amphibians).

Construction activities will temporarily displace more mobile wildlife species from the construction workspace and adjacent areas due to increased human activity and noise levels. Following construction, wildlife species are expected to resume their normal habits consistent with the availability of post-construction habitats.

National Fuel will adhere to the construction, restoration, and mitigation methods identified in the FERC Plan and FERC Procedures, which are incorporated into National Fuel's ESCAMP and the Project-specific E&SCP. National Fuel will restore vegetative cover, stabilize wetland and stream crossings and use resource-specific procedures for constructing through sensitive habitats. Woodland vegetation removed from the temporary workspace and ATWS areas will be replaced initially by non-woody vegetation that may provide food and shelter for wildlife adapted to open habitats. Trees will be allowed to grow back on cleared workspace outside of the permanent pipeline ROW. After construction, wildlife is expected to return and colonize post construction habitats.

The greatest risks for impacts to migratory birds from the proposed Project are associated with loss of forest and shrub habitat resulting from clearing in permanent and temporary workspaces, and risk to ground nesting species due to construction activity. Direct effects due to construction activities include destruction of nests and eggs, mortality of young, loss of habitat, and construction-related disturbance causing reduced nest attendance and foraging time of adults. Indirect effects include reduced nest success due to reduced nest attendance and foraging time, noise and construction activity disturbance causing fleeing behavior resulting in increased vulnerability to predators, and reduction of interior forest habitat area causing increased vulnerability and habitat unsuitability for interior forest dependent species. Cumulative effects include loss of habitat alteration on a landscape scale potentially affecting local bird populations.

To support compliance with the MBTA, the USFWS has standard recommendations for projects to minimize impacts to birds protected by the MBTA. The recommendation most important to Project logistical planning for construction involves a seasonal restriction for construction and ROW maintenance: "Clearing or mowing of natural or semi-natural habitats such as forests, woodlots, reverting fields, fencerows, and shrubby areas should be conducted between September 1 and March 31," which is outside the nesting season for most native bird species. This recommendation for "winter clearing" is intended to avoid take of most breeding birds, their nests, and their young (i.e., eggs, hatchlings). National Fuel has scheduled tree clearing activities to be completed during winter before April 1. However, in certain areas and

circumstances, there may be safety concerns related to conducting clearing activities in winter in this region, especially on particularly steep slopes in snowy/icy conditions. National Fuel will work with the USFWS to determine the prudent and necessary seasonal timing restrictions along with the proposed Project as it coordinates with USFWS regarding the MBTA.

In accordance with restrictions already imposed by FERC for post-construction vegetation maintenance along the permanent ROW, routine vegetation maintenance along the Project ROW will not be conducted between April 15 and August 1 of any year following completion of construction or other ground disturbance, thus avoiding the primary breeding and nesting season for migratory birds.

In addition, National Fuel has enrolled in the Nationwide Monarch Butterfly Candidate Conservation Agreement for Energy and Transportation Lands (Monarch CCAA) (Cardno, Inc. 2020), a program that also benefits migratory birds. As part of National Fuel's existing Vegetative Maintenance Plan and enrollment in the Monarch CCAA, National Fuel's enrolled transmission ROWs are maintained in a 10-foot-wide swath centered over the underground pipeline that is mowed annually, and the remainder of the ROW, up to 65 additional feet on a 75 foot wide ROW (32.5 feet on either side of the 10-foot-wide strip), that is mowed once every 3 years. During non-mowed years (every 2 of 3 years), the remainder of pipeline ROW corridor will be maintained as idle lands. Further, as a conservation measure of the Monarch CCAA, National Fuel applies to native pollinator seed mixes as part of post-construction site restoration when possible. These native pollinator seed mixes provide higher quality nesting habitat for grassland nesters and more abundant prey base for insectivorous birds.

Timber will be removed from the ROW and limbs and shrubs will be chipped to prevent nesting in downed vegetation. Wood chips will be spread off the LOD or hauled off site at landowner request.

The currently anticipated construction schedule assumes National Fuel will begin tree clearing activities outside of MBTA nesting windows for PA (which are April 1 through August 1) by completing tree clearing before April 1. In addition, National Fuel has designed the Project to minimize tree clearing by paralleling existing utility rights of way to the extent practical.

Stream crossings will be completed as quickly as possible and stream habitats restored upon completion of construction. National Fuel will use construction procedures consistent with its ESCAMP (see Appendix E) and the FERC Procedures to minimize sedimentation, turbidity, and other impacts that may temporarily affect stream vegetation and wildlife.

National Fuel will continue to work with local, state, and federal agencies; landowners; and soil conservation authorities to ensure that construction and mitigation procedures are compatible with both site-specific and regional environmental protection objectives, including following recommendations from PA GC to minimize impacts to species within the Racoon Creek Valley & State Park IBA.

Construction and operation procedures will accommodate general and site-specific protective measures for any significant and/or sensitive wildlife habitat identified during the course of

constructing and operating the Project. Seasonal timing to account for reproductive and migratory patterns will be coordinated with state and federal agencies, as required.

3.4 ENDANGERED AND THREATENED SPECIES

This section identifies and discusses the presence of state- and federally-listed plant and animal species potentially located within the general vicinity of the Project. It also identifies significant habitats such as federally-designated critical habitat and rare plant communities known to occur near the Project. Copies of agency correspondence are provided in Appendix C.

3.4.1 EXISTING RESOURCES

3.4.1.1 Federally-Listed Species

The USFWS was provided a Project description and mapping via IPaC, per the PNDI direction. The official IPaC Species List, generated on June 2, 2025, indicated that the Project is within the range of the following seven federally-listed and proposed species:

- Indiana bat (*Myotis sodalist*) – Endangered
- Northern long-eared bat (NLEB) (*Myotis septenrionalis*) – Endangered
- Tricolored bat (*Perimyotis subflavus*) – Proposed Endangered
- Clubshell (*Pleurobema clava*) – Endangered
- Northern riffleshell (*Epioblasma rangiana*) – Endangered
- Rayed bean *(Villosa fabalis*) – Endangered
- Monarch butterfly (*Danaus Plexippus*) – Proposed Threatened

On July 17, 2025, Tetra Tech completed the NLEB and tricolored bat USFWS Determination Key (D-Key) and received the determination of May Affect –Not Likely to Adversely Affect for both bat species (Appendix C). On August 5, 2025, Tetra Tech completed the Northeast Endangered Species D-Key and received the determination of No Effect for clubshell, northern riffleshell and rayed bean, and a May Affect determination for Indiana bat (Appendix C). On August 13, 2025, Tetra Tech received a response from USFWS via email indicating that the Project is not expected to result in negative impacts to Indiana bats and no further review is required (Appendix C).

Tree clearing/cutting is being scheduled between November 15 and March 31 to avoid potential impacts to protected bat species.

3.4.1.2 State-Listed Species

On behalf of National Fuel, Tetra Tech submitted requests to the PNDI regarding the potential of species of concern within the Project area. Based on the results of the search (PNDI-825480 dated June 2, 2025; PNDI-838089 dated May 23, 2025; and PNDI-825477 dated May 23, 2025), the PNDI responded “No Known Impact” for the Mercer CS and Staging Area 9 portions of the Project, and that no further consultation was required via the PNDI receipts (Appendix C). For the pipeline portion of the Project, on June 2, 2025, PA DCNR responded that the Project is not likely to adversely affect any species or areas of concern, and that no further consultation was

required via the PNDI receipt (Appendix C). On June 2, 2025, PA FBC responded that while an element occurrence of a rare, candidate, threatened, or endangered species under their jurisdiction is known from the vicinity of the Project, given the nature of the Project, the immediate location, or the current status of the nearby element occurrence(s), no adverse impacts are expected to the species of special concern (Appendix C). On July 22, 2025, PA GC responded that potential impact was anticipated on two endangered species, little brown bat (*Myotis lucifugus*) and tricolored bat. They advised that a seasonal restriction should be implemented to avoid potential impacts to these species, with tree trimming or cutting limited to November 15 to March 31. They also noted that the Project is located in the Racoon Creek Valley & State Park IBA, and recommended several voluntary measures to minimize impact, including co-locating the pipeline and associated facilities with disturbed areas, minimizing the width of the temporary construction and permanent ROWs, maximizing the rotation of mowing and/or clearing along the permanent ROW, and using specific seed mixes (Appendix C). National Fuel responded with commitments and demonstration of the extent of Project observance of these recommendations in a follow-up letter dated August 6, 2025 (Appendix C).

3.4.2 CONSTRUCTION AND OPERATION IMPACTS

No federally-listed or state-listed T&E species or their critical habitats are expected to be adversely affected by the proposed Project.

After reviewing the relevant information, including National Fuel's commitment to winter clearing, USFWS determined that the Project is not anticipated to adversely affect any federally listed T&E species.

Additionally, PA FBC and PA DCNR indicated that the Project is not likely to adversely affect any state-listed species. The PA GC indicated that the Project had the potential to impact state-listed species, however National Fuel will follow the seasonal restriction to avoid these impacts. Construction and operation of the Project is not likely to impact any state species of concern.

3.5 REFERENCES

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RESOURCE REPORT 4
CULTURAL RESOURCES

4.0 CULTURAL RESOURCES

This Resource Report discusses the status and results of cultural resource reviews and approvals for the Project. Section 106 of the National Historic Preservation Act (NHPA), as amended (16 U.S. Code Part 470 et seq.) requires federal agencies, including FERC, to consider the effects of proposed projects on historic properties listed or eligible for listing on the National Register of Historic Places (NRHP) (including archaeological resources, historic structures, and cultural landscapes) and to afford the Advisory Council on Historic Preservation (ACHP) an opportunity to comment under 36 CFR Part 800 (18 CFR § 380.14).

Regulations for the protection of historic properties in 36 CFR Part 800 describe the process for compliance with Section 106 of the NHPA, including defining the Area of Potential Effect (APE); steps to identify resources and evaluate effects; steps to avoid, minimize, or mitigate adverse effects on historic properties; and consultation with interested parties including the PA SHPO and Native American tribes. The information provided in Resource Report 4 is intended to assist FERC in complying with Section 106 of the NHPA, describes the potential Project construction and operation impacts on cultural resources, and summarizes the results of consultation with federal, state, local, and tribal agencies, as appropriate.

Section 4.1 defines the Project's APE and section 4.2 presents the results of background research. Section 4.3 describes the status of the agency review and comments for the Project. Section 4.4 details the status of cultural resource investigations to be conducted in the area considered for the proposed Project and provides a plan to address the potential for unanticipated discoveries during construction activities. Section 4.5 provides a list of references. Copies of correspondence between National Fuel, agencies, and Native American tribes are provided in Appendix C. National Fuel's Unanticipated Discovery Plan for Cultural Resources and Human Remains is provided as Appendix H.

4.1 AREA OF POTENTIAL EFFECT

The APE includes all areas where the undertaking could cause changes to land, structures, or their uses whether the change would be direct, indirect, beneficial, or adverse. In addition to areas of ground disturbance, this includes all locations from which elements of the undertaking could be visible. As defined in 36 CFR § 800.16(i), an "effect" is an "alteration to the characteristics of a historic property qualifying it for inclusion in or eligibility for the NRHP." The APE is defined based on the potential for effect, which may differ between above-ground resources, including historic structures and cultural landscapes, and subsurface resources. Figures that illustrate the APE for the Project are provided in Appendix A (Maps and Construction Alignment Sheets).

4.1.1 APE FOR ARCHAEOLOGICAL RESOURCES

For archaeological resources, the APE has been determined to include all areas of ground-disturbing activities, including all Project workspaces, access roads, and staging areas. This includes 7.26 miles of the pipeline's 75-foot-wide permanent ROW within a 150-foot-wide survey corridor that includes temporary construction workspace. ATWS will be utilized at specific locations, such as at stream and road crossings (for specialized construction techniques) and in agricultural areas (to allow for topsoil segregation). The pipeline lateral is proposed to be installed mostly via open trench excavation methods. Additionally, National Fuel's existing pigging facility at the east end of the proposed pipeline will be expanded for the installation of pig launcher, gas quality equipment, over-pressure protection and ancillary equipment (Service Creek Tie) in Independence Township, Beaver County, PA. A new M&R (Shippingport M&R Station) will be constructed at the Shippingport Industrial Park that includes the installation of a 24-inch pig receiver, OPP, gas filtration equipment, control valves, measurement equipment, station fencing and security systems, and ancillary equipment at the west end of the proposed pipeline in Shippingport Borough, Beaver County, PA. Crossover valves will be added at National Fuel's existing Mercer CS in Jefferson Township, Mercer County, PA. In addition, nine staging areas totaling approximately 14.21 ac and access roads totaling approximately 3.28 miles (19.87 ac) will be used for construction. In total, for the entire Project, the APE would reflect an approximately 131-ac Project impact area.

4.1.2 APE FOR HISTORIC ARCHITECTURAL PROPERTIES

The APE for historic architectural properties includes areas where direct and indirect Project impacts have the potential to alter character-defining features of a historic property's significance and includes areas that have a visual link to the proposed Project such that their landscape, setting, and viewshed that could change as a result of the Project. Historic architectural properties are defined as any building or structure at least 50 years of age listed or eligible for listing on the NRHP.

Given that National Fuel proposes the construction of underground pipeline, the APE for historic architectural properties for this Project is limited to areas where alterations to the landscape through removal of vegetation or modifications to surface topography lie within view of a historic resource. Tetra Tech has identified the Project's visual APE to include a 1.0-mile radius within which the undertaking may indirectly cause alterations in the character or use of potential aboveground historic properties.

4.2 BACKGROUND RESEARCH

Before initiating consultation with the PA SHPO, Tetra Tech conducted preliminary archaeological and architectural background research to gather information regarding previous cultural resources investigations in the area, as well as previously recorded archaeological sites and historic architectural properties within 1.0-mile of the Project. Tetra Tech used the following methodology to complete the background research:

- Identify any previously recorded archaeological sites and historic architectural properties through background research and state site file searches. Data pertaining to the previously recorded sites, including their locational, functional, and temporal characteristics were reviewed where applicable.
- Review recent cultural resources surveys conducted in the counties where the proposed Project is located.
- Review primary and secondary historic information (e.g., maps, county histories) to identify areas where previously recorded historic structures or landscapes potentially were located.

As part of this work, Tetra Tech consulted PA SHPO site files via PA-SHARE online database (PHMC 2025), in addition to conducting a review of historical documents, USGS topographic maps, aerial imagery, and the NRHP (NRHP 2025). No NRHP-listed or -eligible properties were identified within the proposed Project or within 1.0-mile of the Project.

4.2.1 PREVIOUSLY IDENTIFIED ARCHAEOLOGICAL RESOURCES

A search of available records on PA-SHARE indicated that 29 previously recorded archaeological resources are located within 1.0 mile of the Project (Table 4-1). Of these, seven (7) are not NRHP-eligible, 21 have Undetermined NRHP eligibility, and one (1) is listed as demolished. Twenty-five (25) of the sites are Indigenous and four (4) are historic. Only one (1) of the previously documented archaeological resources was identified within the archaeological APE. This site is designated as the Parrish Site (2014RE00950; 36BV0377) and is a 19th-century historic site determined Not Eligible for NRHP-listing.

Table 4-1. Previously Recorded Archaeological Resources within 1.0 Mile

Resource ID Number	PASS Number	Name	Type	NRHP Eligibility
1950RE00056	36BV0004	Lower Field	Woodland, Village with Burials	Demolished
2013RE00931	36BV0370	Ohio River-Midland Beaver Road 1	Indigenous, Lithic Reduction	Not Eligible
2014RE00841	36BV0375	Oldham-Mutkus	Indigenous, Open Habitation	Not Eligible
2014RE00950	36BV0377	Parrish Site	19th-Century	Not Eligible
2014RE00972	36BV0379	Service Creek	Indigenous, Open Habitation	Not Eligible
2014RE01115	36BV0374	Mutkus	Indigenous, Open Habitation	Not Eligible
2016RE03627	36BV0395	AECOM-BVIn003-ST1	19th-Century	Not Eligible
2018RE03867	36BV0409	Buckeye 3	Indigenous, Open Habitation	Not Eligible
1950RE00058	36BV0003	Old Indian Fort	Archaic, Woodland, Open Habitation	Undetermined
1952RE00041	36BV0049	Upper Parsons & Parish	Woodland, Open Habitation	Undetermined
1958RE00052	36BV0061	John Rought	Woodland, Open Habitation	Undetermined

Resource ID Number	PASS Number	Name	Type	NRHP Eligibility
1963RE00022	36BV0111	South Service Creek	Indigenous, Open Habitation	Undetermined
1963RE00068	36BV0108	Kowan Farm	Archaic, Open Habitation	Undetermined
1963RE00098	36BV0112	Rought Run	Indigenous, Open Habitation	Undetermined
1963RE00120	36BV0110	North Service Creek	Indigenous, Open Habitation	Undetermined
1963RE00144	36BV0092	Industry	Indigenous, Unknown	Undetermined
1978RE01120	36BV0274	Bobish	Archaic, Open Habitation	Undetermined
1978RE01138	36BV0159	n/a	Archaic, Woodland, Open Habitation	Undetermined
1978RE01162	36BV0160	Plaza West	Woodland, Open Habitation	Undetermined
1978RE01289	36BV0161	Ben Parish	Indigenous, Open Habitation	Undetermined
1978RE01298	36BV0078	Parish Farm	Archaic, Woodland, Open Habitation	Undetermined
1984RE03341	36BV0251	Antel	Archaic, Woodland, Surface Scatter	Undetermined
1986RE01615	36BV0267	Old Shippingport Ferry	18th-19th Century	Undetermined
1995RE56542	36BV0027	Shippingport Sand & Gravel	Indigenous, Open Habitation	Undetermined
1995RE57067	36BV0109	n/a	Indigenous, Unknown	Undetermined
1995RE57441	36BV0022	Boy Scout Camp	Archaic, Woodland, Open Habitation	Undetermined
1995RE57816	36BV0089	Petroglyph	Indigenous, Unknown	Undetermined
2016RE02728	36BV/055	n/a	Indigenous, Isolate	Undetermined
2016RE02889	36BV0399	AECOM-BVSh001-HS1	19th-20th Century	Undetermined

Notes:
PASS = Pennsylvania Archaeological Site Survey

4.2.2 PREVIOUSLY IDENTIFIED ARCHITECTURAL RESOURCES

Twenty-six (26) historic properties have been identified within 1.0-mile of the Project's direct APE. No NRHP-listed or eligible properties were identified within the proposed Project. One NRHP-eligible property was identified within 1.0-mile of the Project. Three (3) Not Eligible properties and one (1) Undetermined property were identified within the Project area and the remaining 22 historic properties/structures were identified within 1.0-mile of the Project (Table 4-2). Two cultural "Transportation/Pedestrian Related" resources with undetermined NRHP-eligibility cross the Project.

Table 4-2. Previously Recorded Historic Properties within 1.0 Mile of the Project

Resource ID Number	Address	Name/Type (Year; Description)	NRHP Eligibility
1993RE01080	(Pittsburgh to Ohio State Line)	Pennsylvania Railroad: Main Line	Eligible
2001RE00461	State Route (SR) 68, PA	Bridge	Demolished
2004RE03184	SR 3020, PA	Bridge	Demolished
1983RE02011	N/A	Shippingport Atomic Power Station	Not Eligible
2004RE01662	SR 3022, PA	Bridge	Not Eligible
2008RE00735	N/A	Christler-Marker Cemetery	Not Eligible
2014RE00017	N/A	National Fuel Gas Company: Line N Extension	Not Eligible
2017RE00747	185 Kennedy Rd., Hookstown PA 15050	Lokomski House	Not Eligible
2017RE00991	194 Kennedy Rd., Hookstown PA 15050	Franc House	Not Eligible
2017RE01035	101 Shippingport Rd., Aliquippa PA 15001	Larriza Farm	Not Eligible
2017RE02069	122 Waite Lane, Aliquippa PA 15001	Building	Not Eligible
2017RE02282	169 Moores Rd., Aliquippa PA 15001	Dobich Farmstead	Not Eligible
2017RE02302	1022 Frankfort Rd., Aliquippa PA 15001	Smith House	Not Eligible
2017RE02409	4264 Green Garden Rd., Aliquippa PA 15001	Vic's Van Daddy	Not Eligible
2017RE02803	00000 State Route 68, Industry PA 15052	Tank Farm	Not Eligible
2017RE02845	177 Kennedy Rd., Hookstown PA 15050	Lokomski House 2	Not Eligible
2017RE03242	113 Clearview Dr., Aliquippa PA 15001	James and Gill House	Not Eligible
1995RE47069	Service Creek Rd., PA	Service Creek Rd. Bridge	Undetermined
2015RE00378	2562 Midland-Beaver Rd, PA	N/A	Undetermined
2017RE01102	103 Davis Road, Aliquippa PA 15001	Reddinger Farm	Undetermined
2017RE02482	3802 Green Garden Road, Aliquippa PA 15001	Reddinger House	Undetermined
2017RE02658	4041 Patterson Drive, Aliquippa PA 15001	Langton House	Undetermined
2017RE03261	236 Moores Road, Aliquippa PA 15001	Clancy House	Undetermined
2018RE01232	3865 Green Garden Road, Aliquippa PA 15001	n/a	Undetermined
2018RE03413	Shippingport - HI Road, PA	Shippingport Bridge	Undetermined
2021RE00108	N/A	Christler-Marker Cemetery	Undetermined

Source: PHMC 2025

4.2.3 PREVIOUS CULTURAL RESOURCE INVESTIGATIONS

The background review indicated limited portions of the Project have been previously surveyed by previous cultural resource investigations. There are 11 previous cultural resource investigations within 1.0-mile of the Project's direct APE (Table 4-3).

Table 4-3. Previous Surveys within 1.0 Mile of the Project

SHPO Project Number	Report Title	CRM Firm	Date Completed
1992SR00234	Phase I Cultural Resources Survey, Industry Borough. Sanitary Sewer System, Beaver County, PA	Kelley Lynn Cosgrove	9/1/1992
1995SR00252	Abbreviated Phase I Archaeological Reconnaissance: Proposed Towboat Dock, Shippingport Borough, Beaver County, PA	Aarcon, Inc. Archaeology & Historical Consultants	3/1/1995
1995SR00291	Phase I Cultural Resource Survey, Line N National Fuel Gas Pipeline Replace Project, Centre, Hopewell, Independence Townships, Beaver County, PA	Garrow & Associates, Inc.	12/1/1995
2013SR00163	Phase I Archaeological Survey, Bruce Mansfield-Glenwillow 345kV Transmission Line Rebuild, Shippingport, Ohioville, and Industry Boroughs, and South Beaver Township, Beaver County, PA	Louis Berger Group	11/1/2013
2014SR00207	Phase I Archeological Investigation West Side Replacement and Modernization Project, Venango, Beaver, Allegheny, and Washington Counties, PA	GAI Consultants, Inc.	2/18/2014
2014SR00210	Phase I Archaeological Report, Addendum 2, West Side Expansion and Modernization Project, Venango, Beaver, Allegheny, and Washington Counties, PA	GAI Consultants, Inc.	11/18/2014
2016SR00344	Phase I Archaeological Investigations, Revolution Pipeline Project, Butler, Beaver, Allegheny, and Washington Counties, PA	SWCA, Inc.	8/5/2016
2017SR00146	Record of Disturbance Form, Shell Industry; Shell Petrochemical Plant Development, Industry Borough, Beaver County, PA	Leonard R. Voellinger, RPA / Jacobs	2/20/2017
2017SR00503	Phase I Archaeological Survey Report, Falcon Ethane Pipeline Project, Washington, Allegheny, and Beaver Counties, PA	AECOM, Inc.	1/1/2017
2017SR00505	Phase I Archaeological Survey, Addendum 1, Falcon Ethane Pipeline, Washington, Allegheny, and Beaver Counties, PA	AECOM, Inc.	7/1/2017
2018SR00268	Phase I Archaeological Survey, Buckeye Midland Terminal HDD Pipeline Replacement Project, Industry and Shippingport Boroughs, Beaver County, PA	Apogee, Inc.	9/28/2018

Notes:
CRM = cultural resource management
Source: PHMC 2025

4.3 AGENCY RECOMMENDATIONS

4.3.1 FEDERAL AND STATE AGENCIES

Tetra Tech submitted a letter request dated June 6, 2025, to the PA SHPO via PA-SHARE for review and comment on the proposed Project. The PA SHPO reviewed the proposed Project with respect to its survey files and in accordance with Section 106 of the NHPA of 1966, as amended in 1980 and 1992, and the regulations (36 CFR Part 800) of the ACHP. In correspondence dated June 27, 2025, the PA SHPO responded that the Project will have No Effect on historic architectural properties included in or eligible for inclusion in the NRHP within the direct and indirect APE given the limited aboveground impact of the Project. Therefore, further historical architectural work was not required for the Project.

The PA SHPO recommended a Phase I archaeological survey of the direct APE be performed given the archaeological sensitivity of the Project area. National Fuel conducted a Phase I archaeological survey of the Project area in July 2025. The results of the survey and follow-up SHPO correspondence will be provided to FERC and a No Effect determination is anticipated. Documentation of all correspondence to date can be found in Appendix C (Agency Correspondence).

4.3.2 LOCAL AGENCIES

The PA SHPO responded that the Project will have No Effect on historic architectural properties included in or eligible for inclusion in the NRHP within the direct and indirect APE given the limited above ground impact of the Project. Therefore, further historical architectural work was not required for the Project and the PA SHPO did not instruct National Fuel to coordinate with any county or local historical societies as part of the background review for the Project.

4.3.3 NATIVE AMERICAN TRIBES OR GROUPS

Consultation letters regarding the Project were sent to the relevant Native American authorities in July 2025, and follow-up letters were sent on August 26, 2025. A listing of the groups provided letters, along with responses received as of August 26, 2025, are included in Table 4-4. Inclusion of Native American groups and individuals (including Tribal Historic Preservation Officers [THPOs]) on this list was predicated on information retrieved from the following offices and resources: Bureau of Indian Affairs (2025); National Association of Tribal Historical Preservation Officers (2025); PennDOT Tribal Consultation/Tribal Areas of Interest Database (PennDOT 2025); United States Department of Housing and Urban Development (2025) Tribal Directory Assessment Tool (TDAT); and the NPS Tribal Preservation Program. Copies of all letters and responses received to date are provided in Appendix C.

Because National Fuel proposes to implement an Unanticipated Discovery Plan for Cultural Resources and Human Remains (Appendix H) to allow for appropriate notification and treatment of archeological sites and/or human remains encountered during construction for all portions of the Project, National Fuel believes any additional potential Native American issues of concern that may be associated with the Project will be adequately addressed.

Table 4-4. Summary of Native American Consultation

Absentee-Shawnee Tribe of Indians of Oklahoma (John R. Johnson, Governor; Clayton Martinez, THPO), 2025 S. Gordon Cooper Drive, Shawnee, OK 74801	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Delaware Tribe of Indians (Brad KillsCrow, Chief; Martina Thomas, THPO/Historic Preservation Director), 5100 Tuxedo Blvd., Bartlesville, OK 74006	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Eastern Shawnee Tribe of Oklahoma (Glenna Wallace, Chief) P. O. Box 350, Seneca, MO 64865; (Lora Nuckolls, THPO), 70500 E. 128 Rd., Wyandotte, OK 74370	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Delaware Nation of Oklahoma (Deborah Dotson, Tribal President; Katelyn Lucas, THPO), 31064 State Highway 281, Building 100, PO Box 825, Anadarko, OK 73005	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Osage Nation (Dr. Andrea A. Hunter, Director and THPO; Geoffrey M. Standing Bear, Principal Chief), 100 W. Main, Pawhuska, OK – 74056	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Saint Regis Mohawk Tribe (Beverly Kiohawiton Cook, Chief; Darren Bonaparte, THPO), 71 Margaret Terrance Memorial Way, Akwesasne, NY 13655	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Onondaga Nation (Sidney Hill, Chief; Tony Gonyea, Faithkeeper and THPO), 4040 Route 11 Administrative Building Nedrow, NY 13120	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Seneca Nation of Indians (J. Conrad Seneca, President), 12837 Route 438, Irving, NY 14081; (David Shongo, THPO), 90 O:hi'yoh Way, Salamanca, NY 14779	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Seneca-Cayuga Nation (Chief Charles Diebold), 23701 S. 655 Rd, Grove, OK 74344; (William Tarrant, Cultural Director), 23701 S. 655 Rd, P.O. Box 453220, Grove, OK 74344	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change
Shawnee Tribe (Cassie Harper, Tribal Administrator; Benjamin Joseph Barnes, Chief; Tonya Tipton, THPO), P.O. Box 189, 29 South Highway 69a, Miami, OK 74355	
July 11, 2025 August 1, 2025 August 26, 2025	Submitted Project Notification letter Email response to Sandy Lare, Tetra Tech, received; concurrence that no known historic properties will be negatively impacted by this project and that there is potential for the discovery of unknown resources. Submitted follow-up letter with notification of Project name change
Tonawanda Band of Seneca (Roger Hill, Chief) P.O. Box 795, 7027 Meadville Road, Basom, NY 14013	
July 11, 2025 August 26, 2025	Submitted Project Notification letter Submitted follow-up letter with notification of Project name change

Sources: Bureau of Indian Affairs 2025; National Association of Tribal Historical Preservation Officers 2025; PennDOT Tribal Consultation/Tribal Areas of Interest Database (PennDOT2025); United States Department of Housing and Urban Development (2025) Tribal Directory Assessment Tool; NPS Tribal Preservation Program 2025

4.4 STATUS OF CULTURAL RESOURCES INVESTIGATIONS

4.4.1 OUTSTANDING SURVEYS

As required by Section 106 of the NHPA of 1966, as amended, (16 U.S. Code §470(f)), and its implementing regulations (36 CFR Part 800), which require federal agencies to take into consideration the effects of their actions on historic properties, the PA SHPO recommended a Phase I archaeological investigation be performed within the Project's archaeological APE.

Tetra Tech, on behalf of National Fuel, conducted a Phase I archaeological fieldwork between April and June 2025. Phase I archaeological investigations for the Project were conducted in accordance with the Secretary of the Interior's Standards and Guidelines for Archeology and Historic Preservation (48 Federal Register 44716-42, September 29, 1983 [NPS 2022]). The study also adhered to the NHPA of 1966, as amended; Executive Order 11593; the regulations of the ACHP (36 CFR 800); and research and survey methods set forth by the PA SHPO's Guidelines for Archaeological Investigations in PA (PA SHPO 2021). No archaeological sites or other cultural resources were identified, and no further investigations are recommended in the Phase I Archaeological Investigation report prepared by Tetra Tech (Appendix G) with submission to PA SHPO anticipated in September 2025.

4.4.2 UNANTICIPATED DISCOVERY PLAN

In the event that cultural resources or human remains are revealed during Project construction, National Fuel will follow the procedures outlined in the draft Unanticipated Discovery Plan for Cultural Resources and Human Remains located in Appendix H. Measures to protect unanticipated cultural resources discovered during construction include:

- Immediately stopping work in the vicinity of the discovery;
- Protecting the discovery from further disturbance;
- Restricting personnel from picking up, collecting, disturbing, or destroying artifacts or bones; and
- Notifying appropriate agencies, tribal representatives, and local authorities, as applicable.

4.5 REFERENCES

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RESOURCE REPORT 5
SOCIOECONOMICS

5.0 SOCIOECONOMICS

In accordance with Section 380.12(g) of the Commission's regulations and Commission guidance, Resource Report 5 is required for applications involving significant aboveground facilities such as liquefied natural gas (LNG) facilities or new compressor stations. Based on the proposed Project facilities, Resource Report 5 is not required.

In absentia of Resource Report 5, Resource Report 1 (General Project Description) describes estimated workforce requirements in accordance with FERC guidance provided in 18 CFR § 380.12(c). This general information includes the average workforce requirements, estimated duration of construction activities, and number of personnel to be hired to operate the proposed Project.

RESOURCE REPORT 6
GEOLOGIC RESOURCES

6.0 GEOLOGIC RESOURCES

This Resource Report describes the geologic resources that exist in the vicinity of the Project in Beaver and Mercer counties, PA, and discusses potential impacts on those resources. Section 6.1 describes the geological formations and stratigraphic units traversed by the Project. Section 6.2 identifies the unlikely need for blasting during the Project. Section 6.3 lists the mineral resources of economic value in the vicinity of the Project. Section 6.4 describes geological hazards, surface and subterranean, that may occur near the Project. Section 6.5 describes paleontological resources in the vicinity of the Project. Lastly, Section 6.6 provides a list of references used in preparing this report.

6.1 GEOLOGIC SETTING

The Project area is predominantly located within the Appalachian Plateaus Province, a region characterized by numerous glacially formed river valleys and underlain primarily by Upper Devonian shales and sandstones, with some occurrences of limestone (USGS 2025a). The Mercer CS lies within the glaciated Southern New York Section of the province, whereas the remainder of the Project is situated in the unglaciated Kanawha Section.

The Southern New York Section is underlain by flat-lying to gently dipping Paleozoic sedimentary rocks, including shale, siltstone, sandstone, and some limestone. Its topography reflects both erosional processes and glacial modification, particularly from the Wisconsinan glaciation, which smoothed upland surfaces and deposited glacial till, outwash, and kettle lakes. Elevations in this section generally range from 1,200 to 2,500 feet above mean sea level (amsl). The landscape features broad, rounded uplands and narrow, steep-sided valleys, with dendritic drainage patterns. Structurally, the region is relatively simple, exhibiting minimal folding or faulting and characterized by horizontal to gently dipping strata (USGS 2019).

In contrast, the Kanawha Section is underlain by Middle Pennsylvanian-age sedimentary rocks, most notably the Kanawha Formation, which consists of coal-bearing sequences of sandstone, siltstone, and shale. This formation can reach thicknesses of up to 1,200 feet and includes several named coal beds and marine shale units, such as the Betsie, Dingess, and Winifrede Shale Members. The topography is deeply dissected, with steep ridges, narrow valleys, and elevations ranging from 1,000 to over 3,000 feet amsl. The rugged terrain is shaped primarily by stream erosion. The stratigraphy is complex, with numerous coal seams and interbedded clastic units that have been historically mined and studied for their fossil content and economic significance (Susquehanna River Basin Commission [SRBC] 2006).

6.2 BLASTING

The new pipeline will be installed in a trench with a minimum of 3 feet of cover. The pipeline will generally be installed with 5 feet of cover at waterbody crossings and 4 feet of cover in active agricultural fields. Therefore, the pipeline trench will normally be approximately 5 feet or 60 inches in depth and may reach or exceed 7 feet or 84 inches in depth. The published depth to bedrock ranges from 30 to more than 78 inches in depth within the Project area (U.S. Department of Agriculture, NRCS [USDA/NRCS] 2025), depending on location. There are 17 soil mapping units with a depth to bedrock ≤ 60 inches (described in more detail in Resource Report 7 – Soils). This accounts for approximately 7.10 miles (85.5 percent) of the pipeline length. The ATWS and staging areas contain approximately 13.4 ac of soils with potential for shallow bedrock; however, no major earthwork is anticipated for these areas. The proposed access roads also contain 17.14 ac of soils with shallow bedrock, but no significant trenching or excavation will be needed to use or improve these existing access roads. There are no shallow soils at any of the above-ground facilities.

National Fuel anticipates that shallow bedrock will generally be removed using either an excavator equipped with rock teeth, a dozer drawn ripper, or an excavator equipped with a pneumatic ram, depending on the extent of the rock and its qualities. Although National Fuel does not anticipate the need for blasting, where these methods are either unsuccessful or inadequate, blasting may be required. Blasting controls may be needed to limit stresses on parallel pipelines and other nearby facilities and structures. Field surveys and a review of aerial photographs revealed that there are numerous residential and outbuildings, gas wells, water wells, and other underground infrastructure (e.g., petroleum and natural gas pipelines) within 500 feet of the Project. Currently, some water supply wells or springs have been identified within 150 feet of the pipeline using publicly available map data (see Resource Report 2, section 2.1.6), and direct landowner consultations are also ongoing to identify additional well or spring water supply infrastructure. These types of existing structures and facilities have the potential to be damaged if blasting is required near these areas.

The need for blasting will be determined by the construction contractors on a site-specific basis at the time of construction. If required to construct the Project facilities, blasting will be conducted in compliance with all PA Commonwealth and federal regulations governing the use, storage, and disposal of explosives. National Fuel will utilize licensed blasting contractors that will adhere to Title 25, PA Code, Chapter 211 – Storage, Handling and Use of Explosives (PA Code 2014), Federal regulations referenced in the Federal Explosives Law and Regulations (Bureau of Alcohol, Tobacco, Firearms, and Explosives 2012), and other applicable regulations that address surface blasting, and National Fuel has developed standard blasting guidance (National Fuel Blasting Guidance, Appendix I) for its contractors' use. Additionally, if blasting is required, National Fuel's contractors will develop specific blasting operation and monitoring plans to address site variables such as location, terrain, soil and rock types, type of explosives, charge weight and configuration, depth of charge, spacing between charges, simultaneous detonation or microsecond delays, horizontal and slant distance to the nearest structure, and placement of blasting mats and other controls, if needed. If in-stream blasting is required in PA, an in-stream blasting permit would be obtained from the PA FBC, as required.

6.3 MINERAL RESOURCES

No actively mined mineral resources were observed in or adjacent to the Project right-of-way (ROW) during environmental surveys conducted for the Project. A review of the Project area using the USGS Mineral Resources Online Spatial Data interactive mapper (USGS 2025b) confirmed that no mineral resources are present within the immediate vicinity of the proposed Project area. The nearest active extraction site is a gravel pit located approximately 1.20 miles west of the Mercer CS. Additionally, a geohazard evaluation performed by Tetra Tech (Tetra Tech 2025) (included as Appendix J) identified three outcroppings of the Upper Freeport Coal Seam near stations 30+00, 37+50, and 43+50 along the proposed pipeline route. However, no historical or active mining activities were found within the Project's limits of disturbance or the surrounding 300-foot-wide study area.

6.4 GEOLOGIC HAZARDS

As detailed in the following subsections, the Project area is generally not subject to significant geologic hazards. The risk of earthquakes, soil liquefaction, ground failure due to karst, and volcanism is generally considered low. However, localized areas along the proposed pipeline lateral and access roads exhibit moderate to high landslide potential, which will be addressed through recommended mitigation measures.

6.4.1 EARTHQUAKES AND ACTIVE FAULTS

The Project area was reviewed using 2018 Long-term National Seismic Hazard Map showing peak ground acceleration (pga) having a 2 percent probability of being exceeded in 50 years, for a firm rock site. The map is based on the most recent USGS models for the conterminous U.S. (2018), Hawaii (1998), and Alaska (2007). The models are based on seismicity and fault-slip rates and take into account the frequency of earthquakes of various magnitudes. The map shows the Project area has "very low" risk, having a 2% probability of pga exceeding 4-8 %g in the next 50 years (USGS 2018).

Also, review of the FEMA Earthquake Hazard Program, National Seismic Hazard Maps, which show earthquake ground motions for various probability levels across the U.S., indicates that the proposed Project is located in areas with the lowest seismic hazard class rating (FEMA 2025). In addition, no known quaternary faults (i.e., faults that are found at the Earth's surface and younger than 1.6 million years) have been found in PA (USGS 2025c). Therefore, based on this review, the probability of major movement or damage to the Project facilities as a result of seismic activity is considered to be low.

6.4.2 SOIL LIQUEFACTION

Liquefaction is a phenomenon in which saturated, cohesionless soils, such as fine sands and silts from the Holocene Age, temporarily lose their strength and liquefy when subjected to forces such as intense and prolonged ground shaking. The conditions required to pose a risk of soil liquefaction are not present in the vicinity of the Project, as particle size varies (USDA/NRCS 2025) and the potential for tectonic activity is extremely low. Therefore, the potential for soil liquefaction to present a geologic hazard in the vicinity of the Project is very low.

6.4.3 LANDSLIDE/SLOPE MOVEMENT

National Fuel performed a Geohazard Assessment of the Shippingport Lateral pipeline LOD using desktop review of publicly available information followed by a field reconnaissance, to identify areas of potential concern along the proposed Project LOD (Tetra Tech 2025; see Appendix J). For certain specific segments of the pipeline lateral, the Geohazard Assessment indicated a moderate to high risk of landslides in defined areas and identified specific recommendations for construction and restoration to prevent and minimize the risk of soil instability/slope movement. The desktop review identified and assessed multiple areas of potential concern, including documented old and recent landslides, steep slopes exceeding 3H:1V, and coves underlain by clay—geological features known to be prone to instability. Areas of old landslides and active or recently active landslides as designated by USGS (cited in Tetra Tech 2025) are shown on Figures 1-1 through 1-6 in Appendix J. The Geohazard Assessment identified risk reduction/mitigation measures for implementation in certain specific segments of the pipeline lateral during construction, including trench drainage systems, clay layer removal, and soil compaction techniques to reduce the risk of slope failure during and after construction. For more detailed information the full Geohazard Assessment Memo is included as Appendix J.

6.4.4 GROUND FAILURE

Review of the Project site location was based on Karst Data of the Conterminous Unites States from USGS (USGS 2014, 2020) and ESRI Karst Areas of the United States online mapper (USGS 2004). The entire proposed Project lies within the Appalachian Evaporite Basin. Specifically, the westernmost portion of the pipeline lateral, between MPs 0.00 to 0.55 and 0.70 to 0.83, overlies areas mapped by the USGS as having carbonate rocks at or near the land surface in a humid climate. These conditions significantly increase the potential for karst-related geohazards. Karst terrain, formed through the dissolution of soluble rocks such as limestone and dolomite, is prone to developing features like sinkholes, subsurface voids, and irregular groundwater flow paths. In humid climates, the chemical weathering processes that drive karst formation are more active, which can lead to sudden ground subsidence, structural instability, and challenges in managing water infiltration and drainage.

Given that the proposed Project lies within the Appalachian Evaporite Basin and partially overlays carbonate rocks at or near the land surface in a humid climate, there is a moderate likelihood of ground failure due to karst-related processes. To reduce the risk of ground failure, National Fuel will install appropriate erosion and sediment control BMPs (as outlined for typical BMPs in the Project-specific E&SCP) and will consider additional drainage control systems on a site-specific basis (as recommended in the Geohazard Assessment Memo, Appendix J) to prevent water infiltration that could accelerate rock dissolution. Additionally, National Fuel will monitor ground conditions during and after construction to detect any signs of subsidence or instability, ensuring long-term integrity of the pipeline infrastructure.

6.4.5 VOLCANISM

There is no active volcanism or igneous thermal activity in the vicinity of the Project. Accordingly, the potential for volcanism to present a geologic hazard is considered extremely low (USGS 2025d).

6.5 PALEONTOLOGY

In PA, marine fossils are most prevalent in the Paleozoic rocks in the central regions of the state (Howtofindrocks.com 2022). However, fossils with greater scientific or educational value, such as dinosaurs and early vertebrates, are rare in the state and are generally only known to be found in Mesozoic-age rocks in the southeastern part of the state (Hoskins 1999). The Project is unlikely to disturb or uncover such resources. Should exceptional or significant specimens be unearthed at any point during excavation activities, National Fuel will immediately halt work and contact PA DCNR Geologic Survey and state natural heritage organizations.

6.6 REFERENCES

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RESOURCE REPORT 7
SOILS

7.0 SOILS

This Resource Report identifies the soil resources in Beaver and Mercer counties, PA, that could potentially be affected by the Project. A description of each soil association traversed by the Project is included in section 7.1. Section 7.2 describes the potential impacts of construction and maintenance of the Project on soil resources and mitigation efforts proposed by National Fuel to reduce Project impacts to soil resources. Lastly, section 7.3 lists references used in the preparation of this report.

7.1 EXISTING ENVIRONMENT

Based on review and interpretation of information provided by the NRCS Web Soil Survey (USDA/NRCS 2025a), the Project traverses a total of 16 soil series, in a total of 28 soil mapping units (i.e., in some of the 16-soil series, the Project traverses more than one mapping unit with different slope characteristics). Table 7-1 (at the end of this Resource Report) lists each soil mapping unit, affected acreage, and soil limitations. A map of soil mapping units within the Project area is provided in Appendix A-2.

7.1.1 SOIL DESCRIPTIONS

The Project area encompasses 16 different soil series, as described below. For a breakdown of each soil series, see Table 7-1 at the end of this Resource Report. The following soil descriptions are derived from NRCS Soil Data Mart reports and descriptions (USDA/NRCS 2025b).

Allegheny Silt Loam

Allegheny silt loam (AgB [3 to 8 percent slopes]) is typically found on stream terraces and foot slopes. The parent material consists of loamy alluvium derived from mixed sedimentary rocks. A restrictive layer is not present, but the soil profile includes firm subsoil horizons. The soil is well drained, and water movement is moderate. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. Water saturation is not typically observed within the upper 60 inches. This soil is not hydric.

Atkins Silt Loam

Atkins silt loam (At [0 to 3 percent slopes]) is typically found on nearly level floodplains. The parent material consists of acid alluvium washed from upland soils formed in shale and sandstone. A restrictive layer is not present, but the soil is poorly drained with a seasonal high-water table within 30 centimeters (cm) of the surface. Water movement is very slow. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is frequently flooded. Water saturation occurs near the surface during wet seasons. This soil is hydric.

Canfield Silt Loams

Canfield silt loam (CdA [0 to 3 percent slopes] and CdB2 [3 to 8 percent slopes]) is typically found on till plains and gently sloping uplands. The parent material consists of Wisconsinan age glacial till, sometimes with a loess mantle. A fragipan restrictive layer is present at a depth of 26 to 38 inches. The soil is moderately drained, and water movement in the fragipan is moderately low. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. Water saturation occurs at 10 to 21 inches during November through April. This soil is not hydric.

Clymer Loam

Clymer loam (CmB [3 to 8 percent slopes]) is typically found on mountaintops and upland convex areas. The parent material consists of colluvium and residuum derived from acid sandstone and shale. A restrictive layer is not present, but the soil profile includes a well-developed argillic horizon. The soil is well drained, and water movement is moderate. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. The water saturation is deeper than 60 inches. This soil is not hydric.

Conotton Gravelly Loams

Conotton gravelly loam (CoB [3 to 8 percent slopes] and CoD [15 to 25 percent slopes]) are typically found on outwash plains, stream terraces, and kames. The parent material consists of stratified outwash deposits of sand and gravel from glacial sources. A restrictive layer is not present. The soil is well drained, and water movement is high to very high. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. Water saturation is deeper than 72 inches. This soil is not hydric.

Ernest Silt Loam

Ernest silt loam (ErD [15 to 25 percent slopes]) is typically found on upland hillslopes and footslopes. The parent material consists of colluvium derived from acid shale, siltstone, and sandstone. A fragipan restrictive layer is present at a depth of 30 to 37 inches. The soil is moderately well to somewhat poorly drained, and water movement in the fragipan is moderately low. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is moderate. This soil is not frequently flooded or ponded. Water saturation occurs at 12 to 30 inches from November through April. This soil is not hydric.

Gilpin Silt Loam

Gilpin silt loam (GnD [15 to 25 percent slopes]) is typically found on upland hillslopes and ridges. The parent material consists of residuum derived from interbedded shale, siltstone, and sandstone. A restrictive layer is not present, but the soil is moderately deep to bedrock (20 to 40 inches). The soil is well drained, and water movement is moderate. Available water capacity

to a depth of 60 inches is low to moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. The water saturation is deeper than 40 inches. This soil is not hydric.

Gilpin-Upshur Complex

Gilpin-Upshur complex (GpB [3 to 8 percent slopes], GpC [8 to 15 percent slopes], GpD [15 to 25 percent slopes] and GpF [25 to 60 percent slopes]) is typically found on hillsides and backslopes. The Gilpin component (60%) is derived from acid shale and siltstone and is well drained. The Upshur component (30%) is derived from similar materials and is also well drained. A restrictive layer is not present in either component. Water movement is moderate. Available water capacity is low to moderate. Shrink-swell potential is low. These soils are not frequently flooded or ponded. The water saturation is deeper than 60 inches. These soils are not hydric.

Gilpin-Weikert Channery Silt Loams

Gilpin-Weikert channery silt loams (GsB [3 to 8 percent slopes], GsC [8 to 15 percent slopes], GsD [15 to 25 percent slopes], GsF [25 to 70 percent slopes]) are typically found on upland hillslopes and ridges. The Gilpin component (45–55%) is well drained and moderately deep to bedrock. The Weikert component (30–40%) is somewhat excessively drained and shallow to bedrock (less than 20 inches). Parent material consists of residuum from acid shale, siltstone, and sandstone. A restrictive lithic contact is present in the Weikert component. Water movement is moderate to rapid. Available water capacity is low. Shrink-swell potential is low. These soils are not frequently flooded or ponded. The water saturation is deeper than 40 inches. These soils are not hydric.

Guernsey-Vandergrift Complex

Guernsey-Vandergrift complex (GvC [8 to 15 percent slopes] and GvD [15 to 25 percent slopes]) is typically found on hillslopes and upland shoulders. The Guernsey component is moderately well drained and formed in colluvium and residuum from interbedded siltstone, shale, and limestone. A fragipan is present at depths of 38 to 94 inches. The Vandergrift component is somewhat poorly drained and formed in similar materials. Water movement is moderately low. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. These soils are not frequently flooded or ponded. Water saturation occurs seasonally at 23-inches. These soils are not hydric.

Hazleton Channery Loams

Hazleton channery loam (HaB [3 to 8 percent slopes], HaC [8 to 15 percent slopes] and HaD [15 to 25 percent slopes]) is typically found on summits, shoulders, and upper backslopes of uplands. The parent material consists of residuum from acid gray, brown, or red sandstone. A restrictive lithic contact is present at depths of 100 to 200 cm. The soil is well drained, and water movement is moderately high to high. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. The water saturation is deeper than 60 inches. This soil is not hydric.

Philo Silt Loam

Philo silt loam (Ph [0 to 3 percent slopes]) is typically found on nearly level floodplains. The parent material consists of recent alluvium derived mainly from sandstone and shale. A restrictive layer is not present. The soil is moderately drained, and water movement is moderately high to high. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is occasionally flooded. Water saturation occurs at 40 to 76 cm during wet seasons. This soil is not hydric.

Rayne Silt Loams

Rayne silt loam (RycB [3 to 8 percent slopes] and RycC [8 to 15 percent slopes]) is typically found on upland hillsides and backslopes. The parent material consists of residuum derived from interbedded shale, siltstone, and sandstone. A fragipan restrictive layer is present at a depth of 25 to 36 inches. The soil is moderately drained, and water movement in the fragipan is moderately low. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. Water saturation occurs at 17 to 28 inches during January through April. This soil is not hydric.

Vandergrift-Gilpin Complex

Vandergrift-Gilpin complex (VgD [15 to 35 percent slopes]) is typically found on hillsides and upland slopes. The Vandergrift component is moderately well to somewhat poorly drained and formed in residuum and colluvium derived from interbedded calcareous and non-calcareous shale, siltstone, and limestone. A fragipan is not present, but clay-rich horizons occur at depths of 41 to 147 cm. The Gilpin component is well drained and moderately deep to bedrock. Water movement is moderately low. Available water capacity is moderate. Shrink-swell potential is low. These soils are not frequently flooded or ponded. Water saturation occurs at 40 to 76 cm seasonally. These soils are not hydric.

Wharton Silt Loam

Wharton silt loam (WhC [8 to 15 percent slopes]) is typically found on upland shoulders and backslopes. The parent material consists of residuum derived from interbedded shale, siltstone, and sandstone. A restrictive layer is not present, but the soil profile includes a moderately slowly permeable subsoil. The soil is moderately drained, and water movement is moderately low. Available water capacity to a depth of 60 inches is moderate. Shrink-swell potential is low. This soil is not frequently flooded or ponded. Water saturation occurs at 25 to 50 cm during January through April. This soil is not hydric.

Wharton-Gilpin Silt Loam

Wharton-Gilpin silt loams (WnD [15 to 25 percent slopes]) are typically found on upland hillsides and ridges. The Wharton component is moderately well drained and formed in residuum from shale, siltstone, and sandstone. The Gilpin component is well drained and moderately deep to

bedrock. A fragipan is not present in either component, but the Wharton soil has a moderately slowly permeable subsoil, while the Gilpin soil is shallow to bedrock. Water movement is moderately low. Available water capacity is moderate. Shrink-swell potential is low. These soils are not frequently flooded or ponded. Water saturation occurs seasonally at 25 to 50 cm. These soils are not hydric.

7.1.2 SOIL LIMITATIONS

Soil limitations may include severe erosion potential, medium compaction potential, potential to introduce rock into the soil due to shallow depth to bedrock, and poor revegetation potential.

Approximately 44.51 ac of the Project LOD are prime farmland soils (see Table 7-1). Of that, 16.9 ac within the permanent pipeline lateral ROW and 3.04 ac are within the temporary workspace. A total of 0.53 ac and 0.70 ac of prime farmland soils are present within the permanent additions to the Service Creek Tie and the Shippingport M&R Station, respectively, and 13.38 ac are located in the existing Mercer CS, 5.36 ac are located in ATWS/ staging areas, 2.16 ac are located along permanent access roads, and another 3.08 ac are on temporary access roads. In addition, 28.81 ac of the LOD coincide with farmlands of statewide importance (see Table 7-1). Of that, 15.21 ac are within the permanent pipeline lateral ROW and 3.28 ac are within the temporary workspace, 5.85 ac are located in ATWS/staging areas, 2.48 ac are located along permanent access roads, and another 1.99 ac are on temporary access roads. The prime farmland soils and farmlands of statewide importance are composed of silt loam soils that have medium susceptibility to compaction. National Fuel will segregate and conserve topsoil during construction through active agricultural and pasture areas. Following final grading, National Fuel will ensure that soils are not compacted, and if mitigation is needed, National Fuel will decompact all affected areas. National Fuel will implement additional measures to minimize or avoid significant adverse effects of the Project on soils, as discussed below in section 7.2.

Only one mapped soil unit within the LOD possesses hydric soil characteristics: Atkins silt loam (At [0 to 3 percent slopes]). This soil unit is only found within ATWS-38 and there are 0.02 ac within the Project area. Table 7-1 lists each soil unit's hydric status, as identified by the NRCS (USDA/NRCS 2025a). Field surveys conducted between March and May 2025 identified that numerous wetlands exist within and adjacent to the Project area (see Resource Report 2 for additional wetland information). National Fuel will implement measures to minimize or avoid significant adverse effects of the Project on wetland soils in these areas, as discussed below in section 7.2 and in Resource Report 2.

Approximately 51.83 ac (40 percent of Project acreage) include soils which have severe erosion potential in off-road/trail conditions, and 79.08 ac (60 percent of Project acreage) include severe erosion potential in road/trail conditions (note that the same "area" could have both of these characteristics; i.e., they are not to be added together and compared to entire Project acreage). Table 7-1 identifies soils with potential erosion hazard in off-road/trail and road/trail conditions, as identified by the NRCS (USDA/NRCS 2025a). Because these areas of the Project could undergo some clearing and grading as a result of construction activities, National Fuel will implement measures to minimize or avoid significant adverse effects of the Project on soils

in these areas, as discussed below in section 7.2.

Introducing stones or rocks into surface soil layers during construction can reduce soil moisture-holding capacity, resulting in a reduction of soil productivity and damage to agricultural equipment. The process of excavating soils with shallow bedrock (through ripping or blasting) can potentially introduce rocks into surface soil. Table 7-1 identifies soils with a high risk for introducing stones/rocks into surface soil based on depth to bedrock, as identified by the NRCS (USDA/NRCS 2025). Approximately 54.99 ac of soil within the permanent pipeline lateral ROW contains soils with shallow bedrock (less than 60 inches). If blasting is required, it will be performed according to strict guidelines designed to control energy released, and in compliance with National Fuel's Blasting Guidance (Appendix I).

Soil pH, available water capacity (AWC), and depth to restrictive layer are three of the most critical indicators of revegetation potential, as they directly influence plant establishment, growth, and long-term survival. Soil pH affects the chemical environment of the soil, particularly nutrient availability and microbial activity. When pH levels are too low (acidic) or too high (alkaline), essential nutrients become less available, and toxic elements like aluminum may become more soluble, inhibiting root development and plant health. Available water capacity reflects the soil's ability to retain and supply water to plants between rainfall or irrigation events. Soils with low AWC dry out quickly, placing stress on young plants and reducing the likelihood of successful establishment. Lastly, depth to restrictive layer—such as bedrock, hardpans, or compacted zones—limits root penetration and access to deeper moisture and nutrients. Shallow soils or those with restrictive layers close to the surface can severely constrain root growth and reduce the resilience of vegetation to environmental stress. Together, these three indicators provide a foundational assessment of a soil's capacity to support revegetation efforts. Approximately 47.28 ac within the Project area have revegetation concerns due to the combination of low pH (<5.5), low AWC (<0.2), and shallow depth to restrictive layer (≤30 inches). Table 7-1 identifies soils with revegetation concerns based on these three indicators, as identified by the NRCS (USDA/NRCS 2025a, 2025b). Measures to help improve soil conditions and support successful vegetation establishment are discussed in section 7.2.

7.2 CONSTRUCTION AND OPERATION IMPACTS AND MITIGATION

A total of approximately 131.01 ac of soil will be temporarily disturbed within the proposed Project LOD. The method of excavation used will depend on the specific soil conditions encountered, however it is expected that track excavators will be required. Ditch-line breakers, usually composed of sandbags, will be installed on steep slopes. Where bedrock is encountered, attempt to rip the ditch with a trackhoe. Only if this technique proves unsuccessful will blasting be used. In addition, existing permanent and temporary access roads may be subject to soil disturbances associated with widening and adding aggregate to create a stable surface, as well as from equipment/vehicle use. Excavation and major earth-moving soil disturbances will generally not be required within existing access roads proposed to support Project construction.

Construction of the Project will result in minor, short-term impacts on soils. The primary impact of construction will include grading and excavation disturbances to soils in these areas and the potential erosion hazard created by exposing bare soil. National Fuel will implement erosion and sedimentation control measures consistent with the FERC Plan (FERC 2013a), intended to minimize impacts to soils. National Fuel will implement BMPs for erosion control, restoration, revegetation, and periodic monitoring. These measures are intended to minimize impacts on soils. Topsoil segregation and conservation measures are prescribed in residential areas as well as agricultural and pasture lands, along with rock removal, erosion control, and restoration. Overall impacts to soils affected by the Project will be minimized using these measures, and by the restoration and revegetation of work areas along the pipeline lateral ROW. In addition to fertilizing and seeding, National Fuel will apply soil pH modifiers such as lime to the soils, which tends to reduce soil acidity and thus promote vegetation regrowth. Therefore, no significant difficulties with revegetation are anticipated in the Project area. Following completion of construction, National Fuel will periodically monitor revegetation to ensure successful restoration of the work areas and stabilization of soils disturbed during construction.

Construction also may result in compaction of soils from construction equipment in the work area, possible intermixing of topsoil and subsoil, loss of organic matter, deterioration of soil structure, and soil settling or slumping. Soil compaction can lead to increased runoff and infiltration in urban areas or even adversely affect agricultural crop production. Table 7-1 lists soils within the Project area that are prone to compaction, based on the soil hydrology, soil texture, soil drainage, season of construction, flooding frequency and duration, permeability, and the presence and duration of a seasonally high-water table. Restoration activities could include ripping and breaking up the soil to minimize the occurrence of compacted soils (FERC 2013a). Restoration activities will include careful segregation of the topsoil and subsoil in agricultural and pasture lands, as well as residential areas, to minimize impacts due to soil compaction. If post-construction monitoring and inspection warrant additional restoration efforts, National Fuel will employ mechanical methods to break up the soil to restore the affected areas to preconstruction conditions.

In conjunction with obtaining a PA DEP Erosion and Sedimentation Control General Permit for the Project, National Fuel has developed a Project-specific E&SCP that will be implemented for all areas that will be disturbed during construction. National Fuel's E&SCP will conform to all applicable PA DEP requirements for erosion and sediment control.

The new Service Creek Tie site will be a permanently graded pad area with surface stabilized using gravel that may incorporate GEOWEB geocell in vehicular travel areas to reinforce the gravel stabilization. Areas outside of the new pad area will be revegetated or otherwise restored to previous cover, including the pipeline lateral workspace and the existing temporary and permanent access roads. The 131.01 ac of soils disturbed during construction of the pipeline lateral will be restored to original contours and revegetated. In addition, access roads will be restored as closely as practicable to their original condition and use. With the use of mitigation measures, and through the restoration of work areas, potential impacts to soils in the Project area will be minimized.

As mentioned in section 6.4.3, National Fuel completed a Geohazard Assessment of the Shippingport Lateral pipeline LOD using desktop review of publicly available information followed by a field reconnaissance, to identify potential areas of concern regarding soil/ground instability (Tetra Tech 2025, see Appendix J). The Geohazard Assessment identified risk reduction/ mitigation measures for implementation in certain specific segments of the pipeline lateral during construction, including trench drainage systems, clay layer removal, and soil compaction techniques to reduce the risk of slope failure during and after construction. To reduce the potential for soil erosion and instability, National Fuel will install appropriate erosion and sediment control BMPs (as outlined for typical BMPs in the Project-specific E&SCP) and will consider additional drainage control systems on a site-specific basis (as recommended in the Geohazard Assessment, Appendix J).

Table 7-1. Soils Affected by the Project

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
Pipeline													
0.21	2.34	AgB	Allegheny silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	>78	>78	5.6 - moderately acid	0.23
0.05	0.50	CmB	Clymer loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	44	44	6.1 - slightly acid	0.2
0.03	0.07	CoD	Conotton gravelly loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	>78	>78	5.5 - moderately acid	0.12
0.24	3.11	ErD	Ernest silt loam, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	>78	24	4.7 - very strongly acid	0.22
0.20	0.92	GnD	Gilpin silt loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.21
1.09	4.53	GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
1.35	6.19	GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
1.47	8.90	GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
1.05	5.62	GpF	Gilpin-Upshur complex, 25 to 60 percent slopes	No	Not prime farmland	Well drained	High	Severe	Severe	30	30	4.3 - extremely acid	0.01
0.39	1.94	GsB	Gilpin-Weikert channery silt loams, 3 to 8 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Moderate	33	33	5.3 - strongly acid	0.18
0.03	0.23	GsC	Gilpin-Weikert channery silt loams, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Moderate	33	33	5.3 - strongly acid	0.18
0.25	1.69	GsD	Gilpin-Weikert channery silt loams, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	33	33	5.3 - strongly acid	0.18
1.19	8.06	GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
0.34	1.65	GvC	Guernsey-Vandergrift complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Moderately well drained	Medium	Moderate	Severe	56	56	6 - moderately acid	0.22

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
0.13	0.71	GvD	Guernsey-Vandergrift complex, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	60	60	5.9 - moderately acid	0.22
0.27	1.73	HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18
0.08	0.29	HaC	Hazleton channery loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
0.13	0.39	HaD	Hazleton channery loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
0.02	0.26	Ph	Philo silt loam, 0 to 3 percent slopes	No	All areas are prime farmland	Moderately well drained	Medium	Slight	Slight	>78	>78	5.3 - strongly acid	0.23
1.40	7.54	RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
1.09	4.81	RycC	Rayne silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	45	45	5 - very strongly acid	0.15
0.74	2.66	VgD	Vandergrift-Gilpin complex, 15 to 35 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	71	71	5.3 - strongly acid	0.18
	0.10	WhC	Wharton silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Moderately well drained	Medium	Moderate	Severe	69	69	5.6 - moderately acid	0.23
0.19	0.73	WnD	Wharton-Gilpin silt loams, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	69	69	5.6 - moderately acid	0.18
Aboveground Facilities													
Service Creek Tie													
	0.68	WnD	Wharton-Gilpin silt loams, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	69	69	5.6 - moderately acid	0.18
	0.53	Ph	Philo silt loam, 0 to 3 percent slopes	No	All areas are prime farmland	Moderately well drained	Medium	Slight	Slight	>78	>78	5.3 - strongly acid	0.23

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
Shippingport M&R Station													
	0.03	CoB	Conotton gravelly loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Moderate	79	>78	5.6 - moderately acid	0.12
	0.53	CoD	Conotton gravelly loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	>78	>78	5.5 - moderately acid	0.12
	0.03	AgB	Allegheny silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	>78	>78	5.6 - moderately acid	0.23
Mercer CS													
	1.81	CdA	Canfield silt loam, 0 to 3 percent slopes	No	All areas are prime farmland	Moderately well drained	Medium	Slight	Slight	>78	26	6.3 - slightly acid	0.22
	11.57	CdB2	Canfield silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Moderately well drained	Medium	Moderate	Moderate	>78	26	6.3 - slightly acid	0.22
ATWS													
ATWS-1													
0.07		CoB	Conotton gravelly loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Moderate	79	>78	5.6 - moderately acid	0.12
0.23		CoD	Conotton gravelly loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	>78	>78	5.5 - moderately acid	0.12
0.27		AgB	Allegheny silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	>78	>78	5.6 - moderately acid	0.23
ATWS-2													
0.12		CoD	Conotton gravelly loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	>78	>78	5.5 - moderately acid	0.12
0.25		AgB	Allegheny silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	>78	>78	5.6 - moderately acid	0.23
ATWS-3													
0.01		GsC	Gilpin-Weikert channery silt loams, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Moderate	33	33	5.3 - strongly acid	0.18
0.07		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
ATWS-4													
0.00		GsC	Gilpin-Weikert channery silt loams, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Moderate	33	33	5.3 - strongly acid	0.18
0.12		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
ATWS-5													
0.06		ErD	Ernest silt loam, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	>78	24	4.7 - very strongly acid	0.22
ATWS-6													
0.08		ErD	Ernest silt loam, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	>78	24	4.7 - very strongly acid	0.22
ATWS-7													
0.12		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
0.31		GsB	Gilpin-Weikert channery silt loams, 3 to 8 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Moderate	33	33	5.3 - strongly acid	0.18
ATWS-8													
0.10		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
ATWS-9													
0.04		RycC	Rayne silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	45	45	5 - very strongly acid	0.15
ATWS-10													
0.03		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
ATWS-11													
0.06		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
0.07		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
ATWS-12													
0.12		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
ATWS-13													
0.00		GsD	Gilpin-Weikert channery silt loams, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	33	33	5.3 - strongly acid	0.18
0.03		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
ATWS-14													
0.14		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
0.13		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
0.06		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
ATWS-15													
0.02		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
ATWS-16													
0.01		WhC	Wharton silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Moderately well drained	Medium	Moderate	Severe	69	69	5.6 - moderately acid	0.23
0.05		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
ATWS-17													
0.01		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
ATWS-18													
0.10		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
0.01		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
ATWS-19													
0.40		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
ATWS-20													
0.03		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
0.02		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
ATWS-21													
0.05		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
ATWS-22													
0.01		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
0.05		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
ATWS-23													
0.04		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
0.03		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
ATWS-24													
0.04		HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18
ATWS-25													
0.04		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
ATWS-26													
0.05		HaD	Hazleton channery loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
0.01		HaC	Hazleton channery loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
ATWS-27													
0.11		GvD	Guernsey-Vandergrift complex, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	60	60	5.9 - moderately acid	0.22
0.00		HaC	Hazleton channery loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
0.03		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
ATWS-28													
<0.01		CmB	Clymer loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	44	44	6.1 - slightly acid	0.2
0.03		VgD	Vandergrift-Gilpin complex, 15 to 35 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	71	71	5.3 - strongly acid	0.18
ATWS-29													
0.03		VgD	Vandergrift-Gilpin complex, 15 to 35 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	71	71	5.3 - strongly acid	0.18
0.08		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
ATWS-30													
0.06		VgD	Vandergrift-Gilpin complex, 15 to 35 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	71	71	5.3 - strongly acid	0.18
0.03		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
ATWS-31													
0.10		GvC	Guernsey-Vandergrift complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Moderately well drained	Medium	Moderate	Severe	56	56	6 - moderately acid	0.22
ATWS-32													
0.05		GpF	Gilpin-Upshur complex, 25 to 60 percent slopes	No	Not prime farmland	Well drained	High	Severe	Severe	30	30	4.3 - extremely acid	0.01

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
ATWS-33													
0.07		HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18
ATWS-34													
0.33		GpF	Gilpin-Upshur complex, 25 to 60 percent slopes	No	Not prime farmland	Well drained	High	Severe	Severe	30	30	4.3 - extremely acid	0.01
0.13		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
ATWS-35													
0.08		HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18
ATWS-36													
0.02		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
ATWS-37													
0.01		WnD	Wharton-Gilpin silt loams, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	69	69	5.6 - moderately acid	0.18
0.02		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
ATWS-38													
0.02		At	Atkins silt loam, 0 to 3 percent slopes	Yes	Farmland of statewide importance	Poorly drained	Medium	Slight	Slight	>78	>78	5.7 - moderately acid	0.23
0.20		WnD	Wharton-Gilpin silt loams, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	69	69	5.6 - moderately acid	0.18
0.69		Ph	Philo silt loam, 0 to 3 percent slopes	No	All areas are prime farmland	Moderately well drained	Medium	Slight	Slight	>78	>78	5.3 - strongly acid	0.23
Staging Areas													
Staging Area 1													
0.41		CoB	Conotton gravelly loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Moderate	79	>78	5.6 - moderately acid	0.12
0.26		CoD	Conotton gravelly loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	>78	>78	5.5 - moderately acid	0.12

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
Staging Area 2													
0.54		AgB	Allegheny silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	>78	>78	5.6 - moderately acid	0.23
Staging Area 3													
0.80		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
0.02		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
0.00		RycC	Rayne silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	45	45	5 - very strongly acid	0.15
Staging Area 4													
0.05		WhC	Wharton silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Moderately well drained	Medium	Moderate	Severe	69	69	5.6 - moderately acid	0.23
0.47		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
0.57		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
Staging Area 5													
0.62		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
0.32		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
Staging Area 6													
0.00		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
0.91		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
0.14		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
Staging Area 7													
0.06		GpF	Gilpin-Upshur complex, 25 to 60 percent slopes	No	Not prime farmland	Well drained	High	Severe	Severe	30	30	4.3 - extremely acid	0.01

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
0.27		HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18
Staging Area 8													
0.23		GpF	Gilpin-Upshur complex, 25 to 60 percent slopes	No	Not prime farmland	Well drained	High	Severe	Severe	30	30	4.3 - extremely acid	0.01
Staging Area 9													
1.09		GpF	Gilpin-Upshur complex, 25 to 60 percent slopes	No	Not prime farmland	Well drained	High	Severe	Severe	30	30	4.3 - extremely acid	0.01
0.87		GsC	Gilpin-Weikert channery silt loams, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Moderate	33	33	5.3 - strongly acid	0.18
0.20		GnD	Gilpin silt loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.21
3.41		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
Access Roads													
PAR-1													
	0.20	CoD	Conotton gravelly loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	>78	>78	5.5 - moderately acid	0.12
PAR-2													
	0.65	ErD	Ernest silt loam, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	>78	24	4.7 - very strongly acid	0.22
	1.44	GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
PAR-3													
	0.59	ErD	Ernest silt loam, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	>78	24	4.7 - very strongly acid	0.22
PAR-4													
	0.21	GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
	0.04	RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
	0.08	RycC	Rayne silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	45	45	5 - very strongly acid	0.15

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
PAR-5													
	0.48	GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
	0.64	GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
PAR-6													
	0.85	GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
	0.47	GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
	0.61	RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
	0.44	RycC	Rayne silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	45	45	5 - very strongly acid	0.15
PAR-7													
	0.04	GpF	Gilpin-Upshur complex, 25 to 60 percent slopes	No	Not prime farmland	Well drained	High	Severe	Severe	30	30	4.3 - extremely acid	0.01
	0.14	HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18
PAR-8													
	1.26	HaD	Hazleton channery loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
	0.28	GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
	0.02	GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
PAR-9													
	0.44	HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18
	0.35	HaD	Hazleton channery loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
PAR-10													
	0.34	WhC	Wharton silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Moderately well drained	Medium	Moderate	Severe	69	69	5.6 - moderately acid	0.23
	0.25	GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
	0.89	GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
	0.49	GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
PAR-11													
	0.09	WnD	Wharton-Gilpin silt loams, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	69	69	5.6 - moderately acid	0.18
TAR-1													
0.18		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
TAR-2													
0.18		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
0.99		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
0.05		GsD	Gilpin-Weikert channery silt loams, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Severe	33	33	5.3 - strongly acid	0.18
0.05		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
TAR-3													
0.03		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
0.06		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
0.14		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
TAR-4													
0.13		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
0.13		GsF	Gilpin-Weikert channery silt loams, 25 to 70 percent slopes	No	Not prime farmland	Well drained	Medium	Very severe	Severe	33	33	4.3 - extremely acid	0.01
0.05		GpC	Gilpin-Upshur complex, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Severe	30	30	5.3 - strongly acid	0.17
0.10		RycB	Rayne silt loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	45	45	5 - very strongly acid	0.15
TAR-5													
0.04		WnD	Wharton-Gilpin silt loams, 15 to 25 percent slopes	No	Not prime farmland	Moderately well drained	Medium	Severe	Severe	69	69	5.6 - moderately acid	0.18
0.14		Ph	Philo silt loam, 0 to 3 percent slopes	No	All areas are prime farmland	Moderately well drained	Medium	Slight	Slight	>78	>78	5.3 - strongly acid	0.23
TAR-6													
0.49		GpB	Gilpin-Upshur complex, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	30	30	5.3 - strongly acid	0.17
TAR-7													
0.13		HaD	Hazleton channery loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
0.00		CmB	Clymer loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Moderate	Moderate	44	44	6.1 - slightly acid	0.2
0.87		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
TAR-8													
0.13		HaD	Hazleton channery loam, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
1.25		GpD	Gilpin-Upshur complex, 15 to 25 percent slopes	No	Not prime farmland	Well drained	Medium	Severe	Severe	30	30	5.3 - strongly acid	0.17
TAR-9													
1.94		HaB	Hazleton channery loam, 3 to 8 percent slopes	No	All areas are prime farmland	Well drained	Medium	Slight	Slight	55	55	5.5 - moderately acid	0.18

Approx. Area Affected (ac)		Soil Map Unit	Soil Name	Hydric Status	Farmland Class	Drainage Class	Soil Susceptibility to Compaction[a]	Potential Erosion Hazard		Depth to Bedrock (in)[d]	Depth to Any Soil Restrictive Layer (in)[e]	pH - NRCS pH Classification[f]	Available Water Capacity[g]
TEMP	PERM							Off-Road/Off-Trail[b]	Road/ Trail[c]				
0.66		HaC	Hazleton channery loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Slight	55	55	5.5 - moderately acid	0.18
TAR-10													
0.14		GsC	Gilpin-Weikert channery silt loams, 8 to 15 percent slopes	No	Farmland of statewide importance	Well drained	Medium	Moderate	Moderate	33	33	5.3 - strongly acid	0.18
0.01		WhC	Wharton silt loam, 8 to 15 percent slopes	No	Farmland of statewide importance	Moderately well drained	Medium	Moderate	Severe	69	69	5.6 - moderately acid	0.23

[a] Interpretation ratings are based on soil properties in the upper 12 inches of the profile. Factors considered are soil texture, soil organic matter content, soil structure, rock fragment content, and the existing bulk density.

[b] The ratings in this interpretation indicate the hazard of soil loss from off-road and off-trail areas after disturbance activities that expose the soil surface. The ratings are based on slope, soil erosion factor K, and an index of rainfall erosivity (R). The soil loss is caused by sheet or rill erosion in off-road or off-trail areas where 50 to 75 percent of the surface has been exposed by logging, grazing, mining, or other kinds of disturbance.

[c] The ratings in this interpretation indicate the hazard of soil loss from unsurfaced roads and trails. The ratings are based on soil erosion factor K, slope, and content of rock fragments.

[d] The term bedrock in soil survey refers to a continuous root and water restrictive layer of rock that occurs within the soil profile. If more than one type of bedrock is described for an individual soil type, the depth to the shallowest one is given. If no bedrock is described in a map unit, it is represented by the "greater than 200" depth class.

[e] A "restrictive layer" is a nearly continuous layer that has one or more physical, chemical, or thermal properties that significantly impede the movement of water and air through the soil or that restrict roots or otherwise provide an unfavorable root environment. Examples are bedrock, cemented layers, dense layers, and frozen layers. If more than one type of restrictive layer is described for an individual soil type, the depth to the shallowest one is presented. If no restrictive layer is described in a map unit, it is represented by the "greater than 200" depth class.

[f] Soil reaction (pH) is a measure of acidity or alkalinity. Chemically, it is a measurement of the hydrogen ion activity [H+] in the soil solution. The pH scale ranges from 0 to 14; a pH of 7 is considered neutral. If pH values are greater than 7, the solution is considered basic or alkaline; if they are below 7, the solution is acidic.

[g] Available water capacity (AWC) refers to the quantity of water that the soil is capable of storing for use by plants. It is expressed in centimeters of water per centimeter of soil for each soil layer.

Source: USDA/NRCS 2025a.

7.3 REFERENCES

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RESOURCE REPORT 8
LAND USE, RECREATION, AND AESTHETICS

8.0 LAND USE, RECREATION, AND AESTHETICS

This Resource Report identifies the land use in the Project area and addresses impacts and mitigation during construction and operation. Section 8.1 characterizes current land uses affected by the Project and quantifies approximate impacts by each land use type. Section 8.2 provides information on residential areas, both existing and planned. Section 8.3 identifies special use areas, including public lands, conservation areas, and designated recreation areas. Section 8.4 discusses contaminated or hazardous waste sites and nuclear power reactors in the vicinity of the Project area. Section 8.5 identifies coastal zone management areas. Visual resources are discussed in section 8.6. Section 8.7 discusses ROW applications. References used in compiling this report are provided in section 8.8.

8.1 LAND USE

The Project is located in Independence Township, Raccoon Township, and Shippingport Borough in Beaver County, PA and Jefferson Township in Mercer County, PA. Characterization of land use was identified based on field environmental surveys conducted in March through May 2025, interpretation of recent aerial photography of the proposed Project area, and review of publicly available information. The existing land use categories consist of forest/woodland, shrub/scrub, open land, agricultural land, residential, and developed land. Classification of land use types was prepared in conjunction with information presented in Resource Report 3 – Vegetation and Wildlife.

Table 8-1 presents information on the lengths and percentages of each category of land use traversed by the Shippingport Lateral. Table 8-2 presents information on acreage affected during construction and operation activities. The Project will lead to permanent impacts in forest/woodland and shrub/scrub land uses.

Table 8-1. Land Use Types Traversed by the Shippingport Lateral

Forest/ Woodland		Shrub/Scrub		Open Land		Agricultural Land[a]		Residential		Developed Land		Totals[b]	
Mi	Percent	Mi	Percent	Mi	Percent	Mi	Percent	Mi	Percent	Mi	Percent	Mi	Percent
4.27	59.72	0.01	0.14	2.25	31.47	0.00	0.00	0.50	6.99	0.12	1.68	7.15	100.00

[a] Agricultural Land is noted in Table 8-2 as an impacted land use in permanent right-of-way and additional temporary workspace but is not crossed by the Shippingport Lateral centerline.
[b] Total Distance (7.15 miles) reflects the length of pipeline based on GIS analysis which requires using a 2-dimensional plane for analysis, as opposed to the 3-dimensional civil survey length (7.26 miles) mentioned in Resource Report 1 – General Project Description; 3-dimensional length calculations account for topographical variations present, which increases the actual length of the pipeline.

Table 8-2. Land Use Affected by Construction and Operation of the Project (ac)

Project Component	Forest/ Woodland		Shrub/Scrub		Open Land		Agricultural Land		Residential		Developed Land		Totals		
	Temp[a]	Perm[b]	Temp[a]	Perm[b]	Temp[a]	Perm[b]	Temp[a]	Perm[b]	Temp[a]	Perm[b]	Temp[a]	Perm[b]	Temp[a]	Perm[b,c]	Grand
Shippingport Lateral	3.86	37.18	0.04	0.09	9.33	22.37	0.11	0.13	1.88	4.15	0.67	1.03	**15.89**	**64.95**	**80.84**
Service Creek Tie[d]	0.68	0.66	-	-	0.23	0.54	-	-	-	-	-	-	**0.91**	**1.20**	**2.11**
Shippingport M&R Station[d]	-	-	0.28	0.06	-	-	-	-	-	-	0.28	0.53	**0.56**	**0.59**	**1.15**
Modifications at Mercer CS	0.00	0.94	-	-	0.00	7.77	-	-	-	-	0.00	4.67	**0.00**	**13.38**	**13.38**
Staging Areas	0.02	0.00	3.88	0.00	0.86	0.00	-	-	1.29	0.00	8.16	0.00	**14.21**	**0.00**	**14.21**
Access Roads[d]	2.60	4.10	0.00	0.01	3.50	2.46	-	-	1.57	2.65	0.39	2.04	**8.06**	**11.26**	**19.32**
Totals	**7.16**	**42.88**	**4.20**	**0.16**	**13.92**	**33.14**	**0.11**	**0.13**	**4.74**	**6.80**	**9.50**	**8.27**	**39.63**	**91.38**	**131.01**

Totals may differ slightly from sum of parts due to rounding.
[a] Temporary impacts include Temporary Workspace (TWS), Additional Temporary Workspace (ATWS), and Temporary Access Roads (TARs).
[b] National Fuel's permanent right-of-way (ROW). For the Access Roads ROW, these are generally roads not within National Fuel's permanent pipeline ROW, and acreage presented primarily reflects existing, well-constructed access roads for which new land easements will be acquired enabling National Fuel the right to use during construction (and some roads also for operations and maintenance) of the pipeline lateral. Land uses assigned for Access Roads reflect uses adjacent to existing roads.
[c] Following construction, all areas will be restored, and most preconstruction land uses will be allowed to resume, except no structures or deep rooted vegetation will be allowed. The Permanent total represents the acreage of the new permanent land easement that will encumber the landowners' properties.
[d] Project component includes associated ATWS impacts for temporary impacts. (ATWS-1 is included in Shippingport M&R Station's temporary impacts; ATWS-5, ATWS-9, and ATWS-10 are included in the Access Road temporary impacts, and ATWS-38 is included in Service Creek Tie's temporary impacts.)

8.1.1 PIPELINE FACILITIES

Construction and Rights-of-Way

Construction workspace for the Project will include National Fuel's 75-foot-wide Shippingport Lateral permanent ROW, as well as TWS of typically 25 feet beyond the permanent ROW throughout the Project area, and ATWS of variable width beyond the permanent ROW and TWS at certain locations including stream and road crossings (for specialized construction techniques) and agricultural areas (to allow for topsoil segregation). Land use affected by TWS and ATWS is summarized as temporary in Table 8-2.

Existing Rights-of-Way

The Shippingport Lateral will be located parallel and adjacent to, as well as cross, various pipelines and electrical easements for approximately 5.76 miles, most of its length. Table 8-3 presents a summary of existing ROWs that coincide with or are adjacent to the proposed Shippingport Lateral ROW. Negotiations to use or cross existing ROWs will be complete prior to construction, and National Fuel will complete any necessary encroachment agreements with the pipeline operators and for safety compliance.

Table 8-3. Existing ROWs Parallel and Adjacent to the Shippingport Lateral ROW

Existing Right-of-Way (all in Beaver County, PA)	Approx. MP		Type of Right-of-Way	Width (ft)	Approx. Length Parallel (ft)[a]	Position Relative to Proposed Pipeline	Land Use
	Begin	End					
Unknown	0.00	0.04	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Developed Land
Peoples 4" Steel	0.00	0.04	Pipeline, Easement	Unknown	N/A	Crossed by	Open Land
Unknown	0.00	0.08	Electrical/ Overhead Wire	Unknown	144	East of	Forest/ Woodland, Open Land
Unknown	0.30	0.34	Electrical/ Overhead Wire	Unknown	45	East of	Residential
Unknown	0.57	0.64	Electrical/ Overhead Wire	300	N/A	Crossed by	Forest/ Woodland, Open Land
Unknown	0.80	0.80	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Forest/ Woodland, Open Land
Peoples Gas[b]	1.02	1.36	Pipeline, Easement	Unknown	1,574	West of, Crossed by, and East of	Forest/ Woodland, Open Land
Marathon[b]	1.02	1.52	Pipeline, Easement	Unknown	2,370	West of, Crossed by, and East of	Forest/ Woodland, Open Land, Developed Land
Unknown	1.06	1.10	Electrical/ Overhead Wire	85	N/A	Crossed by	Open Land
Buckeye – ZD717CP[b]	1.06	1.36	Pipeline, Easement	Unknown	1,468.05	West of, Crossed by, and East of	Forest/ Woodland, Open Land
Unknown[b]	1.29	1.44	Electrical/ Overhead Wire	Unknown	650	East of	Forest/ Woodland, Open Land
Dominion[b]	1.32	1.36	Pipeline, Easement	Unknown	186	East of	Forest/ Woodland, Open Land
Shell	1.32	1.40	Pipeline, Easement	Unknown	N/A	Crossed by	Forest/ Woodland, Open Land
BOC	1.40	1.40	Pipeline, Easement	Unknown	N/A	Crossed by	Open Land
Shell 12" SCIO	1.40	1.40	Pipeline, Easement	Unknown	N/A	Crossed by	Open Land
Uknown[c]	1.59	3.71	Electrical/ Overhead Wire	100-125	10,929	Crossed by and East of	Open Land, Developed Land
Buckeye Gasline[c]	1.63	3.18	Pipeline, Easement	Unknown	8,533	West of, Crossed by, and East of	Forest/ Woodland, Open Land, Developed Land
Uknown	1.89	1.89	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Developed Land
Unknown[c]	1.93	1.93	Electrical/ Overhead Wire	Unknown	113	North of	Forest/ Woodland, Open Land
Unknown	2.05	2.05	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Forest/ Woodland, Open Land
Unknown	2.95	2.95	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Open Land
Unknown	3.03	3.03	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Open Land
Unknown	3.03	3.03	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Open Land
Unknown[c]	3.03	3.07	Electrical/ Overhead Wire	Unknown	95	South of	Open Land
Unknown[c]	3.48	4.20	Electrical/ Overhead Wire	200	3,698	South of, Crossed by, North of	Open Land, Agricultural Land, Residential land Developed Land
Unknown[c]	3.90	4.02	Electrical/ Overhead Wire	Unknown	451	North of	Residential
Unknown	3.60	3.64	Electrical/ Overhead Wire	100	N/A	Crossed by	Forest/ Woodland, Open Land

Existing Right-of-Way (all in Beaver County, PA)	Approx. MP		Type of Right-of-Way	Width (ft)	Approx. Length Parallel (ft)[a]	Position Relative to Proposed Pipeline	Land Use
	Begin	End					
Unknown	4.02	4.02	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Developed Land
Unknown[c]	4.02	4.05	Electrical/ Overhead Wire	Unknown	107	North of	Open Land, Residential
Buckeye Gasoline[c]	4.09	4.89	Pipeline, Easement	Unknown	4,239	West of, Crossed by, and East of	Forest/ Woodland, Open Land, Developed Land
Unknown	4.43	4.43	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Developed land
Unknown[c]	4.43	4.51	Electrical/ Overhead Wire	Unknown	193	South of	Forest/ Woodland, Open Land
Buckeye	5.00	5.00	Pipeline, Easement	Unknown	128	East of	Forest/ Woodland, Open Land
Unknown[d]	5.19	7.26	Electrical/ Overhead Wire	100	10,659	West of, Crossed by, and East of	Forest/ Woodland, Open Land, Developed Land
Buckeye Gasline[d]	5.53	6.06	Pipeline, Easement	Unknown	2,706	West of, Crossed by, and East of	Forest/ Woodland, Open Land, Developed Land
Unknown	5.53	5.57	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Developed Land
Unknown[d]	5.57	5.83	Electrical/ Overhead Wire	Unknown	1,358	South of, Crossed by, and North of	Forest/ Woodland, Open Land,
Unknown	6.40	6.40	Electrical/ Overhead Wire	Unknown	N/A	Crossed by	Forest/ Woodland, Open Land
Energy Transfer	6.44	6.55	Pipeline, Easement	Unknown	N/A	Crossed by	Forest/ Woodland, Open Land
Sunoco Liquids Line	6.48	6.48	Pipeline, Easement	Unknown	N/A	Crossed by	Forest/ Woodland, Open Land
Unknown[d]	6.63	6.89	Electrical/ Overhead Wire	85	1,106	North of, Crossed by	Forest/ Woodland, Open Land
Shell Ethane Line[d]	6.63	6.89	Pipeline, Easement	Unknown	1,094	North of, Crossed by	Forest/ Woodland, Open Land
Unknown[d]	6.78	7.20	Electrical/ Overhead Wire	85	1,670	West of, Crossed by, and East of	Forest/ Woodland, Open Land
NFG	7.16	7.20	Pipeline, Easement	Unknown	N/A	Crossed by	Forest/ Woodland
NFG[d]	7.20	7.26	Pipeline, Easement	Unknown	290	Crossed by, East of	Forest/ Woodland, Open Land, Developed Land
Approximate Total Length Parallel to Existing Rights-of-Way (ft)[e]					**30,421 (5.76 miles)**		

[a] Note that multiple easements parallel each other and the proposed Shippingport Lateral easement in some locations (but not always the same distance as each other); therefore, adding all the values in this column would overestimate the length actually paralleled. Colors shaded in this table signify multiple parallel ROWs in same segment (e.g., there may be multiple pipelines and a transmission line corridor all spanning between MP A - MP B, but each of these might not span the same length). Therefore, easements that parallel with each other and the proposed Shippingport Lateral are noted by consecutive footnotes below which provide the total length that the proposed Shippingport Lateral parallels each set of multiple easements.

[b] This easement contributes to a total parallel easement length of approximately 2,324 feet.

[c] This easement contributes to a total parallel easement length of approximately 17,121 feet.

[d] This easement contributes to a total parallel easement length of approximately 10,659 feet.

[e] Total length was calculated by adding values from "Approx. Length Parallel (ft)" column that are not in easements marked by footnotes 2, 3, or 4, to the sum of footnotes 2, 3, and 4.

Additional Temporary Workspaces and Staging Areas

The proposed Project will require the temporary use of nine staging areas and numerous ATWS areas. Land use affected by TWS is summarized in Table 8-2. Table 8-4 describes the acreage, current land use, and reasons for each ATWS and staging area is needed. Following the completion of construction activities, these areas will be returned to their previous land uses.

Table 8-4. Additional Temporary Workspace and Staging Areas Proposed for Use During Construction

Approximate MP/Facility	ATWS or Staging Area Name	County	Reason Needed	Area (ac)	Existing Land Use
ATWS					
0.00	ATWS-1	Beaver, PA	Construction of Shippingport M&R Station	0.56	Shrub/ scrub, Developed Land
0.00-0.08	ATWS-2	Beaver, PA	Crossing Church Ln and Shippingport Hill Rd	0.37	Shrub/ scrub, Open Land, Developed Land
0.61-0.64	ATWS-3	Beaver, PA	Overhead wire crossing, facilitate construction on steep slopes, topsoil segregation	0.08	Forst/ Woodland, Open Land
0.61-0.64	ATWS-4	Beaver, PA	Overhead wire crossing, facilitate construction on steep slopes, topsoil segregation	0.12	Open Land
PAR-3	ATWS-5	Beaver, PA	Crossing Streams S09 and S10	0.06	Residential
0.76-0.80	ATWS-6	Beaver, PA	Crossing Stream S02, topsoil segregation, facilitate construction on steep slopes	0.08	Residential
1.02-1.17	ATWS-7	Beaver, PA	Third-party utility crossings – Buckeye – ZD717CP, Marathon, Peoples Gas; topsoil segregation; facilitate construction on steep slopes	0.42	Forest/ Woodland, Open Land,
1.36-1.44	ATWS-8	Beaver, PA	Third-party utility crossings – Shell, BOC, Shell 12” SCIO; topsoil segregation	0.10	Forest/ Woodland, Open Land,
1.40	ATWS-9	Beaver, PA	Third-party utility crossings – Shell, BOC, Shell 12” SCIO; topsoil segregation	0.04	Open Land, Developed Land
1.70	ATWS-10	Beaver, PA	Topsoil segregation	0.03	Open Land
1.86-1.89	ATWS-11	Beaver, PA	Crossing Wetland W02_PEM, Crossing Frankfort Rd, topsoil segregation	0.14	Developed Land

Approximate MP/Facility	ATWS or Staging Area Name	County	Reason Needed	Area (ac)	Existing Land Use
2.05-2.08	ATWS-12	Beaver, PA	Third-party utility crossing – Buckeye Gasline, topsoil segregation	0.12	Forest/ Woodland, Open Land
2.88	ATWS-13	Beaver, PA	Third-party utility crossing – Buckeye Gasline	0.04	Open Land
2.95-3.03	ATWS-14	Beaver, PA	Crossing Pohls Rd, third-party utility crossing – Buckeye Gasline, topsoil segregation	0.33	Residential
3.52	ATWS-15	Beaver, PA	Third-party utility crossing – Buckeye Gasline	0.02	Forest/ Woodland, Open Land
3.60	ATWS-16	Beaver, PA	Crossing Stream S12, overhead wire crossing, crossing Service Church Rd, topsoil segregation	0.06	Residential
3.60-3.64	ATWS-17	Beaver, PA	Crossing Stream S12, overhead wire crossing, crossing Service Church Rd	0.01	Residential
3.86	ATWS-18	Beaver, PA	Crossing electric easement, topsoil segregation	0.11	Agricultural Land
3.94-4.02	ATWS-19	Beaver, PA	Crossing electric easement, crossing Green Garden Rd, topsoil segregation	0.40	Forest/ Woodland, Residential
4.02	ATWS-20	Beaver, PA	Crossing electric easement, crossing Green Garden Rd, topsoil segregation	0.05	Forest/ Woodland, Residential
4.09-4.13	ATWS-21	Beaver, PA	Third-party utility crossing – Buckeye Gasline	0.05	Open Land
4.13-4.17	ATWS-22	Beaver, PA	Crossing Streams S14 and S15, third-party utility crossing – Buckeye Gasline	0.06	Forest/ Woodland, Open Land
4.20-4.24	ATWS-23	Beaver, PA	Crossing Streams S14 and S15, third-party utility crossing – Buckeye Gasline	0.07	Forest/ Woodland, Open Land
4.43	ATWS-24	Beaver, PA	Crossing Shannon Rd, topsoil segregation	0.04	Open Land
4.77-4.81	ATWS-25	Beaver, PA	Crossing Streams S44 and S45, topsoil segregation	0.04	Forest/ Woodland
5.53	ATWS-26	Beaver, PA	Crossing Clearview Rd, facilitate construction on steep slopes, third party utility crossing – Buckeye Gasline, topsoil segregation	0.06	Open Land, Developed Land
5.57-5.61	ATWS-27	Beaver, PA	Crossing Clearview Rd, third party utility crossing – Buckeye Gasline, topsoil segregation	0.14	Open Land
5.68	ATWS-28	Beaver, PA	Third-party utility crossing – Buckeye Gasline	0.03	Open Land
5.95-6.02	ATWS-29	Beaver, PA	Crossing Stream S36, facilitate construction on steep slopes, topsoil segregation	0.11	Open Land

Approximate MP/Facility	ATWS or Staging Area Name	County	Reason Needed	Area (ac)	Existing Land Use
5.98-6.06	ATWS-30	Beaver, PA	Facilitate construction on steep slopes, topsoil segregation	0.09	Open Land
6.21-6.29	ATWS-31	Beaver, PA	Topsoil segregation	0.10	Open Land
6.36-6.40	ATWS-32	Beaver, PA	Crossing Davis Rd, topsoil segregation	0.05	Open Land
6.44-6.48	ATWS-33	Beaver, PA	Third-party utility crossings – Energy Transfer and Sunoco Liquids Line, facilitate construction on steep slopes, topsoil segregation	0.07	Forest/ Woodland, Open Land
6.59-6.78	ATWS-34	Beaver, PA	Overhead wire crossing, facilitate construction on steep slopes	0.47	Forest/ Woodland, Open Land
6.86-6.89	ATWS-35	Beaver, PA	Third-party utility crossing – Shell Ethane Line, facilitate construction on steep slopes, topsoil segregation	0.08	Forest/ Woodland
7.20	ATWS-36	Beaver, PA	Utility crossing, facilitate construction on steep slopes	0.02	Forest/ Woodland
7.20	ATWS-37	Beaver, PA	Crossing Service Creek Rd, facilitate construction on steep slopes	0.03	Forest/ Woodland
7.23-7.26	ATWS-38	Beaver, PA	Construction of Service Creek Tie, crossing Service Creek Rd	0.91	Forest/ Woodland, Open Land
Staging Area					
0.00	Staging Area 1	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	0.67	Developed Land
0.00-0.08	Staging Area 2	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	0.54	Open Land
1.40-1.52	Staging Area 3	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	0.83	Developed Land
TAR-3	Staging Area 4	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	1.10	Developed Land
1.82-1.89	Staging Area 5	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	0.94	Developed Land
4.02-4.05	Staging Area 6	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	1.05	Residential

Approximate MP/Facility	ATWS or Staging Area Name	County	Reason Needed	Area (ac)	Existing Land Use
4.43-4.51	Staging Area 7	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	0.33	Forest/ Woodland, Open Land
4.51-4.55	Staging Area 8	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	0.24	Residential
TAR-10	Staging Area 9	Beaver, PA	Pipe, equipment, and materials storage, contractor yards, and parking areas during construction	8.51	Shrub/ Scrub, Developed Land

Access Roads

The proposed Project will require temporary and permanent use of existing access roads (some temporary roads will require minor extensions) for access to the Project area. Table 8-5 describes the length, width, current land use, and proposed modifications to all TARs and PARs.

Table 8-5. Access Roads Proposed for Use During Construction

Approximate MP/Facility	Access Road	Length (ft)	Maximum Width (ft)	Land Use / Description	Proposed Modification[a]
TARs[1]					
1.70	TAR-1	160	50	Gravel in forested residential area	Some tree removal, the road will be extended to the ROW. Will be restored to pre-existing conditions.
1.74	TAR-2	1,110	50	Gravel in forested residential area	Some tree removal, the road will be extended to the ROW. Will be restored to pre-existing conditions.
Staging Area 4	TAR-3	201	50	Gravel in commercial area (lot behind gas station)	Will be restored to pre-existing conditions.
2.95-3.03	TAR-4	355	50	Gravel in residential area	Will be restored to pre-existing conditions.
3.56	TAR-5	163	50	Dirt road in forested area	The road will be widened up to 30 feet and temporarily topped with gravel. Will be restored to pre-existing conditions
3.94	TAR-6	427	50	Dirt road in residential driveway/ROW	Will be restored to pre-existing conditions

Approximate MP/Facility	Access Road	Length (ft)	Maximum Width (ft)	Land Use / Description	Proposed Modification[a]
5.00-5.04	TAR-7	863	50	Gravel/grass in residential area/open land	The road will be extended to the ROW, widened to up 30 feet, and temporarily topped with gravel where necessary. Will be restored to pre-existing conditions
5.23	TAR-8	1,245	50	Dirt road in forested area	The road will be extended to the ROW, widened to up 30 feet, and temporarily topped with gravel where necessary. Will be restored to pre-existing conditions
6.44-6.89	TAR-9	2,298	50	2-track in ROW/forested area	The road will be extended to the ROW, widened up to 30 feet, and temporarily topped with gravel. Will be restored to pre-existing conditions
Staging Area 9	TAR-10	155	50	Gravel in industrial area	Will be restored to pre-existing conditions
PARs					
0.00	PAR-1	146	60	Asphalt in industrial area	Will be restored to pre-existing conditions.
0.30-0.34	PAR-2	1,809	50	Asphalt in residential area	Some tree removal. Will be restored to pre-existing conditions.
0.76	PAR-3	513	50	Gravel in residential area	Will be restored to pre-existing conditions.
1.40	PAR-4	285	50	Gravel/grass in light industrial area (gravel storage lot)/ROW	Will be restored to pre-existing conditions.
1.63	PAR-5	972	50	Gravel in residential area	Will be restored to pre-existing conditions.
2.50-2.54	PAR-6	2,063	50	Dirt road in industrial area (junk car lot)	Some tree removal, the road will be extended to the ROW. Portions will be temporarily topped with gravel. Will be restored to pre-existing conditions.
4.51	PAR-7	179	60	2-track gravel in residential area	Some tree removal. Will be restored to pre-existing conditions.
5.30	PAR-8	1,354	50	Gravel in residential area	Some tree removal. Will be restored to pre-existing conditions.
6.44-6.48	PAR-9	657	50	Gravel/grass in ROW	Some tree removal, the road will be extended to the ROW. Portions will be temporarily stoned. Will be restored to pre-existing conditions.

Approximate MP/Facility	Access Road	Length (ft)	Maximum Width (ft)	Land Use / Description	Proposed Modification[a]
7.05	PAR-10	2,265	50	2-track in forested area	Some tree removal, the road will be extended to the ROW. Portions will be temporarily topped with gravel. Will be restored to pre-existing conditions.
7.26	PAR-11	104	30	Gravel in industrial area (pump station)	Will be restored to pre-existing conditions.

[a] All modifications to temporary access roads will be removed post-construction.

8.1.2 ABOVEGROUND FACILITIES

The Service Creek Tie will require approximately 0.66 ac of forest/woodland land use to be permanently converted to developed land use for operation of this facility. The 0.68 ac of forest/woodland land use and 0.77 ac of open land impacted by construction will be allowed to revert to preconstruction conditions.

The Shippingport M&R Station will require approximately 0.06 ac of shrub/scrub land use to be permanently converted to developed land use for operation of this facility. The 0.28 ac of shrub/scrub land and 0.81 ac of developed land temporarily impacted by construction will be allowed to revert to preconstruction conditions.

Work at National Fuel's existing Mercer CS will involve the addition of crossover valves. Approximately 0.94 ac of forest/woodland are located within the 13.38-ac existing station property/LOD are accounted for in the impact table to represent land use types; however, these forested areas will not be temporarily or permanently impacted and will be avoided by Project activities. The approximately 7.77 ac of open land and 4.67 ac of developed land impacted by construction will be allowed to revert to preconstruction conditions.

8.1.3 FACILITIES ABANDONMENT/REPLACEMENT

The Project will not involve abandoning or replacing facilities.

8.1.4 IMPACTS AND MITIGATION MEASURES

The proposed construction area for the Project totals approximately 131.01 ac (Table 8-2). Of this total acreage, permanent land use impacts will consist of the establishment of new permanent easements for the permanent Project facilities, including approximately 64.95 ac in the Shippingport Lateral permanent pipeline ROW, approximately 1.20 ac at the Service Creek Tie-In, 0.59 ac at the Shippingport M&R Station, and approximately 11.26 ac will be maintained as permanent access roads. Within the permanent easements, trees/deep-rooted vegetation and structures and excavations are generally prohibited, to protect the integrity of the pipeline and facilitate inspection and pipeline safety. The entire 13.38 ac currently in permanent use

at the existing Mercer CS will also remain under permanent easements, but the existing land use will not change. Approximately 39.63 ac will be temporarily impacted in TWS, ATWS, temporary impacts at Service Creek Tie and Shippingport M&R Station, staging areas, and temporary access roads. Following construction, all areas used for temporary workspace will be restored, and the previous land use can resume. Table 8-2 provides the acreage of land uses affected by the proposed permanent and temporary construction work areas.

The 14.21 ac of staging areas will be used as pipe, equipment, and materials storage, contractor yards, and parking areas during construction. Following construction, these areas will be returned to preconstruction land use.

Of the 131.01 ac required for the Project, the land uses affected consist of: 50.04 ac of forest/woodland, 4.36 ac of shrub/scrub, 47.06 ac of open land, 0.24 ac of agricultural land, 11.54 ac of residential land, and 17.77 ac of developed land. See Table 8-2 for land use acreages affected by construction of the Project.

Following construction, all areas used for temporary workspace will be restored and allowed to revert to preexisting land uses and preconstruction conditions. The Project will permanently impact approximately 7.16 ac of forest/woodland and 4.20 ac of shrub/scrub. The pipeline lateral centerline will cross approximately 4.27 miles of forested/woodland area and 0.01 mile of shrub/scrub land. Tree clearing will be necessary where the permanent ROW has not been maintained or access roads in forested areas need to be widened, however, some permanent ROW areas and staging areas are already cleared of trees.

No crops were observed in the agricultural land crossed by the Project during surveys. Mitigation methods will include topsoil segregation and replacing drainage tiles. At least 12 inches of topsoil will be segregated in deep soils, and every effort to segregate the entire topsoil layer will be made in soils with less than 12 inches of topsoil. Any drainage tiles that are damaged by construction will be repaired to their original condition or better.

Typical use of open land affected by the Project includes herbaceous uplands, herbaceous wetlands, and areas of maintained lawn. Mitigation methods will include topsoil segregation as described above for agricultural land. Mitigation methods of herbaceous wetlands are discussed in Resource Report 2.

Impacts and mitigation measures for residential land are described in section 8.2.

Developed land includes industrial/commercial land and existing paved roads and includes various areas along the Shippingport Lateral; the Shippingport M&R Station; existing Mercer CS; and Staging Areas 1, 3, 4, 5, and 9. Safe and accessible conditions will be maintained at all road crossings and access points during construction. No industrial or commercial buildings will be impacted by the Project.

8.2 RESIDENTIAL AREAS

8.2.1 PLANNED RESIDENTIAL AND COMMERCIAL AREAS

A review of the Beaver County Focus 2024 Map Pages (Beaver County 2024a) shows "Future Character Areas"; designed to guide the land use, growth, and preservation of Beaver County communities over the full duration of the 2024 Beaver County Focus Comprehensive Plan (Beaver County 2024b); for the townships in the Project area:

- Shippingport Borough to be largely Development Centers and Conservation along waterways, with some rural residential;
- Raccoon Township to be largely hamlets and rural residential with conservation along waterways;
- Independence Township to be rural residential with conservation along the waterways; and
- Hopewell Township to be commercial corridor along Interstate Highway 376 and otherwise conservation along waterways and rural residential.

A review publicly-available 2025 meeting minutes and/or Agendas of the of Shippingport Borough Council, Hopewell Township Board of Commissioners, Independence Township Supervisors, Jefferson Township Board of Supervisors, and Raccoon Township Supervisors did not reveal any apparent specific residential or commercial developments in proximity to the Project (Borough of Shippingport 2025, Hopewell Township 2025, Independence Township 2025, Jefferson Township 2025, Raccoon Township 2025). Not all boroughs or townships had all meeting minutes and/or agendas available online. Work within Mercer County will be entirely within National Fuel's existing Mercer CS and will not impact any proposed developments.

8.2.2 EXISTING RESIDENCES

There are 13 residential structures located within 50 feet of the proposed construction workspace, as determined through field surveys and interpretation of recent aerial photography. Table 8-6 lists these areas and National Fuel's measures to minimize impacts during construction. See Appendix K for site-specific crossing plans for five (5) residences/potential dwelling structures within 25 feet of the proposed construction workspace. Note, one residence is in close proximity to Shippingport Lateral and Staging Area 6 – it is represented twice (once for each facility) in Table 8-6 below.

Table 8-6. Structures within 50 Feet of Construction Work Area and Proposed Mitigation

Approximate MP/Facility[a]	Structure type	Distance and Direction from Construction Workspace Area (ft)[b]	Distance from Construction Centerline of Shippingport Lateral (ft)	Proposed Mitigation[c]
Shippingport Lateral				
0.30-0.34	Woodshed[d]	35 NE	97	e
0.80	Residence	42 W	135	a,b
1.06-1.10	Transmission Tower	37 E	94	f
1.44	Shed[d]	43 SE	88	e
1.93	Collapsed Building	Within	10	e
1.93	Old Wood Shed	Within	28	e
2.95	Metal Frame Greenhouse	Within	37	f
2.95-2.99	Barn[e]	Within	Centerline Crosses Under	e
2.99-3.03	Residence	47 S	47	a,b
3.07	Residence	49 S	100	a,b
3.11	Shed	10SW	56	e,f
3.60	Sheds	47 N	94	e,f
3.67	Transmission Tower	38 N	99	f
3.90	Unidentified Building	42 N	98	e,f
3.90	Block Building	24 N	80	e,f
4.02	Residence[d]	Partially Within N	57	a,b,d,f
4.05	Garage[d]	34 N	123	f
4.43	Residence	49 NE	102	a,b
4.58	Non-Residential Building	14 W	28	e,f
4.85	Building	34 N	87	f
5.30	Metal Storage Building	9 W	23	e,f
6.70	Transmission Tower	12 E	99	f
6.86	Transmission Tower	37 N	71	f
6.93-6.97	Transmission Tower	47 N	72	f
Service Creek Tie				
7.26	Existing Valve Set	Within	117	f
7.26	Residence	42 W	224	a,b
Shippingport M&R Station				
0.00	Energy Infrastructure	Abutting, W	314	f

Approximate MP/Facility[a]	Structure type	Distance and Direction from Construction Workspace Area (ft)[b]	Distance from Construction Centerline of Shippingport Lateral (ft)	Proposed Mitigation[c]
Modifications at Mercer CS				
Mercer CS	Multiple Oil and Gas buildings and Infrastructure	Within	N/A	f
Staging and Storage				
Staging Area 3	Shed[d]	22 S	88	e,f
Staging Area 5	Shed	Within	68	e,f
Staging Area 6	Residence[d]	7 S	57	a,b,d,f
Staging Area 6	Garage[d]	12 S	123	f
Staging Area 8	Metal Building	28 S	152	e,f
Access Roads				
PAR-2	Residence	20 NW	281	a,b,d
PAR-2	Garage	39 NW	157	f
PAR-2	Woodshed[d]	33 NW	97	e,f
PAR-3	Residence	35 S	451	a,b
PAR-3	Metal Shed	12 S	284	e,f
PAR-3	Barn	14 S	122	e,f
PAR-5	Residence	8 S	805	a,b,d,f
PAR-5	Barn	9 S	627	f
PAR-6	Office Trailer	4 N	1,404	e,f
TAR-4	Garage	30 N	137	f
TAR-6	Transmission Tower	30 S	129	f
TAR-7	Residence	Abutting S	803	a,b,d,f
TAR-7	Abandoned/ Demolished Barn	4 S	515	e
PAR-8	Residence	Partially within E	513	a,b,d,f
PAR-8	Shed	27 W	256	e,f
PAR-8	Metal Carport	Partially within W	133	f
PAR-8	Barn	9 SE	133	f
PAR-9	Trailer	43 N	375	e,f
TAR-9	Transmission Tower	Partially within S	374	f
PAR-10	Residence	40 E	1,310	a,b
PAR-10	Uninhabitable trailer	22 W	1,196	e,f
PAR-10	Residence	41 E	1,163	a,b
PAR-10	Non-residential Building	14 W	759	e,f
PAR-10	Open Roofed Shed	Partially within W	321	e,f

Approximate MP/Facility[a]	Structure type	Distance and Direction from Construction Workspace Area (ft)[b]	Distance from Construction Centerline of Shippingport Lateral (ft)	Proposed Mitigation[c]

[a] Approximate Stationing refers to buildings or structures near the Shippingport Lateral, Service Creek Tie, and Shippingport M&R Station. Facilities refer to the nearest Project facility otherwise not covered with stationing. PAR = Permanent Access Road. TAR = Temporary Access Road.

[b] N = North, S = South, E = East, W = West

[c] Key to Proposed Mitigation Measures:

a= National Fuel will restore lawns and residential landscaping within the construction work area immediately or as soon as possible after backfilling the trench.

b = National Fuel will install safety fence along the residence and construction workspace areas extending past either side of the residential structure, and will maintain this fencing throughout the open trench phase of construction.

c = National Fuel will attempt to reduce construction area to maintain a 25-foot construction workspace area for a distance of 100 feet on either side of a residence or structure, where possible.

d = National Fuel will implement a site-specific residential construction plan and will utilize either stove pipe or drag section construction technique, a 15-foot separation distance from the construction workspace will be maintained, orange safety fence will be installed along the construction ROW, and vehicle access to the residence will be maintained at all times during the construction period (See Site-Specific Residential Construction Plans in Appendix K). Five site-specific drawings for structures within 25 feet of the construction workspace are submitted with this Resource Report.

e = National Fuel may remove the structure and compensate landowner; negotiations are pending.

f = National Fuel will attempt to reduce construction ROW around structure.

[d] Structure is within 50 ft of two or more Project facilities (e.g., structure could be within 50 ft of the Shippingport Lateral and a staging area).

[e] Landowner has coordinated with National Fuel to agree upon routing in this area and plans to have the barn demolished before construction of the Project.

National Fuel will coordinate directly with those residents where residences are within 50 feet of the construction work areas before commencing construction activities.

In all residential areas, National Fuel will use general procedures to ensure that potential impacts to residences and public roads as a result of construction activities are minimal or nonexistent. In particular, National Fuel will ensure that traffic flow and emergency vehicle access is maintained on public roads during construction, public roads are kept clean of excess mud and will exercise caution while driving equipment and vehicles on public roads between the construction and support sites. Local officials will be notified in advance of the proposed work dates and any necessary road closures.

8.3 PUBLIC LAND, RECREATION, AND OTHER DESIGNATED OR SPECIAL USE AREAS

8.3.1 PUBLIC OR CONSERVATION LAND

At the western terminus of the Project, the Shippingport Lateral parallels, but does not cross, the Shippingport Ballfield, a neighborhood park owned by Shippingport Borough from approximate MPs 0.04 to 0.19 and is approximately 0.03 mile southeast (approximate MP 0.42) of a separate field of the Shippingport Ballfield (owned by Shippingport Borough and Beaver County) (Pennsylvania GeoData [PA GD] 2025a, USGS 2025). The Project LOD has been routed to avoid impacts to this park. National Fuel will

strictly enforce its construction workspace limits and will place temporary fencing to avoid crossing over into the park during construction.

The Project is approximately 0.05-mile west of the limits of the Raccoon Township Municipal park, a community park owned by Raccoon Township. Staging Area 4 is the closest Project component to this park, and the Shippingport Lateral is approximately 0.08-mile south of the Raccoon Township Municipal Park at its closest point (approximate MP 2.05) (PA GD 2025a, USGS 2025). The Project will not impact this park.

From approximate MPs 3.53 to 3.85, the Shippingport Lateral is approximately 150 to 380 feet north of a property owned by the Ambridge Water Authority, and in some GIS mapped sources, this same property is named Beaver County Park. The Shippingport Lateral was routed to avoid this property and instead crosses a private property north of the reservoir property. See Resource Report 2 – Water Use and Quality for further information regarding the Ambridge Reservoir and associated unnamed tributaries crossed by the Project.

With the exception of the three local parks in proximity to but not crossed by the Project and mentioned above, the Project does not cross or come close to any other public or conservation lands, including Tribal lands, national or state parks or forests, national or state hiking trails, conservation easements, or other protected areas within 0.25 mile of the Project area (American Conservation and Stewardship Atlas 2025, NPS 2025a-c, PA DCNR 2025a, PA DEP 2025a-e, PA GD 2025b, Pennsylvania Spatial Data Access [PA SDA] 2025a-c, Stewardship Lands Inventory 2025, USFWS 2025a, USGS 2025, and WeConservePA 2025). The Project area does not cross any lands that are PA DEP Submerged Lands owned by the Commonwealth; NRCS easement program land; or USFWS land and water conservation fund parcels, land utilization projects, national or national realty approved acquisitions or tracts (PA DEP 2025f; USDA 2025; and USFWS 2025b-e). There are no dams or levees in the Project area (USACE 2025a,b).

8.3.2 NATURAL, RECREATIONAL, OR SCENIC AREAS

There are no National Wild and Scenic Rivers, National Trails, National Natural Landmarks, national wilderness areas, or state wild or natural areas within 0.25 mile of the Project area (NPS 2025b,c; PA DEP 2025g, PA GD 2025c; U.S. Department of Transportation [USDOT] 2025; U.S. Forest Service [USFS] 2025a; USFWS 2025a,f-h).

8.3.3 AGENCY AND LANDOWNER CONSULTATION

There are no other public, recreation, or other designated special use areas that require consultations with federal, state, county, or local agencies or organizations.

8.3.4 IMPACT AND MITIGATION

There are no public, recreation, or other designated special use areas that will be crossed by the Project.

8.4 CONTAMINATED/HAZARDOUS WASTE SITES

The USEPA FRS Interests database was queried to within 0.25-mile of the Project LOD. Eighteen (18) facilities were listed, with 45 separate records (USEPA 2025a). Records with violations appeared to be due to compliance reporting, fees not being paid, and/or CWA "other violations;" however, there were no obvious soil or sediment contamination issues in the reviewed area. The FRS data had points for the center of the Bruce Mansfield Power Plant facility (128 Ferry Hill Rd, Shippingport, PA 15077) just outside of the 0.25-mile buffer at MP 0.00. This facility had enforcement or compliance activity or formal enforcement action. The documentation was for Resource Conservation and Recovery Act, Clean Air Act (CAA), and Safe Drinking Water Act reporting. All listed violations were related to CWA and did not involve obvious ground, soil, or groundwater-related issues (USEPA 2025a). National Fuel will promote and adhere to National Fuel's PPC Plan during construction activities.

There are no records of other active or historic solid or hazardous waste treatment, storage, or disposal facilities, quarries, mines, or other special use sites identified within 0.25 mile of the Project area (PA DCNR 2025b, PA DEP 2025i-q, PA GD 2025d, USACE 2025c-f, USEPA 2025b).There are no records of brownfields or superfund sites in the vicinity of the Project area (USEPA 2025c).

8.4.1 NUCLEAR POWER PLANTS

The nearest nuclear power plant is the Beaver Valley Power Station, Unit 1, located approximately 0.86 mile southwest of the Shippingport Lateral pipeline MP 0.03 (measured from the southernmost edge of Staging Area 2 to the eastern low security perimeter fence of the facility property (adjacent to State Route 168), in Beaver County, PA (U.S. Nuclear Regulatory Commission 2025).

8.5 COASTAL ZONE MANAGEMENT AREAS

The Project area is not within a Pennsylvania Coastal Zone Management zone (NOAA 2024).

8.6 VISUAL RESOURCES

There are no state or national scenic rivers, scenic byways, or other designated scenic areas in the vicinity of the Project area (PA DEP 2025g; USDOT 2025; USFS 2025b, USFWS 2025f,g; and Visit Pennsylvania 2025). Because the Project will involve buried pipeline that will largely parallel existing ROWs and will have minimal visual impacts. The location of the Shippingport M&R Station will be within the existing Shippingport Industrial Park and given its previous industrial use, no new visual impacts are anticipated. The Service Creek Tie will be constructed adjacent to the existing Service Creek pig launcher; and given its current land use, no new visual impacts are anticipated. The modifications at the Mercer CS will be entirely within the existing station footprint and no new visual impacts are anticipated. National Fuel will surround its new aboveground facilities (Shippingport M&R Station and Service Creek Tie) with

fencing for security and will provide visual screening in the form of vertical privacy slats within the fencing, to reduce the visibility of these aboveground facilities.

8.7 APPLICATIONS FOR RIGHTS-OF-WAY AND OTHER LAND USE

The Project does not affect any federal or state managed lands; therefore, no applications for rights-of-way or other proposed land use are required.

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RESOURCE REPORT 9
AIR AND NOISE RESOURCES

9.0 AIR AND NOISE RESOURCES

In accordance with the Section 380.12(k) of FERC regulations (18 CFR § 380.12(k)) and FERC guidance (FERC 2017), Resource Report 9 discusses the effects of the Project on the existing air and noise environment. Section 9.1 assesses air quality, section 9.2 assesses noise, and section 9.3 provides a list of references.

The results of the air and noise quality assessment indicate that construction and operation of the Project will not have a significant adverse impact on ambient air or noise quality in the vicinity of the Project area.

9.1 AIR QUALITY

Air quality impacts associated with construction of the Project will include emissions from fossil-fueled non-road construction equipment, mobile equipment supporting construction, fugitive dust from disturbed areas and equipment and vehicle traffic. Gas within the current piping and components to be modified are not expected to be affected during construction. Air quality impacts during the construction phase will be short-term, localized and are not expected to cause or contribute to any significant degradation of air quality. Air quality impacts associated with operation of the Project will include the addition of two emergency generators; two pig launcher/receivers; a filter; and pipe, valves, and/or other fugitive components. The air quality impacts from these sources will be minimal and not expected to cause or contribute to any significant degradation of air quality.

The assessment of air quality impact from the Project was accomplished through a characterization of existing air quality in the Project site region, an assessment of air quality regulatory requirements for the Project, and an estimate of air pollutant emissions from the Project. The air emissions for construction emissions are presented in total for both counties. The operation emissions are presented as being either in Beaver County or Mercer County.

9.1.1 EXISTING AIR QUALITY AND ATTAINMENT DESIGNATION ON CLIMATE OF THE PROJECT AREA

The USEPA has promulgated National Ambient Air Quality Standards (NAAQS) to protect human health and welfare. The NAAQS include primary standards, which are designed to protect human health, including the health of sensitive subpopulations such as children and those with chronic respiratory problems. The NAAQS also include secondary standards designed to protect public welfare, including economic interests, visibility, vegetation, animal species, and other concerns not related to human health.

NAAQS currently apply to the following criteria pollutants: particulate matter (PM) with a nominal aerodynamic diameter of 10 microns or less (PM_{10}); PM with a nominal aerodynamic diameter of 2.5 microns or less ($PM_{2.5}$); sulfur dioxide (SO_2); nitrogen

dioxide (NO_2); carbon monoxide (CO); ozone; and lead. Each NAAQS is expressed in terms of a concentration level and an associated averaging period. The current NAAQS for these criteria pollutants is summarized in Table 9-1. Footnotes to Table 9-1 explain how compliance with each NAAQS is assessed (USEPA 2025a,b). States and municipalities are free to adopt equivalent or more stringent standards. Pennsylvania has adopted all of the NAAQS in the PA Codes, Rules and Regulations, Title 25, Article III, Chapter 131 as well as State Ambient Air Quality Standards (PA DEP 2025) for beryllium, fluorides, and hydrogen sulfide. Pennsylvania's State Ambient Air Quality Standard for beryllium, fluorides, and hydrogen sulfide will not be applicable as no emissions of these pollutants are expected from the Project.

Table 9-1. Applicable Ambient Air Quality Standards

Pollutant	Averaging Period	Primary Standard	Secondary Standard
CO	8- Hour[a]	9 ppm (10,000 µg/m^3)	--
	1- Hour[a]	35 ppm (40,000 µg/m^3)	--
NO_2	Annual[b]	53 ppb (100 µg/m^3)	53 ppb (100 µg/m^3)
	1-hour[c]	100 ppb (188 µg/m^3)	--
PM_{10}	24-Hour[d]	150 µg/m^3	150 µg/m^3
	Annual[e]	50 µg/m^3	50 µg/m^3
$PM_{2.5}$	Annual[f]	9.0 µg/m^3	15.0 µg/m^3
	24-Hour[g]	35 µg/m^3	35 µg/m^3
SO_2	3-Hour[a]	--	500 ppb (1300 µg/m^3)
	1-hour[k, l]	75 ppb (196 µg/m^3)	--
Ozone	8- Hour (2008 Standard)[h, i]	75 ppb (150 µg/m^3)	75 ppb (150 µg/m^3)
	8- Hour (1997 Standard)[h, j]	80 ppb (157 µg/m^3)	80 ppb (157 µg/m^3)
Lead	Rolling 3-month[b]	0.15 µg/m^3	0.15 µg/m^3

[a] Not to be exceeded more than once per year.
[b] Not to be exceeded.
[c] Compliance based on 3-year average of the 98th percentile of the daily maximum 1-hour average at each monitor within an area.
[d] Not to be exceeded more than once per year on average over 3 years.
[e] Annual PM10 revoked in 2006 per 71 Federal Register 61144, October 17, 2006; but left here for reference only (USEPA 2025c).
[f] Compliance based on 3-year average of weighted annual mean $PM_{2.5}$ concentrations at community-oriented monitors.
[g] Compliance based on 3-year average of 98th percentile of 24-hour concentrations at each population-oriented monitor within an area.
[h] Compliance based on 3-year average of fourth-highest daily maximum 8-hour average ozone concentrations measured at each monitor area.
[i] USEPA is currently reconsidering the 8-hour ozone standard set in March 2008.
[j] The 1997 8-hour ozone standard and associated implementation rules remain in place as the transition to the 2008 standard occurs.
[k] Compliance based on 3-year average of 99th percentile of the daily maximum 1-hour average at each monitor within an area.
[l] The 1-hour SO_2 standard was effective as of August 23, 2010.

ppm = parts per million by volume.
ppb = parts per billion by volume.
µg/m^3 = micrograms per cubic meter.

Pursuant to the 1970 Clean Air Act (CAA), states were required to delineate Air Quality Control Regions (AQCRs) and to adopt State Implementation Plans (SIPs) to provide for attainment of the NAAQS as expeditiously as practical, within certain time limits. The 1977 CAA Amendments, in Section 107, required the USEPA and states to identify by category those AQCRs (or portions thereof) meeting and not meeting the NAAQS. Areas meeting the NAAQS are termed attainment areas, and areas not meeting the NAAQS are termed non-attainment areas. Areas that have insufficient data to make determinations of attainment or non-attainment are unclassified or are not designated but are treated as being attainment areas for permitting purposes. The attainment designation of an area is determined on a pollutant-by-pollutant basis, and for each established primary standard. The Project will be located in Beaver County and Mercer County, PA. Mercer County is in an area classified as in attainment or unclassifiable for all NAAQS compounds. Beaver County is in nonattainment for lead, PM2.5, SO2, and VOC; and attainment or unclassifiable for all other NAAQS compounds. For the NAAQS compounds of lead, PM2.5, and SO2, emissions of these compounds are not anticipated from stationary sources associated with the Project, thus no further discussion is needed. For ozone (with NOx and VOC as precursor compounds); the county is classified as marginal nonattainment, but the classification of marginal nonattainment still has a Title V Major Source Threshold of 100 tons per year (tpy) which is the same as attainment. However, the entire state of PA is located in the Ozone Transportation Region (OTR) (USEPA 2025d), which has Title V Major Source Threshold(s) of 100 tpy for NOx, and 50 tpy for VOC.

The climate in the region of the Project is characterized as arid, with cold winters and moderate summers. The New Castle, PA weather station provides representative current climate conditions near the Project. Common climatological measurements for this station were obtained from the Northeastern Regional Climate Center (NRCC) for the period 2000 to 2024. Over the course of a year, the temperature at the station typically varies from an average low of 39.7 degrees Fahrenheit (°F) to an average high of 62.9°F (NRCC). With a typical snowfall of 0.1 inches per year and an average annual precipitation of 0.12 inches (NRCC).

9.1.2 APPLICABLE AIR REGULATORY REQUIREMENTS

The U.S. and PA air permitting programs regulate stationary sources. Air emissions associated with the Project will be from temporary construction activities that do not require an air permit. Increases in operational emissions are exempt from air permitting under the Pennsylvania air permitting program. The following sections discuss air quality requirements and their applicability to the Project.

New Source Review

Preconstruction air permitting programs that regulate the construction of new stationary sources of air pollution are commonly referred to as New Source Review (NSR). NSR can be divided into two groups: major NSR and minor NSR. Major NSR includes two programs: Prevention of Significant Deterioration (PSD) and

Nonattainment NSR. Both major NSR programs are established at the federal level and typically implemented by a state or local permitting body with USEPA review when these local authorities have either delegated jurisdiction by USEPA or have USEPA-approved SIP programs. The minor NSR permitting program regulates pollutants from sources that do not require PSD or Nonattainment NSR permits. The purpose of minor NSR permits is to prevent the construction of sources that would interfere with attainment or maintenance of a NAAQS or violate the control strategy in nonattainment areas. Also, minor NSR permits often contain permit conditions to limit the sources' emissions to avoid the need for PSD analysis or Nonattainment NSR.

The Project does include stationary sources, but these sources are limited to two emergency generators; two pig launcher/receivers; a filter; and pipe, valves, and/or other components; and ultimately, the sources are exempt from air permitting.

New Source Performance Standards

The USEPA has promulgated New Source Performance Standards (NSPS) based on specific emission source categories, and they are organized by subparts of 40 CFR Part 60. Depending upon the type of emission source and applicable subpart, these standards may include emission limits, work practice standards, and requirements for monitoring, recordkeeping, and reporting. NSPS apply to new, modified or reconstructed stationary sources that meet criteria established in 40 CFR Part 60.

40 CFR 60 Part Subpart JJJJ

The new emergency generator engines at the Shippingport M&R Station and Service Creek Tie will be subject to Subpart JJJJ. Therefore, National Fuel will comply as required.

40 CFR Part 60 Subpart OOOOb

Amendments to the current NSPS under 40 CFR 60 Subpart OOOOb were promulgated on March 8, 2024. The final rule became effective May 7, 2024. On July 31, 2025, an interim final rule was published to the Federal Register which extends deadlines for key Subpart OOOOb requirements by 540 days. NSPS Subpart OOOOb applies to affected facilities in the crude oil and natural gas source category that commence construction, modification, or reconstruction after December 6, 2022. However, as indicated below, the proposed Project will not be subject to Subpart OOOOb. Facilities that meet the definition of affected facilities under NSPS Subpart OOOOb include the following:

- Oil and natural gas wells;
- Centrifugal compressors with wet seals or dry seals and reciprocating compressors;
- Process controllers at production well sites, centralized production facilities, natural gas processing plants and compressor stations within the crude oil and natural gas source category;

- Storage vessels with potential volatile organic compound (VOC) emissions greater than 6 tons per year (tpy) or methane (CH_4) emissions greater than 20 tpy;
- A group of equipment within a process unit at an onshore natural gas processing plant;
- Pneumatic pumps at production well sites, centralized production facilities, onshore natural gas processing plants or compressor stations;
- Fugitive equipment components at onshore natural gas processing plants, well sites, centralized production facilities and compressor stations; and
- Sweetening units.

The Project will not include any affected facilities with requirements related to well sites, natural gas processing plants, centrifugal compressors with wet seals, centrifugal compressors with dry seals, or storage vessels with VOC emissions greater than 6 tpy applicable to NSPS Subpart OOOOb.

The project will not result in a net increase of process controllers and will not exceed the 50% reconstruction cost threshold for process controllers; therefore, the process controllers are not subject to NSPS OOOOb.

For fugitive equipment components, the only potentially subject site is the Mercer CS. The rule includes modifications at compressor stations. The Project includes fugitive emissions components located at the Mercer CS. However, the proposed Project at the Mercer CS will not result in a net increase of horsepower at the Mercer CS, thus, neither the Project nor the Mercer CS are subject to the fugitive emissions standards in NSPS OOOOb.

National Emission Standards for Hazardous Air Pollutants

The USEPA has promulgated National Emission Standards for Hazardous Air Pollutants (NESHAPs) based on specific industrial and emission source categories. NESHAP applicability also depends on the major and area source designation of the facility in terms of Hazardous Air Pollutants (HAPs). A major source of HAPs is a facility that has the potential to emit 10 tons per year (tpy) or more of a single HAP, or 25 tpy or more of a combination of HAPs. Facilities that have potentials less than these are classified as area sources. Depending upon the specific variables of the facility (including major/area status), the applicable standards may include employing control devices (emission limits), work practice standards, and requirements for monitoring, recordkeeping and reporting.

40 CFR Part 61 Subpart J

NESHAP Subpart J (40 CFR Part 61 Subpart J) governs fugitive component leaks of benzene. This rule does not apply here because the natural gas (pipeline quality natural gas) at the facility will contain less than 10% benzene. Therefore, this rule does not apply.

40 CFR Part 63 Subpart HHH

Maximum Achievable Control Technology (MACT, part of the NESHAP Program) Subpart HHH (40 CFR Part 63 Subpart HHH) governs Natural Gas Transmissions and Storage facilities that are a major source of HAPs. Here, though the facilities that are part of this Project may be considered Natural Gas Transmissions and Storage facilities, no emissions meet the major source threshold as part of this Project. Therefore, this rule does not apply.

40 CFR Part 63 Subpart ZZZZ

Maximum Achievable Control Technology (MACT, part of the NESHAP Program) Subpart ZZZZ pertains to stationary engines located at either minor or major sources of HAP emissions. Therefore, these regulations are potentially applicable to the generator engines associated with this Project and National Fuel will comply as required.

State Permit Program and Requirements

The Project will include emissions from natural gas-fired emergency generators, pig launchers/receivers, a filter, and fugitive components. All emissions will consist of minor emissions from these activities and will be exempt from permitting and a Request for Determination. Specifically, volatile organic compounds (VOC) emissions from a project at each site that are less than 2.7 tons per year are exempt from air permitting per Pennsylvania Code Title 25 §127.14(a)(8). Therefore, no air permit is required for this Project. Emissions from these activities are presented in Table 9-2 with detailed emissions calculations presented in Appendix L-1.

Table 9-2. Project Operational Emissions (in tons)

Project Location	CO	NO_x	SO_2	VOC	HAP	PM_{10}	$PM_{2.5}$	CO_2	CH_4	N_2O	CO_2e
Shippingport M&R Station (Beaver County)	0.30	0.00	0.00	0.86	0.01	0.00	0.00	2.13	96.45	--	2,415
Service Creek Tie (Beaver County)	0.30	0.00	0.00	0.80	0.01	0.00	0.00	2.10	90.39	--	2,262
Modifications at Mercer CS (Mercer County)	--	--	--	0.84	0.01	--	--	0.45	95.61	--	2,391
Project Operation Totals	**0.61**	**0.01**	**0.00**	**2.950**	**0.04**	**0.00**	**0.00**	**4.68**	**282.45**	**--**	**7,068**

Notes:
"--" indicates emissions are <0.01 ton.
CO_2e – carbon dioxide equivalent
Sums in table are based on Excel spreadsheet/multiple decimal places and may differ from sums added from table due to rounding.

Clean Air Act General Conformity

The requirements outlined in the CAA's general conformity regulations (40 CFR 93, Subpart B) apply to areas designated as "nonattainment" or "maintenance" areas with respect to the NAAQS. The Project will be located in Beaver County and Mercer County, PA. Mercer County is designated as "unclassifiable/attainment" with respect to all criteria pollutants (USEPA 2025d). Mercer County is in an area classified as in attainment or unclassifiable for all NAAQS compounds. Beaver County is in nonattainment for lead, $PM_{2.5}$, SO_2, and VOC; and attainment or unclassifiable for all other NAAQS compounds (USEPA 2025d). The entire state of PA is considered to be in the marginal OTR for ozone. A Project located in a nonattainment area is subject to conformity requirements if potential emissions exceed a pollutant specific threshold as specified in §93.153. Table 9-3 identifies the conformity thresholds for the Project that are designated as nonattainment for one or more pollutants.

Table 9-3. Conformity Thresholds (tpy)

Project Location	AQCR	Nonattainment Pollutants	NO_x	VOC	CO	SO_2	PM_{10}	$PM_{2.5}$	Pb
Beaver County	179	Ozone (marginal/OTR) Lead (nonattainment) $PM_{2.5}$ (moderate) Sulfur Dioxide (nonattainment)	100	50	NA	100	NA	100	25
Mercer County	178	Ozone (OTR)	100	50	NA	NA	NA	NA	NA

For moderate and marginal ozone nonattainment areas in the OTR and areas in the OTR, the oxides of nitrogen (NO_x) and VOC emission major source thresholds in regards to conformity are 100 tpy and 50 tpy, respectively. For areas designated as nonattainment areas for lead (Pb), the major source threshold in regards to conformity is 25 tpy. For areas designated as moderate areas for $PM_{2.5}$, the major source threshold

in regards to conformity is 100 tpy. For areas designated as nonattainment areas for SO_2, the major source threshold in regards to conformity is 100 tpy.

As presented in the Operations Generated Air Emissions section, all project emissions are less than the conformity thresholds. Therefore, General Conformity Regulations do not apply to this Project.

Operations Generated Air Emissions

Overall emissions from stationary sources are expected from operations of the Project. Specifically, emissions from the emergency generators, the pig launcher/receivers, and the fugitive components will occur during normal operations. A summary of operation emissions is presented in Table 9-2, and detailed calculations are provided in Appendix L-1.

Operation Emissions from Fugitive Components and Pigging Operations

Emission calculations associated with installation of valves, connectors, flanges, and other fugitive components associated with the Project are presented in Table 9-2. The calculation methodology was based on the guidance USEPA-453/R-95-017, USEPA Office of Air Quality Planning and Standards, November 1995 gas service emission factors are from Table 2-4: Oil and Gas Production Operations Average Emission Factors. Pigging emission estimates were based on designed volumes of the pig launcher. Additionally, the carbon dioxide equivalent (CO_2e) emissions were calculated from the CH_4 and nitrous oxide (N_2O) emissions using a global warming potential (GWP) 1 for CO_2, 28 for CH_4, and 265 for N_2O.

Construction Generated Air Emissions

Construction of the Project components will result in temporary emissions directly from mobile source combustion units and fugitive particulate matter due to the use of construction equipment. Construction equipment will include large earth-moving equipment (i.e., backhoes, excavators), skid loaders, trucks, and other mobile sources. Construction equipment may be powered either by diesel or gasoline engines and will contribute to overall emissions of oxides of nitrogen (NO_X), CO, VOCs, PM_{10}, $PM_{2.5}$, SO_2, and greenhouse gases (GHG). A listing of the types, size, and number of planned equipment, along with detailed emission calculations and emission factor sources, is included in Appendix L-2. Emissions from construction equipment and vehicle exhaust will be minimized by federal design standards imposed at the time of manufacture and will comply with USEPA mobile and nonroad emission regulations. An estimate of the emissions from all construction activities from the Project is summarized in Table 9-4.

Table 9-4. Project Construction Activity Combined Emissions (in tons)

Project Emissions Type	CO	NO_x	SO_2	VOC	HAP	PM_{10}	$PM_{2.5}$	CO_2	CH_4	N_2O	CO_2e
Non-Road Equipment Engines	9.09	9.63	0.88	1.59	0.06	0.36	0.35	2,725.45	0.14	0.00	2,729.46
On-Road Engines	1.97	1.98	0.01	0.30	0.34	0.11	0.08	1,065.88	0.01	0.00	1,066.30
Fugitive Dust						7.21	0.72				
Tie In Venting				0.18	0.00			0.10	20.50	0.00	573.96
Project Construction Totals	**11.06**	**11.61**	**0.89**	**2.07**	**0.40**	**7.67**	**1.15**	**3,791.43**	**20.65**	**0.00**	**4,369.72**

Notes:
"0" indicates emissions are <0.01 ton.
Sums in table are based on Excel spreadsheet/multiple decimal places and may differ from sums added from table due to rounding.
Emissions are in tons for the entire project that is anticipated to occur in 2026.

Fugitive Dust

Construction activities will generate temporary emissions of fugitive dust due to earth disturbances and equipment movement (i.e., land clearing, grading, excavation, and vehicle traffic on both paved and unpaved roads). The amount of fugitive dust generated will be a function of the specific construction activities, silt and moisture content of the soil, frequency of precipitation during construction activities, vehicle traffic frequency and type, and roadway characteristics. Fugitive dust emissions increase with higher silt content in the soil and decrease with moisture content as water acts as a suppressant. The assumptions, data, and emissions factors used to estimate the emissions from construction activities are provided in Appendix L-2.

Combustion Equipment Engines

Emissions of NO_X, CO, VOCs, PM_{10}, $PM_{2.5}$, SO_2, and GHGs from construction equipment engines used during Project construction have been estimated based on the anticipated types of nonroad and on-road equipment and their estimated levels of use. Emission factors for vehicles were obtained using the publicly available USEPA approved sources or other generally acceptable emissions factors. The assumptions, data, and emissions factors used to estimate the emissions from construction activities, along with a more comprehensive list of construction equipment and associated emissions, are provided in Appendix L-2.

9.1.3 MITIGATION OF AIR EMISSIONS

Mitigation for Fugitive Dust from Construction

During construction, no materials or property will be handled, transported, stored, used, constructed, altered, repaired, or demolished without taking reasonable precautions to prevent particulate matter from becoming airborne. Reasonable precautions to control fugitive dust emissions may include the following:

- Use of water for control of dust in the construction operations, the grading of roads, or the clearing of land;

- Use of water for control of dust in the construction operations, the grading of roads, or the clearing of land;
- Application of water on dirt roads, materials stockpiles, and other surfaces that may create significant airborne dust;
- Paving/grading of roadways and maintaining them in a clean condition;
- Removal of spilled or tracked dirt or other materials from paved streets and of dried sediments resulting from soil erosion; and/or
- Reduction of the speed of vehicular traffic to a point below which significant dust emissions are created.

The Construction Superintendent and environmental inspectors will share the authority to determine if/when water needs to be reapplied for dust control and to determine if/when additional mitigation will be needed.

Mitigation for Emissions from Combustion Equipment Engines

Exhaust emissions from diesel-fueled and gasoline-fueled construction equipment and vehicle engines will be minimized by federal design standards imposed at the time of manufacture of the equipment and will comply with USEPA mobile and nonroad emission regulations (40 CFR Parts 85, 86, and 89). The emissions from exhaust systems of construction equipment will be mitigated through operational measures (e.g., limiting idling).

9.2 NOISE

9.2.1 INTRODUCTION AND SUMMARY OF ACOUSTICAL TERMINOLOGY

A sound source is defined by a sound power level, which is the rate at which acoustic energy is radiated outward and is expressed in units of watts. A sound pressure level is a measure of this fluctuation at a given receiver location and can be obtained through the use of a microphone or calculated from information about the source sound power level and the surrounding environment. Sound power cannot be measured directly, but rather, it is calculated from measurements of sound intensity or sound pressure at a given distance from the source. The perception of sound as “noise” is influenced by several technical factors such as intensity, sound quality, tonality, duration, and the existing background levels.

Sound levels are presented on a logarithmic scale to account for the large range of acoustic pressures that the human ear is exposed to and is expressed in units of decibels (dB). Broadband sound includes sound energy summed across the frequency spectrum. In addition to broadband sound pressure levels, analysis of the various frequency components of the sound spectrum is completed to determine tonal characteristics. The unit of frequency is Hertz (Hz) and typically the frequency analysis examines 9 octave (or 27 1/3 octave) bands ranging from 22 Hz (low) to 11,200 Hz (high), encompassing the entire human audible frequency range.

VOLUME 1 – PUBLIC

National Fuel®
Supply Corporation

CONCISE ENVIRONMENTAL REPORT

NATIONAL FUEL GAS SUPPLY CORPORATION
SHIPPINGPORT LATERAL PROJECT

FERC DOCKET NO. CP25-_____-000

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CFR Part 204 and Transportation Equipment Noise Emission Controls in 40 CFR Part 205. The USEPA considers a L_{dn} of 55 dBA to be the maximum sound level that will not adversely affect public health and welfare by interfering with speech or other activities in outdoor areas. In general, a L_{dn} of 55 dBA is representative of a suburban residential environment (USEPA 1974).

9.2.2.2 State of Pennsylvania

There are currently no known noise regulations at the state level related to this Project.

9.2.2.3 Beaver County

No noise regulations have been identified that would be applicable to the Project with or without numerical decibel limits.

9.2.2.4 Mercer County

No noise regulations have been identified that would be applicable to the Project with or without numerical decibel limits.

9.2.2.5 Independence Township

Section 200-49(C) of the Township Code of Ordinance regulates noise from stationary equipment providing a daytime (7:00 am – 10:00 pm) limit of 75 dBA and a nighttime (10:00 pm – 7:00 am) limit of 70 dBA. Those levels cannot be exceeded (1) within 25 feet of any receiving building located in any district, or (2) at any point along the boundary line between the source property and the receiving property in the A-1 and R-1 Districts.

9.2.2.6 Jefferson Township

No noise regulations have been identified that would be applicable to the Project with or without numerical decibel limits.

9.2.2.7 Raccoon Township

Within the Raccoon Township Code of Ordinances, there is a supplementary ordinance in Section 185-52(C) pertaining to stationary equipment identical to that of Independence Township, including the same daytime and nighttime noise limits of 75 and 70 dBA, respectively.

In addition, in Section 123-3(C) of the Township code states the following for “natural gas processing plants” (which may or may not apply which may apply to meter stations, depending on municipal interpretation):

“The noise level does not exceed a noise standard of 60 dBA at the nearest property line or the applicable standard imposed by federal law, whichever is less.”

9.2.2.8 Borough of Shippingport

No noise regulations have been identified that would be applicable to the Project with or without numerical decibel limits.

9.2.3 EXISTING NOISE ENVIRONMENT

The areas surrounding Shippingport, PA are primarily industrial with some residential areas as well. The typical noise sources that contribute to the ambient soundscape may be from construction, machinery, and vehicles. Existing ambient sound levels may be relatively elevated in the surrounding lands. Nearby Service Creek, there is primarily agricultural land with surrounding residential areas and light commercial areas. There are several parks and creeks nearby which may include conservation land type areas. Existing ambient sound levels are expected to be relatively low here, although sound levels surrounding the Project area would be higher near the surrounding roadways such as PA Routes 18, 168, 68, and Interstate Highway 376.

Existing ambient sound levels within the Project area, particularly away from Shippingport, are expected to be relatively low, although may be sporadically elevated in localized areas due to roadway noise or periods of human activity. Agricultural operations would seasonally contribute to sound levels in the area associated with crop harvests and production. Background sound levels are expected to vary both spatially and temporally depending on proximity to area sound sources such as roadways and natural sounds. Typically, background sound levels are quieter during the night than during the daytime, except during periods when evening and nighttime insect noise may contribute to the soundscape, predominantly in the warmer seasons.

9.2.4 NOISE IMPACT EVALUATION

Potential noise effects from the Project could be caused by short-term increases in noise during construction and long-term increases in noise due to operation of the Project. In order to assess compliance at NSAs in accordance with the FERC noise criterion and other applicable local standards, separate acoustic analyses were conducted with respect to noise generated from operational activities at the noted facilities that will be developed as part of this proposed Project. The following subsections further describe these analyses with respect to each station.

9.2.4.1 Construction Effects and Mitigation

Installation of new equipment would consist of some earth work (e.g., site grading, clearing, and grubbing related to station modifications and equipment installation). Construction activities will be performed with standard heavy equipment such as an excavator, grader, pickup trucks, and other mobile sources. Many construction machines operate intermittently and the types of machines in use at a construction site change with the construction phase.

Pipeline construction noise-related impacts for the Project are expected to be short in duration at a given location and will therefore have minimal effect. Individuals at nearby residences and buildings will hear the construction-related noise, but the overall impact will be short-term. Pipeline construction will not result in the generation of, or exposure of persons to, excessive noise or vibration levels for extended periods. National Fuel is not proposing to install the pipeline using horizontal directional drilling at any location.

Construction activities will vary depending on the work at each facility. Maximum sound levels associated with the estimated grouping of construction equipment that may be used at each facility are provided in Table 9-6 at given reference distances from each site. It is important to note that each piece of equipment presented in the table is not used in each phase of construction. Further, any equipment used is generally not operated continuously. To arrive at the average sound level that may occur during a typical workday, usage factors for each type of construction equipment were applied to the maximum sound levels. The usage factors account for the fact that equipment is not always operated at full throttle conditions and is not used for an entire workday. The usage factors for each piece of equipment, the estimated construction sound levels adjusted to reflect a typical workday, and the estimated construction sound levels at reference distances are also provided in Table 9-6.

Table 9-6. Construction Noise Levels (dBA)

Equipment Type	Maximum Sound Level at 50 Ft	Usage Factor	50 Foot Sound Level Adjusted for Usage Factor	Reference Distances				
				100 Ft	250 Ft	500 Ft	1,000 Ft	2,000 Ft
Air Compressor	81	0.40	77	71	63	57	51	45
Bobcat	80	0.40	76	70	62	56	50	44
Booster/Pumps	77	0.50	74	68	60	54	48	42
Chipper/Shredder	90	0.16	82	76	68	62	56	50
Compactor, Vibratory	80	0.20	73	67	59	53	47	41
Concrete Mixer Truck	85	0.40	81	75	67	61	55	49
Dozers	85	0.40	81	75	67	61	55	49
Dump Truck	84	0.40	80	74	66	60	54	48
Excavator	85	0.40	81	75	67	61	55	49
Front End Loaders	80	0.40	76	70	62	56	50	44
Generators	82	0.50	79	73	65	59	53	47
Grader	85	0.40	81	75	67	61	55	49
Hydrotest Truck	55	0.40	51	45	37	31	25	19
Pickup Trucks	55	0.40	51	45	37	31	25	19
Pipe Beveling Machine	90	0.20	83	77	69	63	57	51
Pipe Tractor Trailer	84	0.40	80	74	66	60	54	48
Portable Pipe Bender	86	0.16	78	72	64	58	52	46

Equipment Type	Maximum Sound Level at 50 Ft	Usage Factor	50 Foot Sound Level Adjusted for Usage Factor	Reference Distances				
				100 Ft	250 Ft	500 Ft	1,000 Ft	2,000 Ft
Side Boom	85	0.20	78	72	64	58	52	46
Skid Steer Loader	80	0.40	76	70	62	56	50	44
Tandem Truck	55	0.40	51	45	37	31	25	19
Trenching Machines	80	0.20	73	67	59	53	47	41
Water / Fuel Truck	55	0.40	51	45	37	31	25	19
Welding Machine	70	0.40	66	60	52	46	40	34
X-Ray Truck/Machine	55	0.40	51	45	37	31	25	19
Vacuum Truck, triaxle	55	0.40	51	45	37	31	25	19
Farm Tractor 70 HP	84	0.40	80	74	66	60	54	48
Combined Noise Level (dBA)				86	78	72	66	60
Source: Federal Highway Administration 2006.								

9.2.4.2 Aboveground Facility Operation Effects and Mitigation

Operational sound associated with the Project was evaluated employing a computer simulation. The DataKustic GmbH (2024) CadnaA, the computer-aided noise abatement program (v 2024) was used for the acoustic modeling analysis. CadnaA conforms to Organization for International Standardization (ISO) 9613-2, "Acoustics – Attenuation of Sound during Propagation Outdoors" (ISO 1996). The method evaluates A-weighted sound pressure levels under meteorological conditions favorable to propagation from sources of known sound emission. The calculation of sound propagation from source to receiver locations consists of 1/1 octave band algorithms that incorporate the following physical effects:

- Geometric spreading wave divergence;
- Reflection from surfaces;
- Atmospheric absorption at 10 degrees Celcius and 70 percent relative humidity;
- Screening by topography and obstacles;
- Terrain complexity and ground effects;
- Sound power at multiple frequencies;
- Source directivity factors;
- Multiple noise sources and source type (point, area, and/or line);
- Height of both sources and receptors;
- Averaging predicted sound levels over a given time period; and
- Site specific long term meteorological conditions.

Terrain conditions, vegetation type, ground cover, the density and height of foliage can also influence the absorption that takes place when sound travels over land or water. Topographical information was imported into the acoustic model using the official

USGS digital elevation dataset to accurately represent terrain in three dimensions. In addition, the ISO 9613-2 standard accounts for ground absorption by assigning a numerical coefficient of G=0 for acoustically hard, reflective surfaces and G=1 for absorptive surfaces and soft ground. If the ground is hard-packed dirt, typically found in industrial complexes, pavement, or bare rock, or for sound traveling over bodies of water, the absorption coefficient is defined as G=0 to account for reduced sound attenuation. In contrast, ground covered in vegetation, including suburban lawns, will be acoustic absorptive and aid in sound attenuation, i.e., G=1.0. For the Mercer CS acoustic modeling analysis, a mixed (semi-reflective) ground absorption value of G=0.5 was selected for all offsite areas, which consist largely of rural grasslands with scrub-shrub habitat, and a hard reflective ground absorption value of G=0.0 was selected for on-site paved areas.

CadnaA allows for three basic types of sound sources to be introduced into the model: point, line, and areas sources. Point sources can be used for small sources or for larger sources with proportional dimensions that are located away from the relevant receptors. Line sources are used for linear-shaped sources such as pipelines or transmission lines. Finally, area sources can be vertical, such as openings in a building or noise-radiating façade, or horizontal such as a roof, or a number of fixed or moving sources distributed on the ground. The Project-specific sounds sources were included in the model as a combination of point (e.g., valves), area (e.g., horizontal filter separator), and line (e.g., flow control runs) sources. Octave band sound power data from the equipment manufacturer were used as inputs to the model wherever possible. In the absence of manufacturer data, reasonable and appropriate assumptions were derived from engineering guidelines, literature, and measurement data.

9.2.4.2.1 Shippingport M&R Station

The Shippingport M&R Station is proposed to be added to the Shippingport Industrial Park at the west end of the proposed pipeline lateral in Shippingport Borough, Beaver County, PA. Equipment at the M&R station will include a pig receiver, horizontal filter separator, piping and associated valves, measurement equipment, an RTU building, and an emergency generator. The emergency generator is only expected to be active during brief periods of maintenance and readiness testing, and during atypical operation. For the purpose of assessing Project compliance, it is assumed that the Project will generally operate normally.

Reference sound power levels input to CadnaA were provided by equipment manufacturers, based on information contained in reference documents or developed using empirical methods. The source levels of the installed equipment is expected to have similar characteristics like the sound data used in the acoustic modeling analysis; however, it is possible that the final warranty sound data could vary slightly. Representative octave band center frequencies were derived from standardized engineering technical guidelines based on measurements from similar equipment types. Table 9-7 summarizes the equipment sound power level data used as inputs to the acoustic modeling analysis for the M&R station. For the purpose of the analysis, it

was assumed that all equipment would potentially operate at full load during both daytime and nighttime hours.

Table 9-7. Shippingport M&R Station Equipment Sound Level

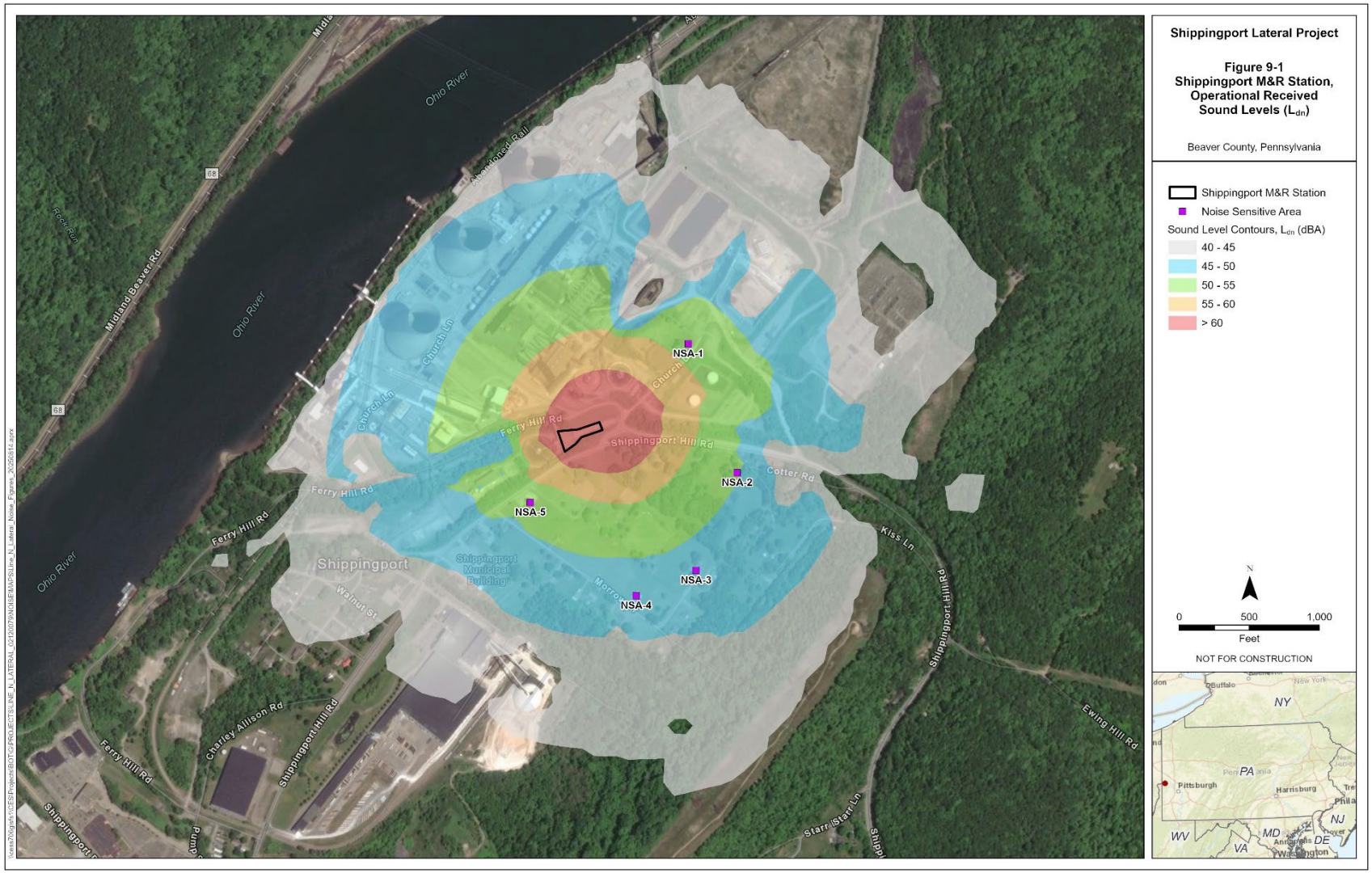
Sound Source	Sound Power Level at Octave Band Frequency									Broadband Level
	31.5	63	125	250	500	1k	2k	4k	8k	dBA
24-inch Receiver (at 3 ft)	29	97	100	96	89	89	96	89	82	86
Remove Control Valve (at 3 ft)	86	85	82	77	71	66	64	63	62	75
Horizontal Filter Separator	88	84	89	88	86	89	89	87	82	95
Coriolis Meter Run (at 3 ft)	71	73	75	76	76	75	74	73	71	81
Flow Control Run	105	98	92	83	90	93	77	76	70	94
RTU Building (HVAC)	-	88	77	77	79	80	76	70	64	84
RTU Building (Exhaust)	67	67	75	66	63	60	59	63	60	69
Emergency Generator	-	112	102	97	93	90	87	85	84	97

Computer modeling was performed to calculate noise levels that will be generated by the operation of the new equipment. The results of the analysis are tabulated at the closest NSAs in each geographical direction relative to the Shippingport M&R Station in Table 9-8 and also presented visually in the form of a sound contour plot. Figure 9-1 displays operational broadband (dBA) sound levels presented as color-coded isopleths in in 5 dBA increments projected on a scaled USGS orthophoto map. Results are independent of the existing acoustic environment, representative of facility-generated sound levels only. The sound contour isopleths are plotted at a height of 1.52 meters above ground level, about the height of the ears of a standing person.

Table 9-8. Shippingport M&R Station Acoustic Modeling Results

NSA	Distance to NSA (ft)	Direction to NSA	Predicted Received Sound Level, L_{dn} (dBA)	Compliance with the FERC 55 dBA L_{dn} Noise Criterion?
1	752	NE	53	Yes
2	917	E	50	Yes
3	1,183	SE	48	Yes
4	1,164	S	48	Yes
5	461	SW	52	Yes

Figure 9-1. Shippingport M&R Station, Operational Received Sound Levels (L_{dn})



9.2.4.2.2 Service Creek Tie

Installation of pig launchers, gas quality equipment and ancillary equipment (Service Creek Tie) adjacent to National Fuel's existing Service Creek pigging facility at the east end of the proposed pipeline in Independence Township, Beaver County, PA. Equipment at the Service Creek Tie including a Clam-on Meter, RTU and Odorizer Buildings, a launcher, and an emergency generator. The generator is only expected to be active during brief periods of maintenance and readiness testing, and during atypical operation. For the purpose of assessing Project compliance, it is assumed that the Project will generally operate normally.

Table 9-9 summarizes the equipment sound power level data used as inputs to the acoustic modeling analysis for the Service Creek Tie. It was assumed that all equipment would potentially operate at full load during both daytime and nighttime hours.

Table 9-9. Service Creek Tie Equipment Sound Levels

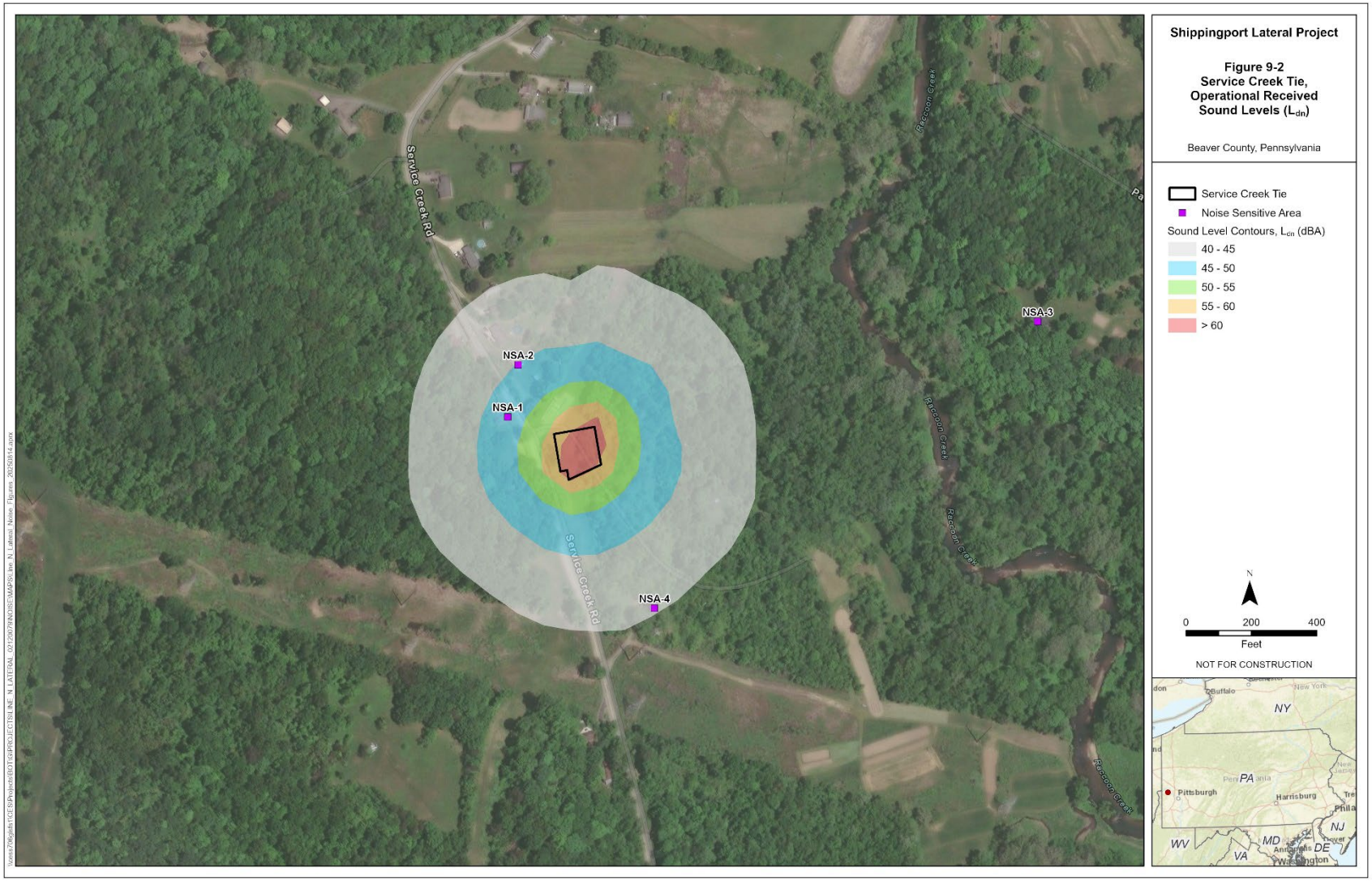
Sound Source	Sound Power Level at Octave Band Frequency									Broadband Level
	31.5	63	125	250	500	1k	2k	4k	8k	dBA
Clamp-on Meter	97	92	91	93	92	95	85	85	80	97
RTU Building and Odorizer Building (HVAC)	-	88	77	77	79	80	76	70	64	84
RTU Building and Odorizer Building (Exhaust)	67	67	75	66	63	60	59	63	60	69
Launcher (at 3 ft)	29	97	100	96	89	89	96	89	82	86
Emergency Generator	-	112	102	97	93	90	87	85	84	97

Initial results indicated potential compliance issues with the 55 dBA L_{dn} FERC noise criterion that were primarily related to the sound emissions from the launcher and clamp-on meter. By assuming the associated aboveground piping and infrastructure would have damping material or equal noise abatement technique applied, compliance could be successfully attained at all the closest NSAs. The results of the analysis are tabulated at the closest NSAs in each geographical direction relative to the Service Creek Tie in Table 9-10 and the corresponding sound contour plot is presented as Figure 9-2.

Table 9-10. Service Creek Tie Acoustic Modeling Results

NSA	Distance to NSA (ft)	Direction to NSA	Predicted Received Sound Level, L_{dn} (dBA)	Compliance with the FERC 55 dBA L_{dn} Noise Criterion?
1	162	NW	48	Yes
2	220	N	45	Yes
3	1,393	E	31	Yes
4	459	S	40	Yes

Figure 9-2. Service Creek Tie, Operational Received Sound Levels (L_{dn})



9.2.4.2.3 Modifications at Mercer CS

Minor modifications will also be made to National Fuel's existing Mercer CS in Jefferson Township, Mercer County, PA. The modifications include the addition of crossover valves, control equipment, and auxiliary equipment, which are considered low-level sound sources especially in comparison to other equipment onsite at the Mercer CS. Therefore, it is not anticipated that there would be any appreciable change in received sound levels at NSAs as a result of the Mercer CS modifications.

9.3 REFERENCES

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RESOURCE REPORT 10
ALTERNATIVES

10.0 ALTERNATIVES

This Resource Report discusses alternatives to the proposed Project. Section 10.1 describes the No-Action alternative. System alternatives are discussed in section 10.2, and section 10.3 discusses alternative energy sources. Lastly, section 10.4 discusses alternative routes and locations for the Project.

10.1 NO-ACTION ALTERNATIVE

The No-Action Alternative involves consideration of the potential benefits and adverse impacts if the proposed Project were not approved and constructed. If the Project were not constructed, the potential environmental impacts associated with construction and operation of the Project would not occur. However, the stated purpose of the Project (described in Resource Report 1, section 1.1.1) would not be met. Therefore, for the reasons discussed below, the No-Action Alternative was not considered acceptable.

The Project will allow National Fuel to provide safe, reliable, and efficient natural gas transportation service to the Shippingport Power Station (SPS). The Project Shipper has requested that National Fuel construct the Project to enable the transportation and delivery of gas to such facility.

Without the Project, the Project Shipper would be forced to seek natural gas transportation through other means, such as other future pipeline expansion projects with unpredictable schedules, undetermined environmental impacts, and uncertain economics, as there are no known alternative existing pipeline systems with adequate capacity to move the requested natural gas volumes into the required Project delivery point.

10.2 SYSTEM ALTERNATIVES

System alternatives to the proposed Project were not analyzed, due to the nature of the Project. The formerly decommissioned Bruce Mansfield coal power generating facility is being converted to the SPS natural gas power generation plant and the Project Shipper has requested National Fuel to transport and deliver gas for its use. National Fuel believes its existing Line N20 transmission pipeline is the closest capable operating natural gas pipeline system to be able to extend to serve the SPS with the least new infrastructure required to efficiently support the Project shipper's needs.

10.3 ALTERNATIVE ENERGY SOURCES

This Project will not affect the availability or potential usage of alternative energy sources. Non-gas energy alternatives do not meet the purpose and need of the Project and, therefore, would not obviate the need for the Project. Therefore, evaluation of alternative energy sources was not applicable to this Project.

10.4 ALTERNATIVE SITES/LOCATIONS

10.4.1 ALTERNATIVE LOCATIONS FOR PIPELINE

Alternative pipeline route locations were initially considered but eliminated from analysis because better alternatives evolved that were feasibly available and reduced certain impacts, resulting in the proposed pipeline routing. This Project was designed such that the pipeline is located over stable soils, topography is conducive to pipeline construction, avoids sensitive resources, co-located in existing utility corridors, and avoids landowners who did not grant permission for Project field survey to be completed on their property, where possible. This route and aboveground facility locations were also chosen because of their proximity to previously disturbed areas, including existing industrial areas, and paralleling or co-locating in existing power line and pipeline corridors.

Ten pipeline route alternatives are presented in this Alternatives Analysis. Route Alternatives 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are depicted on Figure 10-1 at the end of the Resource Report, are discussed below, and are compared in Tables 10-1 through 10-7.

Pipeline Route Alternatives 1 and 2

Route Alternatives 1 and 2 deviate from the preferred route near the western terminus (approximate MPs 0.00 to 0.64 for Route Alternative 1 and MPs 0.19 to 0.64 for Route Alternative 2) (Figure 10-2, Sheet 1 of 7). Both route alternatives and the corresponding segment of the preferred Shippingport Lateral ROW avoid the Shippingport Ballfield (noted in Resource Report 8, section 8.3.1). However, both route alternatives would require more tree clearing, would cross more roads and more soils classified as prime farmland and farmland of statewide importance than the preferred route. The preferred route also avoids the public land owned by Shippingport Borough, which Route Alternative 1 traverses. Additionally, Route Alternative 1 is underlain with more potential karst geology than the corresponding segment of the preferred Shippingport Lateral ROW (Table 10-1). Finally, National Fuel considered several minor variations of the preferred route on one property (not shown or analyzed here) at the request of a landowner, who had specific preferences for the route on his property.

Table 10-1. Environmental Comparison of Major Route Alternatives 1 and 2 to the Corresponding Segment of the Preferred Route

Data Source	Route Feature	Corresponding Segment Preferred Shippingport Lateral ROW	Alternative 1	Alternative 2
National Fuel Gas Supply Corporation	Total Length (mi)	0.62	0.69	0.65
	Total Parcels (no.)	16	28	22
Microsoft	Total Structures within 100 ft (no.)	1	1	0
Pennsylvania Department of Transportation (PA DOT)	Total Roads Crossed (no.)	1	2	2
PA Ch. 93 Streams	Total Streams (no.)	0	0	0
	Warm Water Fishes Streams (no.)	0	0	0
	Cold Water Fishes Streams (no.)	0	0	0
	High Quality- Cold Water Fishes (no.)	0	0	0
	Trout Streams (stocked or natural) (no.)	0	0	0
NWI Wetlands	PFO (ft)	0	0	0
	PSS (ft)	0	0	0
	PEM (ft)	0	0	0
National Land Cover Database (NLCD)[a]	Developed (mi)	0.34	0.32	0.19
	Forest (mi)	0.26	0.35	0.41
	Herbaceous (mi)	0.03	0.02	0.05
Web Soil Survey	Steep Slopes (>25%) (mi)	0.10	0.09	0.09
Public Lands	(Fed/State/Local) (ft)	0	0	0
Recreation	Trails (no.)	0	0	0
	Parks, ballfields, etc. (no.)	0	0	0
Mining Areas	Mining areas (ft)	0	0	0
Prime Farmland[a]	All areas are prime farmland (mi)	0.26	0.49	0.27
	Farmland of statewide importance (mi)	0.00	0.06	0.06
	Not prime farmland (mi)	0.37	0.15	0.13
Karst	Carbonate rocks at or near the land surface in a humid climate (mi)	0.55	0.62	0.39

[a] Due to rounding, some mileages may not sum to total length of the alternative or the corresponding segment of the preferred Shippingport Lateral ROW.

Sources: Microsoft 2022, Multi-Resolution Land Characteristics Consortium (MRLC) 2025, NRCS 2025, PA DCNR 2017, Pennsylvania Department of Transportation (PA DOT) 2025, PA GD 2025a,b, Pennsylvania State University 2025, USFWS 2025, USGS 2025a,b.

Pipeline Route Alternative 3

Route Alternative 3 deviates from the preferred route from approximate MP 0.30 to 1.44 (Figure 10-1, Sheet 2 of 7). Although the preferred route is underlain with more karst than Route Alternative 3, Route Alternative 3 was not chosen because it would require more tree clearing, cross more land with slopes of 25% or greater, and disturb more soils classified as prime farmland and farmland of statewide importance (Table 10-2). Finally, National Fuel considered several minor variations of the preferred route on one property (not shown or analyzed here) at the request of a landowner, who had specific preferences for the route on his property.

Table 10-2. Environmental Comparison of Major Route Alternative 3 to the Corresponding Segment of the Preferred Route

Data Source	Route Feature	Corresponding Segment Preferred Shippingport Lateral ROW	Alternative 3
National Fuel Gas Supply Corporation	Total Length (mi)	1.10	1.07
	Total Parcels (no.)	10	16
Microsoft	Total Structures within 100 ft (no.)	0	0
Pennsylvania Department of Transportation	Total Roads Crossed (no.)	1	1
PA Ch. 93 Streams	Total Streams (no.)	1	1
	Warm Water Fishes Streams (no.)	1	1
	Cold Water Fishes Streams (no.)	0	0
	High Quality- Cold Water Fishes (no.)	0	0
	Trout Streams (stocked or natural) (no.)	0	0
NWI Wetlands	PFO (ft)	0	0
	PSS (ft)	0	0
	PEM (ft)	0	0
NLCD[a]	Developed (mi)	0.22	0.07
	Forest (mi)	0.85	0.98
	Herbaceous (mi)	0.04	0.02
Web Soil Survey	Steep Slopes (>25%) (mi)	0.23	0.31
Public Lands	(Fed/State/Local) (ft)	0	0
Recreation	Trails (no.)	0	0
	Parks, ballfields, etc. (no.)	0	0
Mining Areas	Mining areas (ft)	0	0
Prime Farmland[a]	All areas are prime farmland (mi)	0.21	0.37
	Farmland of statewide importance (mi)	0.24	0.30
	Not prime farmland (mi)	0.65	0.40
Karst	Carbonate rocks at or near the land surface in a humid climate (mi)	0.38	0.27

[a] Due to rounding, some mileages may not sum to total length of the alternative or the corresponding segment of the preferred Shippingport Lateral ROW.

Sources: Microsoft 2022, MRLC 2025, NRCS 2025, PA DCNR 2017, PA DOT 2025, PA GD 2025a,b, Pennsylvania State University 2025, USFWS 2025, USGS 2025a,b.

Pipeline Route Alternative 4

Route Alternative 4 deviates from the preferred route from approximately MP 3.94 to 5.08 (Figure 10-1, Sheet 3 of 7). Although the preferred route crosses more soils classified as prime farmland and more land with slopes of 25% or greater (see Resource Report 6, section 6.4.3), Route Alternative 4 was not selected due to more road crossings, crossing a stream designated HQ-CWF, more tree clearing, crossing more soils classified as farmland of statewide importance, and due to landowner denial of survey permission (Table 10-3).

Table 10-3. Environmental Comparison of Major Route Alternative 4 to the Corresponding Segment of the Preferred Route

Data Source	Route Feature	Corresponding Segment Preferred Shippingport Lateral ROW	Alternative 4
National Fuel Gas Supply Corporation	Total Length (mi)	1.08	0.94
	Total Parcels (no.)	17	24
Microsoft	Total Structures within 100 ft (no.)	3	1
Pennsylvania Department of Transportation	Total Roads Crossed (no.)	2	4
PA Ch. 93 Streams	Total Streams (no.)	0	1
	Warm Water Fishes Streams (no.)	0	0
	Cold Water Fishes Streams (no.)	0	0
	High Quality- Cold Water Fishes (no.)	0	1
	Trout Streams (stocked or natural) (no.)	0	0
NWI Wetlands	PFO (ft)	0	0
	PSS (ft)	0	0
	PEM (ft)	0	0
NLCD[a]	Developed (mi)	0.16	0.04
	Forest (mi)	0.79	0.80
	Herbaceous (mi)	0.13	0.11
Web Soil Survey	Steep Slopes (>25%) (mi)	0.37	0.24
Public Lands	(Fed/State/Local) (ft)	0	0
Recreation	Trails (no.)	0	0
	Parks, ballfields, etc. (no.)	0	0
Mining Areas	Mining areas (ft)	0	0
Prime Farmland[a]	All areas are prime farmland (mi)	0.23	0.15
	Farmland of statewide importance (mi)	0.05	0.28
	Not prime farmland (mi)	0.8	0.52
Karst	Carbonate rocks at or near the land surface in a humid climate (mi)	0	0

[a] Due to rounding, some mileages may not sum to the total length of the alternative or the corresponding segment of the preferred Shippingport Lateral ROW.

Sources: Microsoft 2022, MRLC 2025, NRCS 2025, PA DCNR 2017, PA DOT 2025, PA GD 2025a,b, Pennsylvania State University 2025, USFWS 2025, USGS 2025a,b.

Pipeline Route Alternative 5

Route Alternative 5 deviates from the preferred route from approximately MP 4.43 to 5.19 (Figure 10-1, Sheet 4 of 7). Although the preferred route crosses more soils classified as prime farmland, Route Alternative 5 was not selected due to crossing a stream designated HQ-CWF (and based on field surveys completed on both routes, the alternative route crosses two (2) more wetlands and several larger stream crossings [in a valley area] compared to the preferred route (which is routed on higher ground). Route Alternative 5 would require more tree clearing, crossing more land with slopes of 25% or greater, and disturbing more soils classified as farmland of statewide importance (Table 10-4).

Table 10-4. Environmental Comparison of Major Route Alternative 5 to the Corresponding Segment of the Preferred Route

Data Source	Route Feature	Corresponding Segment Preferred Shippingport Lateral ROW	Alternative 5
National Fuel Gas Supply Corporation	Total Length (mi)	0.72	0.69
	Total Parcels (no.)	6	9
Microsoft	Total Structures within 100 ft (no.)	2	0
Pennsylvania Department of Transportation	Total Roads Crossed (no.)	0	0
PA Ch. 93 Streams	Total Streams (no.)	0	1
	Warm Water Fishes Streams (no.)	0	0
	Cold Water Fishes Streams (no.)	0	0
	High Quality- Cold Water Fishes (no.)	0	1
	Trout Streams (stocked or natural) (no.)	0	0
NWI Wetlands	PFO (ft)	0	0
	PSS (ft)	0	0
	PEM (ft)	0	0
NLCD[a]	Developed (mi)	0.02	0.02
	Forest (mi)	0.61	0.66
	Herbaceous (mi)	0.09	0.01
Web Soil Survey	Steep Slopes (>25%) (mi)	0.27	0.28
Public Lands	(Fed/State/Local) (ft)	0	0
Recreation	Trails (no.)	0	0
	Parks, ballfields, etc. (no.)	0	0
Mining Areas	Mining areas (ft)	0	0
Prime Farmland[a]	All areas are prime farmland (mi)	0.02	0.01
	Farmland of statewide importance (mi)	0.05	0.07
	Not prime farmland (mi)	0.65	0.61
Karst	Carbonate rocks at or near the land surface in a humid climate (mi)	0	0

[a] Due to rounding, some mileages may not sum to the total length of the alternative or the corresponding segment of the preferred Shippingport Lateral ROW.

Sources: Microsoft 2022, MRLC 2025, NRCS 2025, PA DCNR 2017, PA DOT 2025, PA GD 2025a,b, Pennsylvania State University 2025, USFWS 2025, USGS 2025a,b.

Pipeline Route Alternative 6

Route Alternative 6 deviates from the preferred route from approximate MP 5.23 to 6.36 (Figure 10-1, Sheet 5 of 7). Although the preferred route crosses a waterbody designated WWF and HQ-CWF (see Resource Report 3, section 3.1.2), requires more tree clearing (see Resource Report 3, section 3.3.2), and more soils classified as prime farmland and farmland of statewide importance, Route Alternative 6 was not chosen due to crossing more land with slopes of 25% or greater, whereas the preferred route was selected for better topographic constructability (Table 10-5). In addition, the preferred route was routed along the edge of an agricultural field instead of through it (as in the Alternative Route) to reduce impact on agricultural activities, in response to a landowner request.

Table 10-5. Environmental Comparison of Major Route Alternative 6 to the Corresponding Segment of the Preferred Route

Data Source	Route Feature	Corresponding Segment Preferred Shippingport Lateral ROW	Alternative 6
National Fuel Gas Supply Corporation	Total Length (mi)	1.12	0.98
	Total Parcels (no.)	14	16
Microsoft	Total Structures within 100 ft (no.)	1	0
Pennsylvania Department of Transportation	Total Roads Crossed (no.)	1	1
PA Ch. 93 Streams	Total Streams (no.)	2	1
	Warm Water Fishes Streams (no.)	1	1
	Cold Water Fishes Streams (no.)	0	0
	High Quality- Cold Water Fishes (no.)	1	0
	Trout Streams (stocked or natural) (no.)	0	0
NWI Wetlands	PFO (ft)	0	0
	PSS (ft)	0	0
	PEM (ft)	0	0
NLCD[a]	Developed (mi)	0.03	0.02
	Forest (mi)	0.55	0.37
	Herbaceous (mi)	0.55	0.59
Web Soil Survey	Steep Slopes (>25%) (mi)	0.10	0.13
Public Lands	(Fed/State/Local) (ft)	0	0
Recreation	Trails (no.)	0	0
	Parks, ballfields, etc. (no.)	0	0
Mining Areas	Mining areas (ft)	0	0
Prime Farmland[a]	All areas are prime farmland (mi)	0.05	0.00
	Farmland of statewide importance (mi)	0.38	0.20
	Not prime farmland (mi)	0.70	0.78
Karst	Carbonate rocks at or near the land surface in a humid climate (mi)	0	0

[a] Due to rounding, some mileages may not sum to total length of the alternative or the corresponding segment of the preferred Shippingport Lateral ROW.

Sources: Microsoft 2022, MRLC 2025, NRCS 2025, PA DCNR 2017, PA DOT 2025, PA GD 2025a,b, Pennsylvania State University 2025, USFWS 2025, USGS 2025a,b.

Pipeline Route Alternatives 7 and 8

Route Alternatives 7 and 8 deviate from the preferred route near the eastern terminus (approximate MP 6.74 to 7.26 for Route Alternatives 7 and 8) (Figure 10-1, Sheet 6 of 7). Although the preferred route crosses comparable lengths or more soils classified as prime farmland or farmland of state importance and forested areas, it was still selected due to a better constructability approach to the Service Creek Tie, given alternative slope variations and potential land instability/geohazard considerations (Table 10-6)

Table 10-6. Environmental Comparison of Major Route Alternatives 7 and 8 to the Corresponding Segment of the Preferred Route

Data Source	Route Feature	Corresponding Segment Preferred Shippingport Lateral ROW	Alternative 7	Alternative 8
National Fuel Gas Supply Corporation	Total Length (mi)	0.50	0.44	0.51
	Total Parcels (no.)	8	8	8
Microsoft	Total Structures within 100 ft (no.)	0	0	0
Pennsylvania Department of Transportation (PA DOT)	Total Roads Crossed (no.)	1	1	1
PA Ch. 93 Streams	Total Streams (no.)	0	0	0
	Warm Water Fishes Streams (no.)	0	0	0
	Cold Water Fishes Streams (no.)	0	0	0
	High Quality- Cold Water Fishes (no.)	0	0	0
	Trout Streams (stocked or natural) (no.)	0	0	0
NWI Wetlands	PFO (ft)	0	0	0
	PSS (ft)	0	0	0
	PEM (ft)	0	0	0
National Land Cover Database (NLCD)[a]	Developed (mi)	0.03	0.02	0.02
	Forest (mi)	0.41	0.37	0.42
	Herbaceous (mi)	0.06	0.05	0.07
Web Soil Survey	Steep Slopes (>25%) (mi)	0.23	0.23	0.22
Public Lands	(Fed/State/Local) (ft)	0	0	0
Recreation	Trails (no.)	0	0	0
	Parks, ballfields, etc. (no.)	0	0	0
Mining Areas	Mining areas (ft)	0	0	0
Prime Farmland[a]	All areas are prime farmland (mi)	0.12	0.04	0.07
	Farmland of statewide importance (mi)	0.07	0.04	0.07
	Not prime farmland (mi)	0.32	0.35	0.37
Karst	Carbonate rocks at or near the land surface in a humid climate (mi)	0	0	0

[a] Due to rounding, some mileages may not sum to total length of the alternative or the corresponding segment of the preferred Shippingport Lateral ROW.

Sources: Microsoft 2022, MRLC 2025, NRCS 2025, PA DCNR 2017, PA DOT 2025, PA GD 2025a,b, Pennsylvania State University 2025, USFWS 2025, USGS 2025a,b.

Pipeline Route Alternatives 9 and 10

Route Alternatives 9 and 10 deviate from the preferred route near the eastern terminus (approximate MP 7.00 to 7.20 for Route Alternatives 9 and 10) (Figure 10-1, Sheet 7 of 7). Route Alternatives 9 and 10 were not chosen due to crossing more land with slopes of 25% or greater and more soils classified as prime farmland. Additionally, Route Alternative 10 was not chosen due to more tree clearing and more soils classified as farmland of statewide importance (Table 10-7). The preferred route was also chosen due to a better constructability approach to the Service Creek Tie, given alternative slope variations and potential land instability/geohazard considerations.

Table 10-7. Environmental Comparison of Major Route Alternatives 9 and 10 to the Corresponding Segment of the Preferred Route

Data Source	Route Feature	Corresponding Segment Preferred Shippingport Lateral ROW	Alternative 9	Alternative 10
National Fuel Gas Supply Corporation	Total Length (mi)	0.17	0.22	0.27
	Total Parcels (no.)	4	4	5
Microsoft	Total Structures within 100 ft (no.)	0	0	0
Pennsylvania Department of Transportation (PA DOT)	Total Roads Crossed (no.)	0	0	0
PA Ch. 93 Streams	Total Streams (no.)	0	0	0
	Warm Water Fishes Streams (no.)	0	0	0
	Cold Water Fishes Streams (no.)	0	0	0
	High Quality- Cold Water Fishes (no.)	0	0	0
	Trout Streams (stocked or natural) (no.)	0	0	0
NWI Wetlands	PFO (ft)	0	0	0
	PSS (ft)	0	0	0
	PEM (ft)	0	0	0
National Land Cover Database (NLCD)[a]	Developed (mi)	0.01	0.04	0.08
	Forest (mi)	0.13	0.11	0.19
	Herbaceous (mi)	0.02	0.06	0.01
Web Soil Survey	Steep Slopes (>25%) (mi)	0.07	0.13	0.13
Public Lands	(Fed/State/Local) (ft)	0	0	0
Recreation	Trails (no.)	0	0	0
	Parks, ballfields, etc. (no.)	0	0	0
Mining Areas	Mining areas (ft)	0	0	0
Prime Farmland[a]	All areas are prime farmland (mi)	0.03	0.05	0.04
	Farmland of statewide importance (mi)	0.06	0.04	0.10
	Not prime farmland (mi)	0.07	0.13	0.13
Karst	Carbonate rocks at or near the land surface in a humid climate (mi)	0.00	0.00	0.00

[a] Due to rounding, some mileages may not sum to total length of the alternative or the corresponding segment of the preferred Shippingport Lateral ROW.

Sources: Microsoft 2022, MRLC 2025, NRCS 2025, PA DCNR 2017, PA DOT 2025, PA GD 2025a,b, Pennsylvania State University 2025, USFWS 2025, USGS 2025a,b.

10.4.2 ALTERNATIVE LOCATIONS FOR SERVICE CREEK TIE

One alternative site location was considered for the Service Creek Tie (Figure 10-2 at the end of this Resource Report). However, the preferred Service Creek Tie is sited within National Fuel's existing Service Creek pigging facility at the east end of the proposed Shippingport Lateral, requiring minimal new land encumbrances and no impacts to sensitive resources. This is considered to be ideal from the standpoint of operations, land acquisition, and impact minimization.

10.4.3 ALTERNATIVE LOCATIONS FOR SHIPPINGPORT M&R STATION

No alternative locations were considered for the Shippingport M&R Station. Shippingport M&R Station is sited at the Shippingport Industrial Park and is the delivery point for this Project (MP 0.00). The developed nature of this location (existing pad at the proposed M&R station location) requires minimal land disturbance, minimal new land encumbrances, and no impacts to sensitive resources. This location is considered to be ideal from the standpoint of operations and impact minimization.

10.4.4 ALTERNATIVE LOCATIONS FOR MODIFICATIONS AT MERCER CS

No alternative locations were considered for the crossover valves to be installed at the Mercer CS. Mercer CS is an existing compressor station requiring no new land acquisition, no new land encumbrances, and no impacts to sensitive resources. This is considered to be ideal from the standpoint of operations and impact minimization.

Figure 10-1. Alternative Routes Analysis
Sheet 1

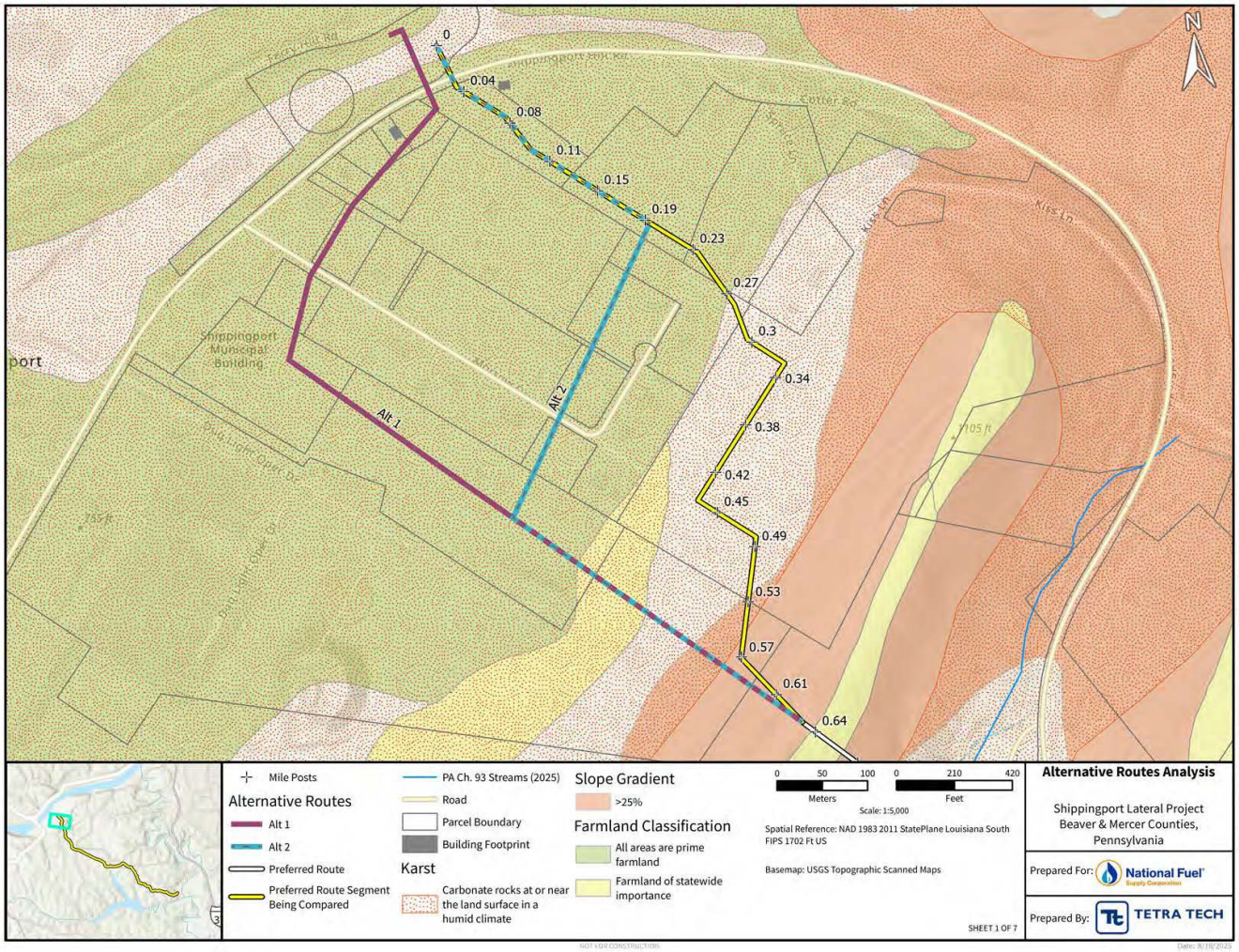


Figure 10-1. Alternative Routes Analysis
Sheet 2

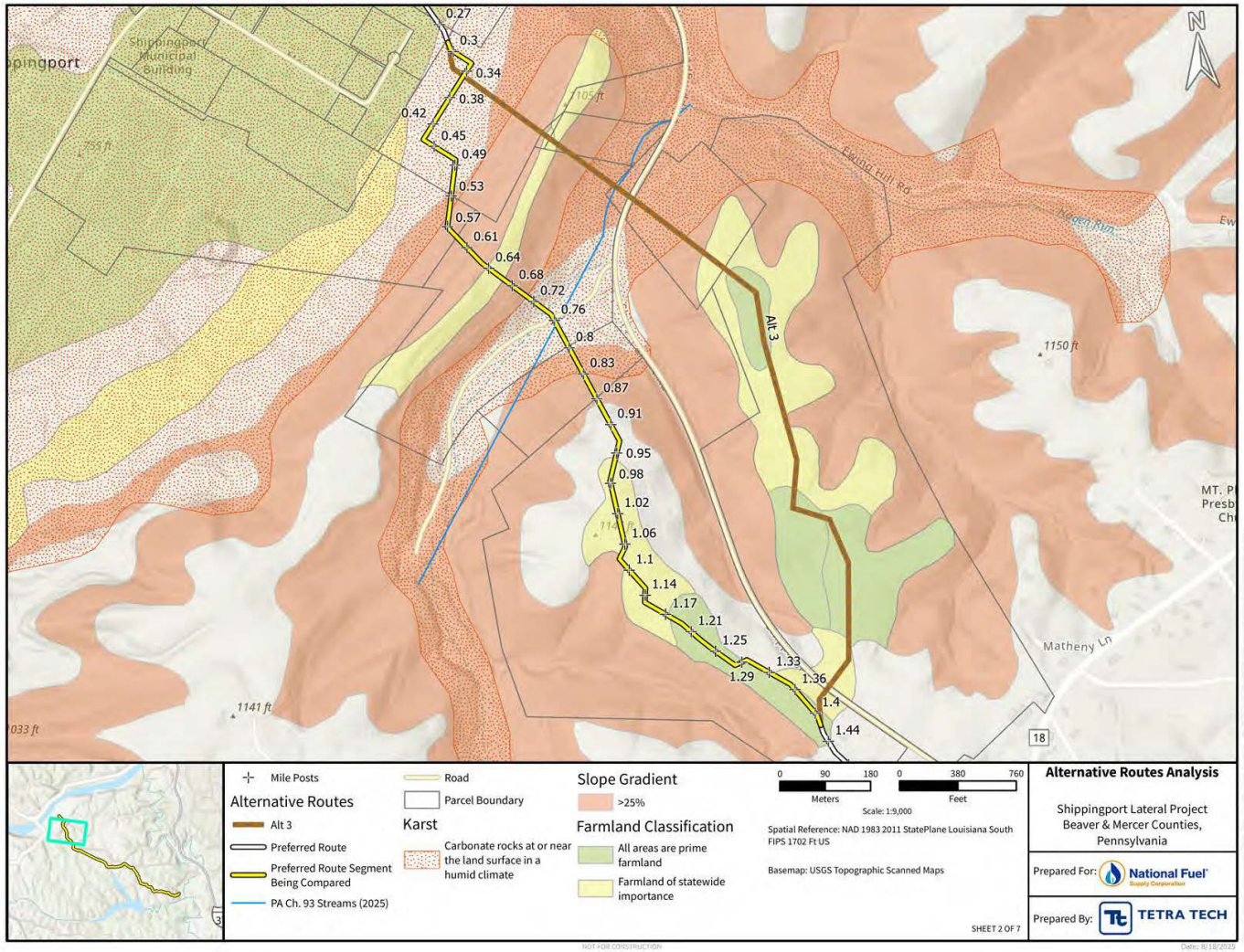


Figure 10-1. Alternative Routes Analysis
Sheet 3

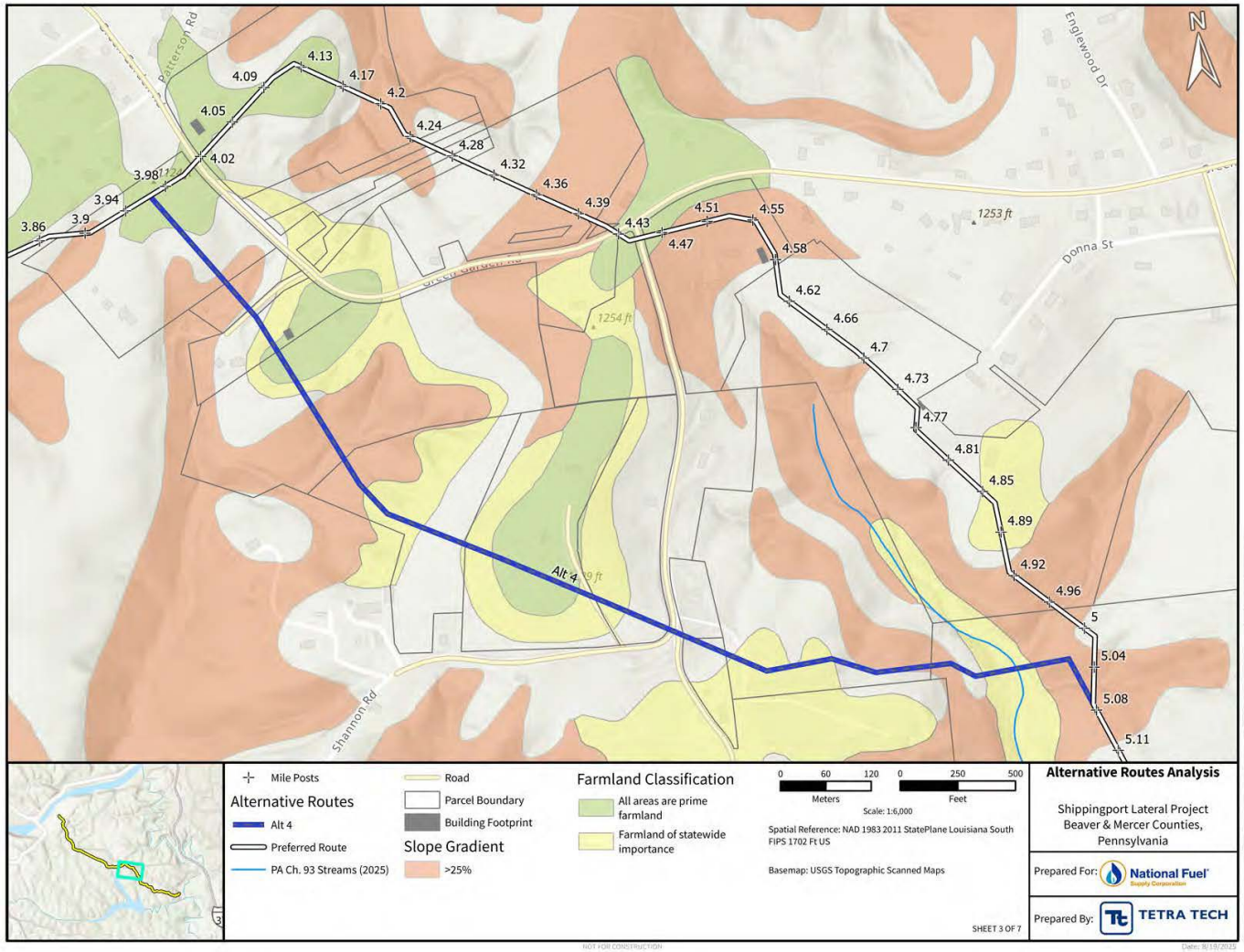


Figure 10-1. Alternative Routes Analysis
Sheet 4

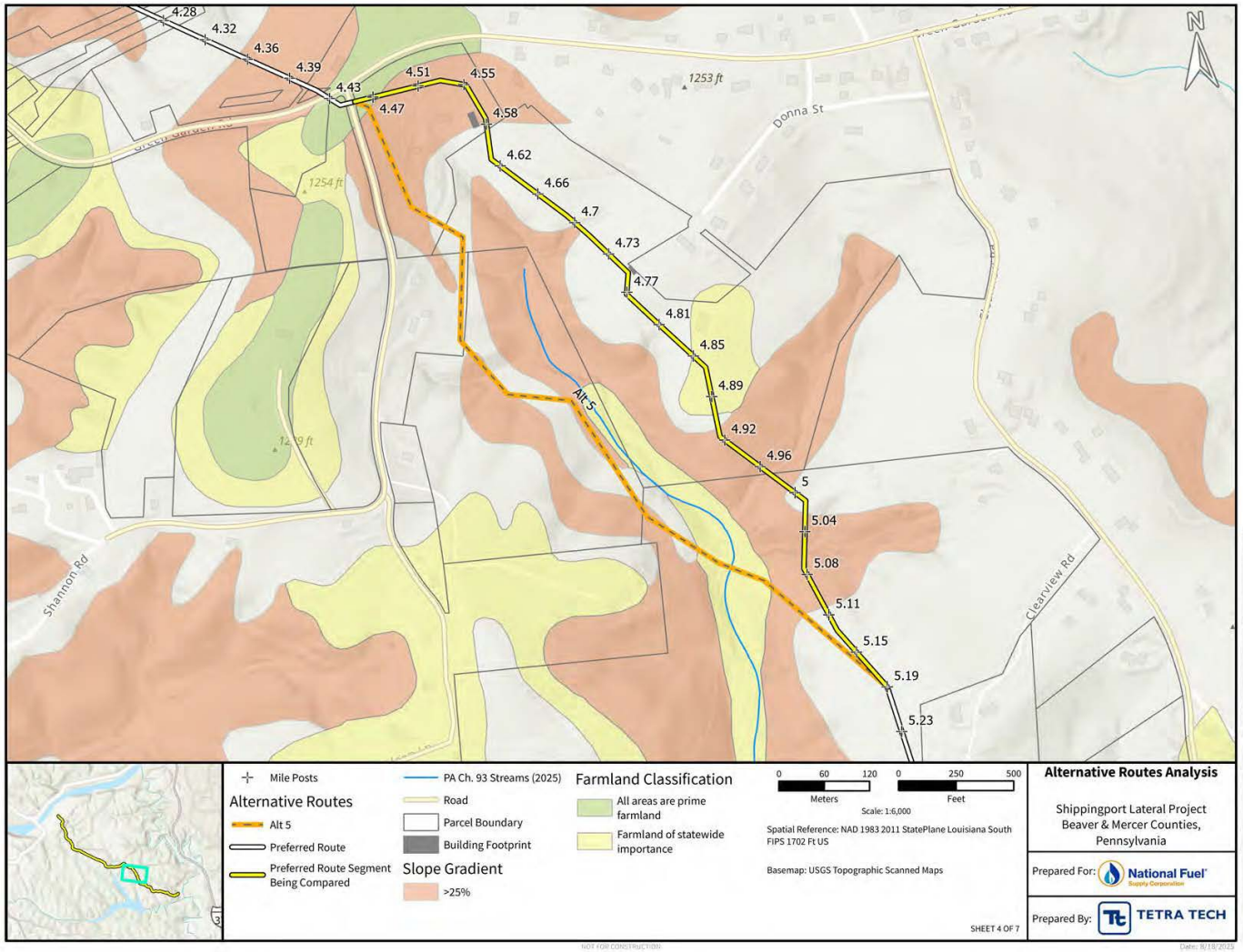


Figure 10-1. Alternative Routes Analysis
Sheet 5

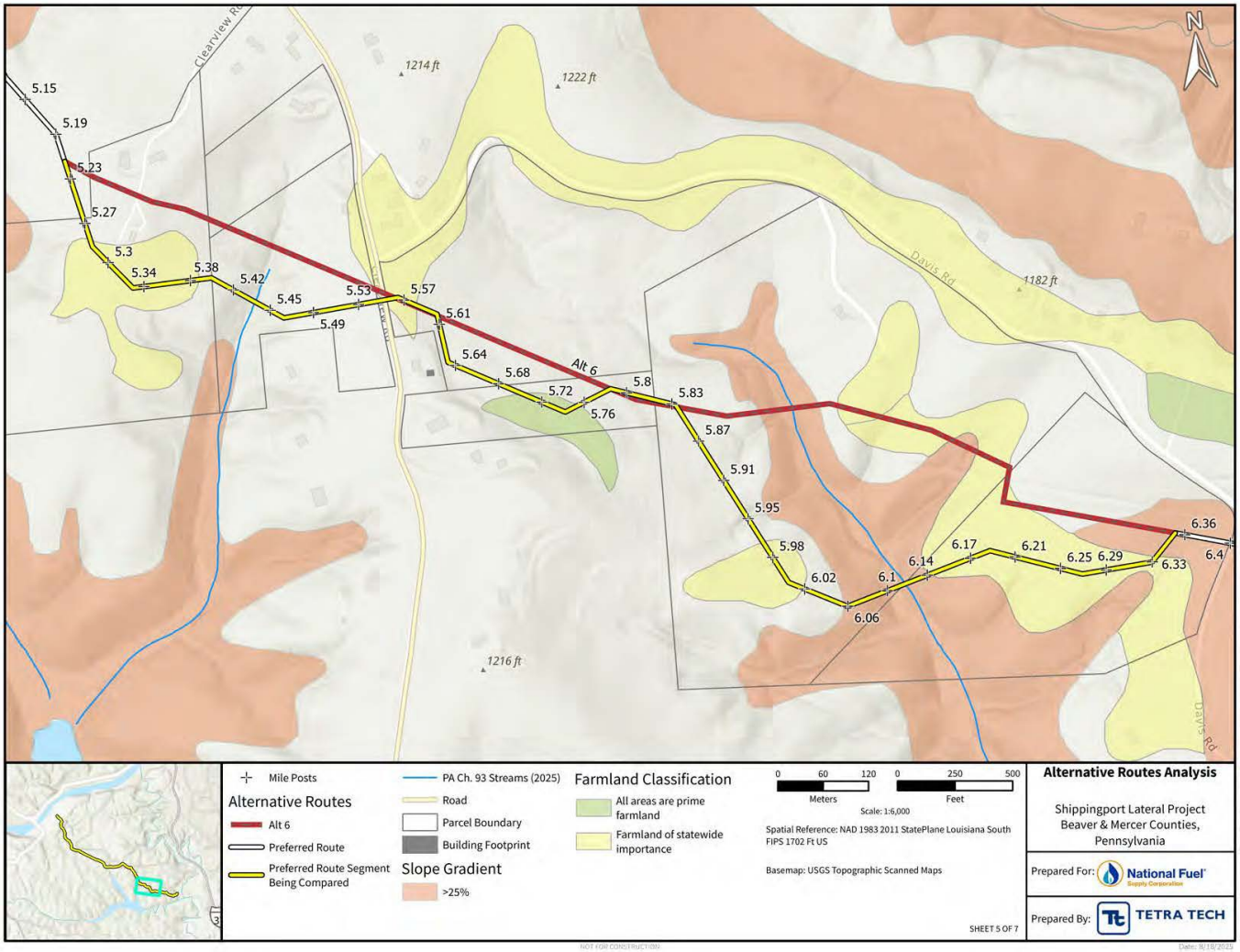


Figure 10-1. Alternative Routes Analysis
Sheet 6

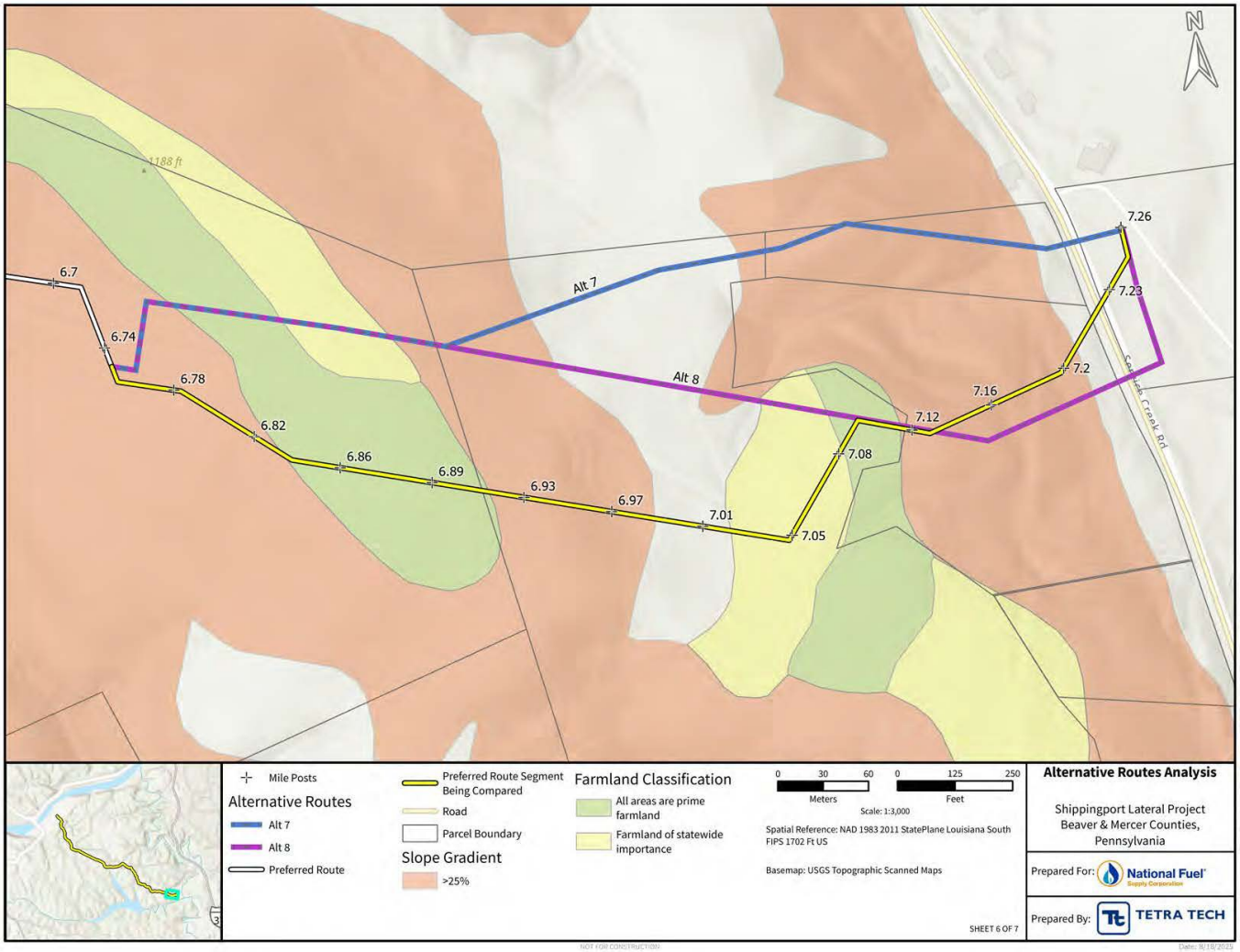


Figure 10-1. Alternative Routes Analysis
Sheet 7

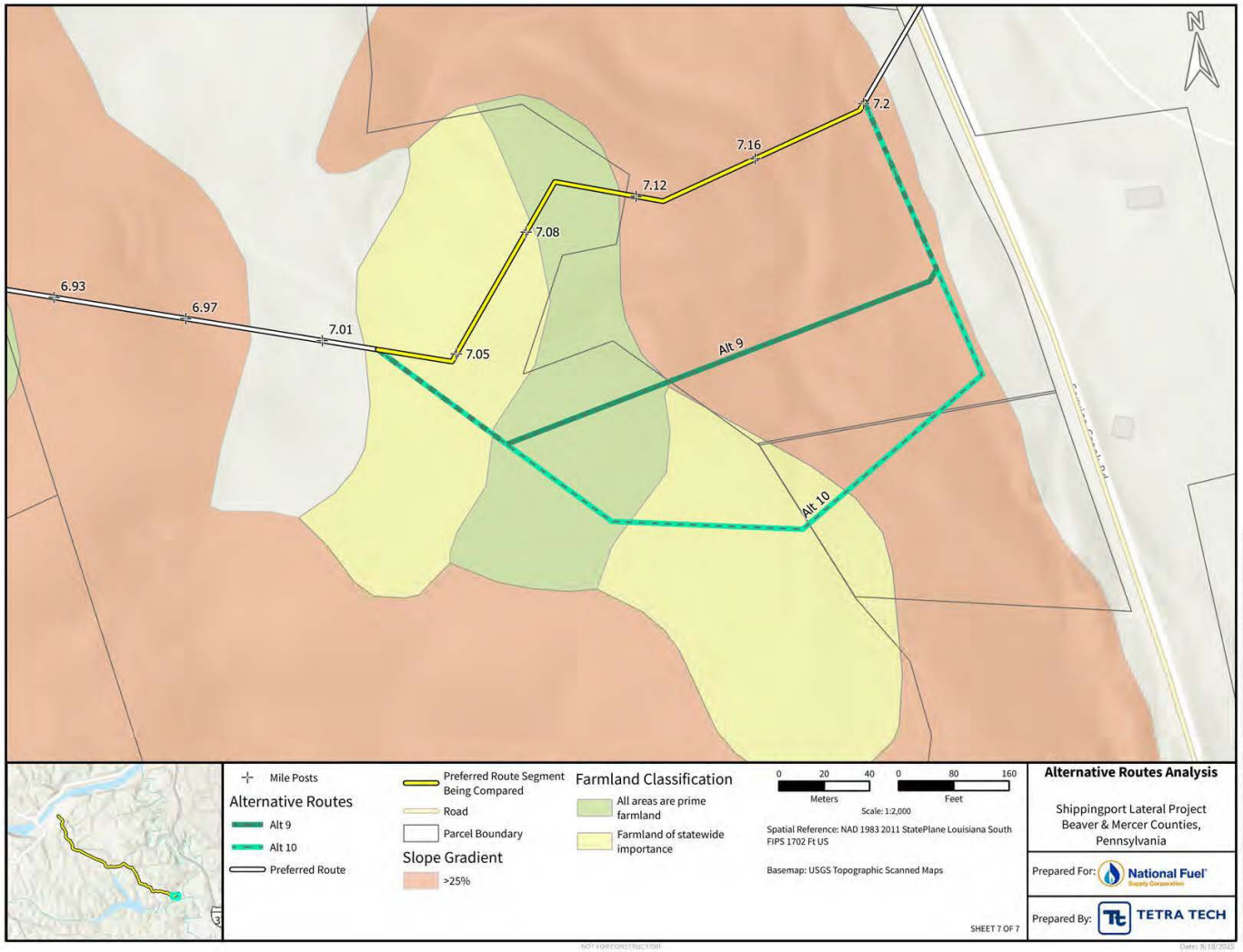
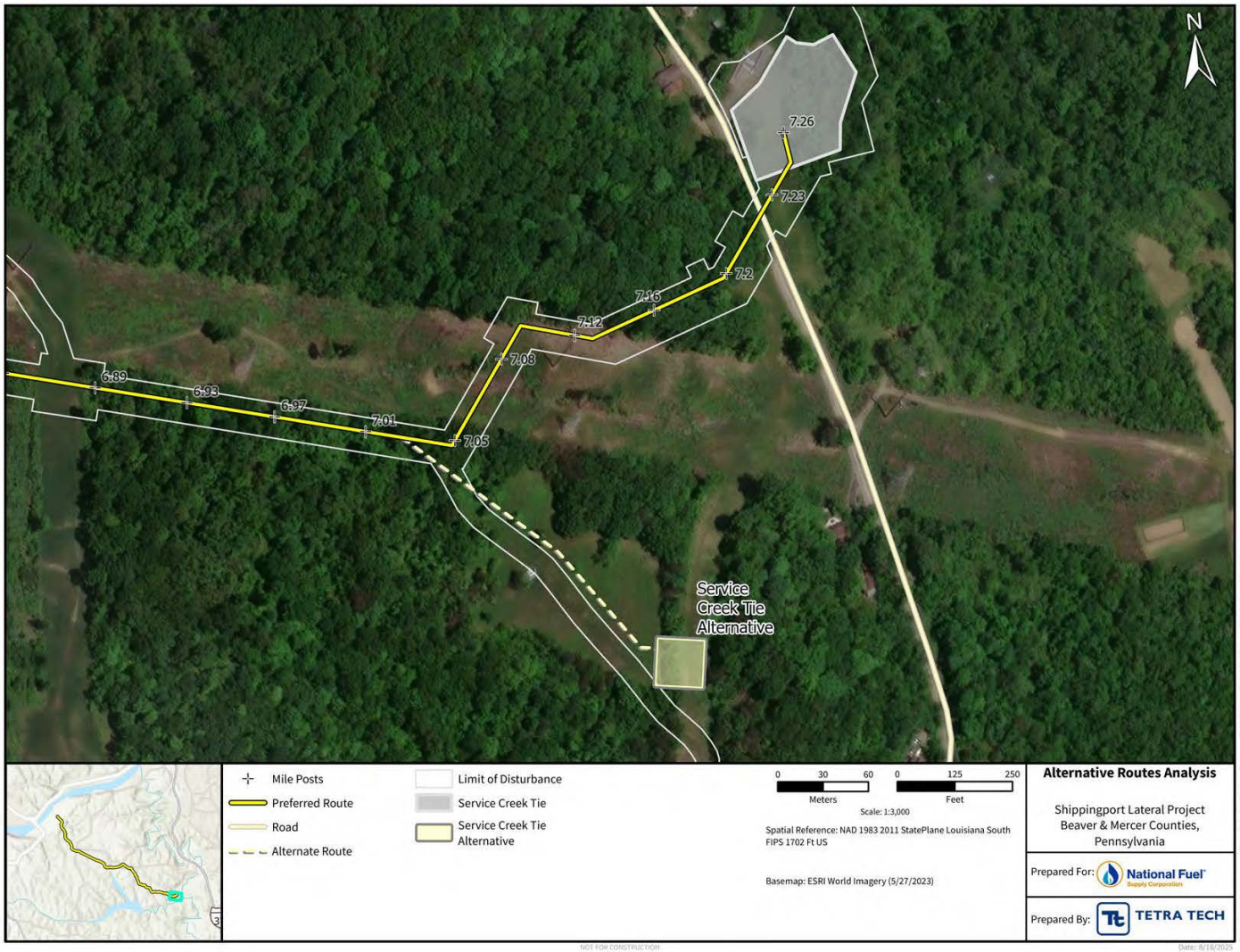


Figure 10-2. Alternatives Sites for Service Creek Tie



10.5 REFERENCES

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RESOURCE REPORT 11
RELIABILITY AND SAFETY

11.0 RELIABILITY AND SAFETY

The Project does not involve new or re-commissioned liquefied natural gas (LNG) facilities. Therefore, in accordance with FERC's guidance in 18 CFR 380.12(m), an extensive discussion regarding safety and reliability is not required.

Nonetheless, the proposed facilities described herein will be designed, constructed, tested, operated, and maintained to conform with or exceed the requirements of 49 CFR Part 192, Transportation of Natural and Other Gas by Pipeline: Minimum Federal Safety Standards, and other applicable governmental regulations. National Fuel will install the pipeline with a minimum of three feet depth of cover, which is compliant with the USDOT Pipeline and Hazardous Materials Safety Administration minimal pipeline burial requirements. Additional cover may be required in agricultural areas, at waterbodies, road crossings or other areas as necessary to maintain the integrity of the pipeline. The new pipe will be protected from corrosion using magnesium anodes for cathodic protection, which will be located entirely within the pipeline ROW. In addition, Project construction activities will be performed in accordance with National Fuel's ongoing health and safety management program to address worker safety.

RESOURCE REPORT 12
PCB CONTAMINATION

12.0 PCB CONTAMINATION

In accordance with FERC guidance provided in 18 CFR § 380.12(n), Resource Report 12 is required for applications involving the replacement, abandonment by removal, or abandonment in place of pipeline facilities determined to have polychlorinated biphenyls (PCBs) in excess of 50 parts per million (ppm) in pipeline liquids.

12.1 PCB CONTAMINATION STATEMENT

The Project involves the installation of new facilities and does not require replacement, abandonment by removal, or abandonment in place of pipeline facilities. This Project may require the removal and disposal of minimal lengths of pipe (such as approximately 20 feet) to connect the new facilities to existing facilities. Any pipe to be replaced will be removed and disposed of at a licensed waste management facility.

Based on current and recent historical PCB testing National Fuel has done in its system in the region, there are no indications of PCBs equal to or greater than 50 ppm in this part of National Fuel's system. Furthermore, since neither National Fuel nor its predecessors used PCB-containing oils in their operations, no sources [as described at 40 CFR §761.30(i)(1)(iii)(A)(3)] of PCBs would be present. Therefore, a detailed pipeline removal and disposal plan for PCB-contaminated pipeline facilities is not required for the proposed Project.

RESOURCE REPORT 13
ADDITIONAL INFORMATION RELATED TO LIQUEFIED NATURAL GAS (LNG) PLANTS

13.0 ADDITIONAL INFORMATION RELEVANT TO LNG PLANTS

The Project does not involve new or re-commissioned LNG facilities. Therefore, in accordance with FERC's guidance in 18 CFR § 380.12(o), Resource Report 13 addressing additional information related to LNG facilities is not required.